

The Effort Economy:
Investigating the Relationship between Effort Intensity and Effort
Cost

by
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Abstract

Mental effort is a valuable resource, and spending mental effort is often associated with a costly and aversive experience. Most researchers assume the cost of mental effort is evaluated by comparing some level of mental effort intensity to rest, but the present thesis challenges this idea and instead hypothesizes that effort intensity is compared to some reference point of effort intensity, leaving open the possibility that effort intensity less than a reference point might not carry a cost but be experienced positively. Three experiments built off a paradigm from the mental effort literature and adapted into a decision making framework were conducted to investigate whether mental effort intensity is evaluated relative to a reference point. Experiment 1 employed a close replication of previous research and failed to find evidence for a reference point of effort intensity. Experiments 2 and 3 modified this basic replication to make costs more explicit, and Experiment 3 specifically framed effort in terms of costs and benefits relative to a reference point. Experiment 2 failed to find evidence for a reference point of effort intensity, whereas Experiment 3 found strong evidence in favor of a reference point of effort intensity, but the direction was opposite from that predicted. The strong but reversed finding in Experiment 3 suggests that people can evaluate effort relative to a reference point, but that they might do so opposite from how they evaluate other resources, such as money.

The Effort Economy: Investigating the Relationship between Effort Intensity and Effort Cost

Two office workers, Jack and Rebecca, are working on different projects at the same company. Jack's project is easier, and he leaves the office at 3 PM every day; Rebecca's is more difficult, and she does not leave until 7 PM every day. Jack and Rebecca then get reassigned to the same project, and they both leave at 5 PM every day. Even though they have performed the same amount of work for the same time, Jack leaves the office feeling exhausted, while Rebecca leaves feeling excited and energized.

The present thesis investigates how people experience mental effort. On one hand, it is possible that people evaluate mental effort by thinking about how much harder it is to be working on a current task rather than simply resting. On the other hand, as demonstrated by the example above, it is possible that people evaluate mental effort by comparing their current effort level to how hard they *were* working or how hard they *expected* to be working. The overarching goal of the present research was to test the hypothesis that mental effort can be evaluated relative to some effort level other than rest. It is often assumed that effort spent is experienced as a costly loss of resources (Kurzban, Duckworth, Kable, & Myers, 2013). However, one aspect of the present hypothesis is that—in contexts where one is spending less effort than expected (as in Rebecca's case)—spending effort might actually feel like a *gain*.

The Introduction begins by looking at past research on the experience of mental effort, which has generally assumed that effort is evaluated relative to rest. The focus then shifts to how people think about other types of resources (e.g., money), and how judgments in these areas are thought to be made relative to the status quo or expectations

about resources. The present series of experiments tested for the presence of relative judgments in the mental effort domain using methods that drew from both traditional mental effort research and research on how people think about resources more generally.

Effort Intensity and Effort Cost

The constructs of *effort intensity* and *effort cost* arise in the literature where mental effort is considered to be a resource (Kurzban et al., 2013; Shenhav et al., 2017) and are at the foundation of the present work. Effort intensity can generally be thought of as how hard one is working at a particular task. Effort intensity is dissociable from constructs such as attention, motivation, and task difficulty (for reviews, see Inzlicht, Shenhav, Olivola, 2018; Shenhav et al., 2017; Westbrook & Braver, 2015). Consider how attention can operate without effort (e.g., noticing a deer jump out in front of one's car), how one can be motivated to rest, or how one can fail to work hard on a difficult task (Westbrook & Braver, 2015). At the end of the opening example above, both Jack and Rebecca are exerting the same level of effort intensity on their mutual project.

Effort cost can generally be thought of as the valence of the experience associated with investing effort intensity. The term effort cost is presently used synonymously with the concept *disutility* from economics. When discussing how relative judgments might influence evaluating effort intensity, the term *effort utility* is used in place of effort cost to capture the idea that effort intensity might also be associated with positive experiences. Past research generally assumed that spending effort intensity carries only inherent cost or disutility (Kurzban et al., 2013; Inzlicht et al., 2018; Shenhav et al., 2017). But I argue that—when one spends less effort intensity than expected, for example—effort intensity can be associated with positive utility. Jack and Rebecca are receiving negative and

positive utility, respectively, even though they are spending the same amount of effort intensity.

Effort Costs as Absolute

Past research on mental effort has generally assumed that the effort cost is an absolute function of effort intensity (for a review, see Kool & Botvinick, 2018). In other words, people will evaluate some level of effort intensity by comparing that level to zero, where no work is being done. This notion of an absolute function is in contrast to a *reference dependent* one, where a given level of effort intensity is evaluated relative to some other level of effort intensity that is liable to change based on experience or expectations—the notion of reference dependence is discussed further in the second half of the Introduction. The present section will review evidence and ideas that contributed to the assumption made in previous mental effort literature that the effort cost is an absolute function of effort intensity.

There has been a great deal of work addressing *why* it is that effort is experienced as costly (for a review, see Kurzban et al., 2013). The present focus is instead on investigating the conditions under which effort is costly and possible conditions where it may not be costly. The general aim of the present research is to investigate if and when effort intensity evaluation is reference dependent and possible consequences of reference dependent judgment.

The Demand Selection Task

The idea that—all things being equal—people prefer to spend less effort rather than more is not a new one (i.e., the law of less work; Hull, 1943). One challenge with testing this claim empirically is that effort tends to be confounded with other aversive

variables. For example, effortful tasks tend to be more error prone and often require more time to complete (Dunn, Lutes, & Risko, 2016). Given that the law of less work emerged in the 1940s, it is perhaps surprising that it was not rigorously in the realm of mental effort tested until recently.

Kool, McGuire, Botvinick, and Rosen (2010) tested the hypothesis that mental effort is experienced as aversive using a novel paradigm called the demand selection task (DST). Their goal was to test this hypothesis while controlling for typically confounding factors, such as errors and time on task. The present experiments built closely off the DST. In the original DST paradigm, participants were first introduced to a simple, effort-demanding task. They then performed the DST, which generally consisted of many repetitions of two-part trials. The first part consisted of a choice between two options. The second part consisted of performing the effort-demanding task associated with the chosen option. Unknown to the participants, one option was associated with a harder version (i.e., high-demand option) of the effort-demanding task than the other option (i.e., low-demand option; Kool et al., 2010).

Kool et al.'s (2010) Experiment 1 can be examined more closely to illustrate an example of the DST. Two decks of cards were presented on a computer screen. Participants were instructed to choose between them in whatever way seemed preferable, but to avoid selection strategies based on superficial features of the deck, such as its location on the screen. After a deck was selected, a card from that deck flipped upward to reveal a digit presented in one of two font colors. Participants were to perform a categorization task on the digit, and the font color cued participants as to which categorization task to perform. One color signaled participants to categorize the parity of

the digit; the other signaled them to categorize its magnitude. Switching between categorization tasks is more effort demanding than performing the same task (Monsell, 2003), and one deck was associated with 0.8 probability of switching (the high-demand option) while the other was associated with 0.2 probability of switching (the low-demand option).

If people are averse to spending mental effort, they ought to choose the low-demand option more often than the high-demand option, and this is exactly what happened (Kool et al., 2010). After about 50 DST trials, the low-demand option was selected about 60% of the time, and the low-demand bias was present for 36 of 43 participants. A total of six experiments controlled for time on task, error-proneness of tasks, and types of effort-demanding tasks. The results converged on a single idea: people experience effort intensity as costly and are motivated to avoid it.

Experiencing Effort Intensity. The effort cost in the context of the DST comes from first-hand experience with effort intensity. Participants preferred the low-demand option because they could *feel* the difference in intensity. The experiential component of effort intensity is a primary way in which effort costs are different from, say, monetary costs. This point will recur throughout the Introduction as it is central to the underlying logic of the experiments.

Effort Discounting

The effort cost is a deeply latent, phenomenal construct, making it difficult to pin down empirically (Botvinick & Cohen, 2014). In addition to the DST, other attempts to index the effort cost via revealed preferences have involved offsetting the cost with reward, and these approaches have proven fruitful. Effort discounting is generally the

approach to studying the effort cost by observing the extent to which subjective value for a reward changes as effort intensity is increased or decreased.

Consider doing long division problems for multiple hours. How much money would one need to offer before doing this work becomes attractive: \$20, \$2,000, or something in between? If one knew the exact payment where the average person would be indifferent between accepting or rejecting the offer to do long division (i.e., their indifference point), this dollar amount would be a measure of the aversiveness of doing long division. Tasks for which people need less money to reach indifference are assumed to be less aversive, while tasks requiring more money are more aversive.

The approach of titrating reward to find the value of effort has become popular recently, and there is evidence to suggest that the indifference point correlates with constructs like age and trait cognitive engagement (Westbrook & Braver, 2015; Westbrook, Kester, & Braver, 2013). The process of effort discounting also appears to hold in contexts where the reward is non-monetary, such as points in a game, and has been successfully studied in simple task switching paradigms similar to those used in the DST (Braun & Arrington, 2018).

The goal of much effort discounting research has been to chart out the relationship between how aversive a task feels and how effort-demanding the task is—in other words, the relationship between effort intensity and effort cost. Three possible models explain the change in effort cost for a one-unit increase in effort intensity: a growing increase in cost (i.e., convex), a diminishing increase in cost (i.e., concave), and a fixed increase in cost (i.e., linear; see Figure 1; for a review, see Kool & Botvinick,

2018). These models will collectively be referred to as the *traditional* models, and they are part of what the present experiments aimed to test.

Evidence in support of specific traditional models has been messy, but has led to one general conclusion: The relationship between effort intensity and its associated cost is likely to be more complex than linear (Navon & Gopher, 1979). Most studies found evidence in favor of the convex model (Bialaszek, Marcowski, & Ostaszewski, 2017; Braun & Arrington, 2018; Klein-Flügge, Kennerley, Saraiva, Penny, & Bestmann, 2015; Kool & Botvinick, 2014), but there are also studies that found evidence in favor of the concave model¹ (Chong et al., 2017; Nishiyama, 2016; Ostaszewski, Babel, & Swebodzinski, 2013), while some found equal support for both (Nishiyama, 2014).

Experiencing Effort Intensity. Researchers using the effort discounting approach have differed with respect to how they have implemented effort intensity in the context of choice. The DST forces participants to experience effort intensity after each selection (Kool et al., 2010). In contrast, some experiments ask participants to make choices between effort-demanding options where the effort intensity of each option is fully *imagined* and described to participants (Ostaszewski et al., 2013). Other experiments are in between, where the effort intensity associated with choices is only realized on a randomly-selected minority of trials (Klein-Flügge et al., 2015), or where

¹ Effort discounting research typically does not involve modeling effort intensity against its cost, but rather it tends to involve modeling subjective value of reward as a function of effort intensity. The body of findings being referred to as supporting the concave model in the text, for example, are findings that support the idea that increases in smaller amounts of effort are associated with larger devaluations of reward than increases in larger amounts of effort. This pattern is conceptually compatible with a concave relationship between effort cost over effort intensity, even though the function of subjective value over effort intensity is in fact hyperbolic in shape.

effort intensity is only experienced at the beginning and end of the experiment and selections are made in between (Chong et al., 2017).

Some concept of effort intensity needs to exist in one's mind before that concept can take on a valence (i.e., effort utility) in the mind. In a perfect world, we might stick an "effort utility meter" into one's head, while the person is exerting effort, to measure how the effort intensity is being experienced. But, unfortunately, the best utility meter available is to interrupt effort intensity and confront the person with choices. How best to establish effort intensity and intersperse choice is unclear. When trying to measure the effort cost or utility associated with painting a fence, for example, does it matter whether a person's preferences are assessed when only having a description of how much work it will be vs. when the person is currently working on the fence?

The present thesis aimed to cover both ends of the issue of experiencing effort when testing the relationship between effort intensity and effort cost. The goal was to test this relationship both in contexts where effort intensity immediately followed choice (as in the original DST; Kool et al., 2010), and in contexts where there was more separation between choice and effort exertion.

Absolute Zero. The three traditional models all make a common assumption. A given level of effort intensity on the x-axis of Figure 1, for example, is always evaluated relative to the zero point on the x-axis (i.e., absolute zero), where no work is being done. Evaluating effort intensity relative to absolute zero leaves no room for considering how relative changes in effort intensity can lead to different effort experiences (i.e., cost or utility). It also means that a given level of effort intensity is associated with a fixed cost, regardless of how one arrived at that level of effort intensity.

Imagine Jack and Rebecca before being reassigned to a mutual project. On the convex model, for example, Rebecca should experience a much higher cost than Jack for exerting a higher level of intensity, which is consistent with the example. However, after being reassigned to a mutual project, Jack and Rebecca arrive at the same point at the x-axis of effort intensity (on Figure 1) and should consequently experience the same degree of effort cost. This violates people's intuitions about the opening example, as Rebecca has a more pleasurable experience than Jack despite doing the same level of work. A model is reviewed in the following section that can explain these discrepant experiences.

Value as Subjective

Looking at research on how people think about resources more generally is helpful for gaining perspective on how people might think about mental effort. For example, with monetary costs and benefits, do people always have the same satisfaction from owning the same amount of money, or does it matter how one arrived at a current level of wealth? Consider the following two problems (taken from Kahneman & Tversky, 1979).

Problem 1: In addition to whatever you own, you have been given \$1,000. You are now asked to choose between:

A: 50% chance of receiving \$1,000 and 50% chance of receiving \$0

B: Receiving \$500 for sure

Most people (84%; Kahneman & Tversky, 1979) are *risk averse* in this context and will choose for sure gain of option B over the chance of a larger gain of option A.

Problem 2: In addition to whatever you own, you have been given \$2,000. You are now asked to choose between:

C: 50% chance of losing \$1,000 and 50% chance of losing \$0

D: Losing \$500 for sure

Most people (69%; Kahneman & Tversky, 1979) are *risk seeking* in this context and will choose option C. Notice, however, that these problems are identical in terms of their final states. If the endowments in the problems are added to the outcomes, the problems reduce to:

A and C: 50% chance of gaining \$1,000 and 50% chance of gaining \$2,000

B and D: Gaining \$1,500 for sure

A traditional economic model (i.e., expected utility theory; Bernoulli, 1738/1954) posits that utility is a concave function of wealth. Because states of wealth are defined in absolute terms, this theory predicts that a given end state of wealth should be preferred to the same extent regardless of how one arrived at that state of wealth. The example above clearly violates this assumption, in that the same end states are preferred differently depending on initial endowments and framing of prospects (Kahneman & Tversky, 1979).

The relationship between wealth and utility described in expected utility theory is analogous to that between effort intensity and effort cost in the traditional models discussed in the previous section. Some absolute state of effort intensity is associated with some fixed amount of effort cost, regardless of how one arrived at that level of effort intensity. But, just as the anecdotal example of Jack and Rebecca clashes with traditional effort models, so too do examples of choice under risk (like the one above) clash with expected utility theory.

Prospect Theory

A famous revision of expected utility theory, *prospect theory*, was proposed to better describe how people make decisions under risk (Kahneman & Tversky, 1979). Prospect theory takes into consideration the tendency for people to evaluate prospects relative to some reference point. A *prospect* is generally thought of as an offer that yields a certain outcome (or outcomes) with a certain probability (Kahneman & Tversky, 1979). The tendency to evaluate prospects relative to a reference point explains why two identical end states might be experienced differently. Prospect theory proposes that a choice between prospects unfolds over two general stages: *editing* and *evaluation*. Editing involves establishing a reference point from which to evaluate prospects; evaluation involves determining and maximizing utility across prospects. While editing is important for present purposes, evaluation will be discussed first as there has been significantly more work investigating this stage.

The general form of evaluation in prospect theory is given by Equation 1 (Kahneman & Tversky, 1979):

$$V = \sum_{i=1}^N \pi(p_i) v(x_i) \quad (1)$$

Value for an outcome [$v(x_i)$] is weighted by its subjective probability [$\pi(p_i)$] and summed across all outcomes in a prospect (N) to give the total subjective value of a prospect (V). The value function [$v(x_i)$] is the most important of the terms in Equation 1 for present purposes. The x-axis of the value function is defined relative to a reference point (see Figure 2). In other words, instead of the x-axis being absolute states of wealth (as in expected utility theory), prospect theory proposes that the x-axis is coded as gains

or losses from the reference point (Kahneman & Tversky, 1979). This function consists of two central features: diminishing sensitivity and loss aversion.

Diminishing Sensitivity. Which of the two differences below feels larger:

Difference A: Gaining \$5 or gaining \$500

Difference B: Gaining \$10,005 or gaining \$10,500

What about for losses:

Difference C: Losing \$5 or losing \$500

Difference D: Losing \$10,005 or losing \$10,500

Prospect theory proposes that gains and losses closer to the reference point (i.e., Differences A and C) are associated with larger changes in value than those further from the reference point (i.e., Differences B and D; Kahneman & Tversky, 1979). In other words, people are more sensitive to a change in smaller amounts of money than they are to the same change in larger amounts. The reference point here can be assumed to be one's current level of wealth. Diminishing sensitivity is modeled by the value function taking a concave shape above the reference point and convex shape below it (see Figure 2). In mathematical terms, the point at which the function changes shape from convex to concave is known as an *inflection point* in the function. This inflection point implies that the rate of change in value as a function of money decreases both to the left and to the right of the inflection point. In theoretical terms, the decreasing in rate of change to the left and right of the inflection (i.e., reference) point models diminishing sensitivity to changes in money as both gains and losses become more extreme.

The inflection point in prospect theory's value function is a critical component that distinguishes it from expected utility theory. Similar to prospect theory, expected

utility theory also predicts diminishing sensitivity with increases in wealth, but, because the function's shape is consistently concave (i.e., no inflection point) over states of wealth, it makes the opposite prediction with respect to losses. Imagine someone with a net worth of \$100,000 facing the losses described above. Expected utility theory predicts that the difference in wealth between \$99,995 and \$99,500 (i.e., Difference C) should be associated with a smaller change in utility than the difference between \$89,995 and \$89,500 (i.e., Difference D), which is opposite the prediction from prospect theory.

Loss Aversion. Loss aversion is generally the idea that losing \$500 is more aversive than gaining \$500 is desirable. More precisely, some amount of loss is associated with greater disutility than the same amount of gain is associated with positive utility (Kahneman & Tversky, 1979). This is modeled by the slope of the function below the reference point being steeper than that above it (see Figure 2).

Decision Under Risk. Diminishing sensitivity and the probability weighting function together explain why risk preferences reverse for gains vs. losses. Just as in expected utility theory, a concave value function over states of wealth (i.e., gains) gives rise to the prediction that people will be risk averse. The probability weighting function $[\pi(p_i)]$ leads people to be *more* risk averse for moderate or extreme probabilities as these probabilities are discounted (Kahneman & Tversky, 1979). All that is important about the probability weighting function for present purposes is that moderate probabilities tend to be underweighted.

Imagine a choice between a gamble where there is a 50% chance to gain \$100 (or gain nothing) vs. gain \$50 for sure. Recall that, in prospect theory, value for a prospect is found by $[V = \sum v(X_i)\pi(p_i)]$. Even without the influence of the probability weighting

function [i.e., $\pi(0.5) = P(0.5)$], due to the concave shape of the value function over gains, a sure gain of \$50 still has a higher value than the gamble [i.e., $v(\$100)P(0.5) < v(\$50)P(1)$]. Adding the influence of the probability weighting function makes the gamble even *less* attractive and makes people more risk averse because [$\pi(0.5) < P(0.5)$].

The opposite is true for losses. Imagine that there is a 50% chance to lose \$100 (or lose nothing) vs. losing \$50 for sure. Due to the convex shape of the value function for losses, a sure loss of \$50 carries more disutility than the gamble. The probability weighting function again adds to this bias, making people underweight the probability and leading them to estimate the gamble more favorably. The change in shape of the value function around the reference point explains why people will tend to be risk seeking when faced with losses but risk averse when faced with gains (Kahneman & Tversky, 1979).

Loss aversion, diminishing sensitivity, and risk preferences are concepts that can be used to describe how evaluating prospects occurs relative to a reference point. In the examples above, this reference point is some reference level of wealth.² The present series of experiments tested whether effort intensity is evaluated relative to some reference level of effort intensity, and these concepts (i.e., loss aversion, diminishing sensitivity, and risk preferences) were relied upon to test this hypothesis.

² Prospect theory has been extended beyond monetary domains. For example, people are loss averse when considering different types of vacations (Kahneman & Tversky, 1979) and when evaluating treatments for hypothetical diseases (Kahneman & Tversky, 1981). However, the present discussion focuses on monetary examples to draw an analogy to mental effort.

Experiencing Costs. In paradigms used to investigate prospect theory and loss aversion, there was often no experience of costs prior to choice (Kahneman, Knetsch, & Thaler, 1990; Kahneman & Tversky, 1979; Novemsky & Kahneman, 2005; Tversky & Kahneman, 1991). For example, in the seminal studies on prospect theory, choices were based on fully hypothetical scenarios (Kahneman & Tversky, 1979)—meaning participants were asked to make choices between prospects while simply imagining that the outcomes were real. Note how this is starkly different from, say, the original DST (Kool et al., 2010; Experiment 1), where an experience of effort cost followed *each* choice.

The present hypothesis posits that evaluation of effort intensity is reference dependent *regardless* of how the experience of intensity is coupled with choice. This means that people should compare effort intensity to a reference when they have extensive or ongoing experience with the effort intensity, and they should also do so when the effort intensity is simply described to them. There just needs to be *some* representation of effort intensity and reference effort intensity in order for comparison to take place. To what extent experience is necessary to construct these representations was tested in the present experiments.

The Reference Point

A lesser known aspect of prospect theory is how the reference point is determined during the initial editing phase. To borrow an example from Kahneman (1992), if a person expects a \$5,000 raise but only receives \$3,000, is this viewed as a gain of \$3,000 or a loss of \$2,000? Many authors agree that there is a shortage of research on the reference point (Hack & Bieberstein, 2015; Köszegi & Rabin, 2006), and this section

covers two areas of research on the reference point that are particularly important for present purposes: how information is combined to compute the reference point (i.e., segregation vs. integration), and whether the reference point is based on one's current or expected state (i.e., status quo vs. expectation).

Segregation vs. Integration. Imagine you own a stock and are deciding whether to sell it. You need some basis from which to compare the stock's current price in order to know whether receiving its current payout will satisfy you. To get an estimate of whether you are getting a good deal, you can compare the stock's current price against the price at which you purchased the stock, against how the stock performed this quarter, against any price point in between, or against some aggregation of these price points.

When evaluating prospects, there are often many possible reference points from which to compare a particular end state (Kahneman, 1992). There are two general schools of thought on how past information is used to construct a reference point. One idea is that information is *aggregated* to form a reference point (Helson, 1964; Lattin & Bucklin, 1989; Putler, 1992; Winer, 1986). For example, when evaluating a gamble, people might aggregate over the outcomes and probabilities to use the expected value of the gamble as a reference point (Hack & Bieberstein, 2015).

Alternatively, instead of aggregating past information, people will rely on single, salient data points to construct a reference point (i.e., the segregated hypothesis). Evidence supporting this claim comes from a study where participants made selling decisions about a stock given its price over time (Baucells, Weber, & Welfens, 2011). This study investigated whether selling decisions were more likely to be affected by discrete data points of the stock's price history (e.g., starting price, highest price, lowest

price) or an aggregation of its price history. Examples of aggregation include the average of the price across time, or a recursive aggregation whereby more recent price points are weighted disproportionately more strongly than more distant price points.

Results showed that selling decisions were influenced to the greatest extent by the stock's starting price, while decisions were influenced to a lesser extent by the most recent price (Baucells et al., 2011). The authors argued that sellers were most likely to feel as though they have made a “gain” from the stock if the price has moved above the purchase (i.e., initial) price, regardless of how the stock has performed over time. The same was true for feelings of loss if the price falls below the purchase price (Baucells et al., 2011). The authors went on to argue that the reference point was likely anchored to the endpoints of an information timeline rather than information in the middle or any aggregation of the information.

The notion that initial information is heavily weighted when constructing a reference point was a guiding idea for the present experiments. If people rely on reference points for evaluating effort intensity, it was assumed that this reference point was established early on with one's initial experience or exposure to an effort-demanding task and that subsequent exposure had a smaller impact on the reference point.

Status Quo vs. Expectations. It was initially thought that people took their status quo as a reference point—for example, as described in the section on the “evaluating” stage above, determining the value for an outcome involves comparing that outcome to one's current level of wealth. But, as suggested by Kahneman's (1992) example of a worker anticipating a raise, one might experience loss if receiving an outcome that was less than *expected*, even if the outcome increases one's current level of wealth.

There is generally more evidence in support of the idea that the reference point hinges on expectations rather than the status quo. For example, the endowment effect is the finding that people tend to value a good or item in their possession more than if that same good were not in their possession (Thaler, 1980). It was thought that this was because the status quo of owning the good is adopted as the reference point, and the possibility of losing the good is more aversive than gaining the good is pleasurable (when it is not in one's possession; Tversky & Kahneman, 1991).

The endowment effect was curiously absent, however, in cases where a seller was striving to reduce inventory or a buyer was spending according to a budget (Novemsky & Kahneman, 2005; Tversky & Kahneman, 1991). Additionally, novice, but not expert, card traders exhibited the endowment effect when valuing their cards (List, 2004). These findings were initially thought to reflect boundaries of loss aversion (Novemsky & Kahneman, 2005), but were later taken as evidence of the reference point being set by expectations rather than the status quo (Kőszegi & Rabin, 2006). For example, the reason that expert card traders do not experience the endowment effect is because they *expect* to eventually forgo their newly acquired cards. There is also evidence to suggest that people will adopt the expected value of a gamble as an expectations-based reference point rather than the status quo of what is earned from the gamble (Hack & Bieberstein, 2015). In situations where people were deciding how long to work on a task, they were sensitive to *expectations* about how much they would be compensated for the work. The degree to which they chose to work longer than what they expect to be paid for negatively correlated with their personal degree of loss aversion (Abeler, Falk, Goette, & Huffman, 2011).

Taken together, these findings suggest that expectations play a larger role than the status quo in constructing the reference point. Applied to the mental effort domain, this could mean that simply having an expectation about engaging in high effort intensity would be sufficient to make a lower level of effort intensity feel like a gain. If a student is told to write a 15-page paper but the teacher later reduces the load to 10 pages, does the student experience this level of effort as a gain of 5 pages or as a loss of 10? The findings above might suggest that this student would use 15 pages as a reference from which to evaluate 10 pages, consequently experiencing the writing of the 10-page paper as a gain.

Effort and Risk

There has been one prior study that has implemented risk with effort-based outcomes in a way that was similar to how the present experiments did so (Apps, Grima, Manohar, & Husain, 2015). This study ended up concluding that people are risk averse for all effort-based prospects—but analyzing their results in terms of the prospect hypothesis leads to a different conclusion. Participants in Apps et al. (2015) faced repeated choices between accepting a fixed level of effort intensity vs. a gamble between a higher or lower level of effort intensity. The outcome of the sure option was the average of the outcomes of the gamble, and the gamble's outcome was realized at .5 probability. Participants tended to prefer the safe, moderate option over the gamble (Apps et al., 2015). However, there was no explicit consideration of reference effort intensity, and it is difficult to infer what a reference intensity might have been in this study.

Because the sure option was always fixed at the same level of effort intensity, one might assume—if effort intensity evaluation is reference dependent—that the level of the sure option would serve as the reference. Under these conditions, the results are

accounted for by a prospect model of effort evaluation. Because losses loom larger than gains, the value of the gamble would be more negative than the weighted sum of its outcomes (i.e., the sure option). This would be analogous to a choice between (1) accepting a gamble between receiving vs. paying \$20 at .5 probability vs. (2) rejecting the gamble (i.e., outcome of \$0 at 1.0 probability). Prospect theory predicts that, under these conditions, people would be risk averse, which is consistent with the results of Apps et al.'s (2015) study. Experiments 2 and 3 extended the logic of Apps et al. (2015) by testing risk preferences after establishing a reference point of effort intensity.

Integrating Mental Effort and Prospect Theory

The core goal of the present research was to take ideas from prospect theory and test to what extent they extend to a mental effort context. One of the central ideas from prospect theory is that people do not evaluate prospects in absolute terms, but rather they evaluate prospects by considering outcomes as gains or losses from a reference point (Kahneman & Tversky, 1979). Much work in the mental effort domain assumes that effort intensity is evaluated in absolute terms, where effort intensity is not thought of as gains or losses but rather as different degrees of cost or loss (Bialaszek et al., 2017; Kool & Botvinick, 2018; Kool et al., 2010; Ostaszewski et al., 2013). The core goal of the present research is to extend prospect theory's notion of subjective evaluation to mental effort.

Prospect theory further claims that the shape of the value function is different depending on whether an outcome is considered to be a gain or a loss (Kahneman & Tversky, 1979). People are more sensitive to gains and losses near the reference point than they are to gains and losses far from the reference point. For example, going from

gaining \$50 to gaining \$100 feels like a larger increase than going from gaining \$20,050 to gaining \$20,100. This prediction of prospect theory is called diminishing sensitivity, and it is modeled by the value function taking a concave shape above the reference point (i.e., for gains) and a convex shape below it (i.e., for losses). Diminishing sensitivity is also one of the theory's features (in addition to the probability weighting function) that gives rise to the asymmetrical risk preferences around the reference point. People are more likely to be risk averse when choosing between prospects where all outcomes are evaluated as gains and are more likely to be risk seeking when making such a choice when all outcomes are evaluated as losses.

The final idea from prospect theory that was tested in an effort context in the present experiments is loss aversion, or the idea that losses are associated with greater disutility than gains are associated with positive utility. This prediction is modeled by the value function taking a steeper slope for losses than for gains. If one is used to leaving the office at 5 PM, for example, loss aversion would imply that having to leave at 7 PM would be more aversive than leaving at 3 PM is desirable.

Note that loss aversion does not directly contribute to prospect theory's prediction of reversed risk preferences for gains and losses. This prediction is instead driven by the non-linearity implied by both diminishing sensitivity and the probability weighting function. Imagine removing diminishing sensitivity and the probability weighting function from prospect theory and leaving only loss aversion. Instead of having convex and concave shapes on either side of an inflection point, the value function would be linear, with a certain positive slope for gains and a steeper, negative slope for losses. Such a function implies risk neutral attitudes toward a choice between prospects that are

matched in expected value when the outcomes of the prospects are either all losses or all gains. For example, prospect theory with *only* loss aversion would predict that $V(-\$100, 0.5; \$0, 0.5) = V(-\$50, 1)$ and that $V(\$100, 0.5; \$0, 0.5) = V(\$50, 1)$. Loss aversion most notably comes into play when directly choosing between prospects with positive *and* negative outcomes [e.g., $V(\$50, 0.5; -\$50, 0.5) < V(\$0, 1)$]. In the previous example, loss aversion biases choice toward the safe option of \$0 because -\$50 is felt as more costly than \$50 is desirable. Only by adding in the non-linear influences of diminishing sensitivity and / or subjective probability weighting does prospect theory predict asymmetrical risk preferences of the type $V(-\$100, 0.5; \$0, 0.5) > V(-\$50, 1)$ vs. $V(\$100, 0.5; \$0, 0.5) < V(\$50, 1)$. The major focus of two of the present experiments was to test for diminishing sensitivity by testing whether preference reversals of the type described in the previous sentence occur in mental effort contexts.

The specific mental effort context that was employed throughout the present experiments was task switching. Task switching was selected out of a variety of possible effort-demanding tasks for several reasons. One reason is that it is a task that has been widely used and validated, both generally for investigating cognitive control and attentional mechanisms (Monsell, 2003), and also specifically for investigating phenomena surrounding mental effort (Kool & Botvinick, 2018; Kool et al., 2010). Another reason for selecting task switching is that the paradigm allows for parametric manipulation of effort demand—probability of switching between tasks can range continuously from zero (easiest) to one (hardest; but see the Pilot Experiment below). This flexibility in varying effort demand is in contrast to an effort-demanding task such as the N-back task (Westbrook et al., 2013), for example, where there are very few levels

of difficulty available with which to confront participants. One final reason for selecting task switching is simply that it is a highly-used paradigm in our laboratory—we have well-established procedures in our laboratory for data quality control, such as trimming error trials or removing participants who do not follow directions.

Hypotheses

The present experiments tested predictions of prospect theory in a mental effort context. In the context of the present experiments, the term *prospect model* was used to generally refer to the collection of ideas from prospect theory (i.e., diminishing sensitivity, loss aversion) that were tested in the present experiments (the term *prospect model* was used similarly to the term *prospect theory*, but the term *prospect theory* was reserved for generally referring to the original conception of prospect theory by Kahneman & Tversky, 1979). The term *prospect hypothesis* was reserved for referring to the exact empirical prediction that the prospect model makes in the context of a given experiment.

Experiment 1 was a close replication of Kool et al.'s (2010) DST but several components were added to the core paradigm. Because the present hypotheses are concerned with testing the *shape* of the relationship between effort intensity and effort cost (i.e., linear vs. concave vs. convex vs. prospect), using only the low- and high-demand effort intensities as in the original DST was not sufficient. More effort intensity levels were introduced into the DST in order to test the relationship between effort intensity and effort cost. The other component that was added was the concept of a reference level of effort intensity. Participants learned about an effort intensity that was described to be typical or expected (i.e., 0.5 probability of switching in a task switching

context). They subsequently made DST-like choices between a deck at the reference-level of effort intensity (i.e., the reference deck) vs. a deck that switched tasks more or less often than the reference deck (i.e., the critical deck). If people evaluate effort intensity prospects relative to a reference point, then critical decks demanding more effort intensity than reference should be evaluated as losses while the opposite evaluated as gains.

The prospect hypothesis predicts that effort intensity is evaluated relative to a reference point, and that sensitivity to changes in effort intensity diminishes further away from the reference point. Therefore, on the prospect hypothesis, people should be disproportionately more sensitive to differences in effort intensity when the critical deck is more similar in effort intensity to the reference deck than when it is more different. Imagine someone with a reference point of leaving work at 3 PM every day. This person confesses that she would pay \$50 to avoid having to leave work at 5 PM. This person is then asked how much she would pay to avoid leaving work at 7 PM. The prospect hypothesis predicts that—despite the fact that four extra hours (i.e., 7 PM) is twice as much as two extra hours (i.e., 5 PM)—this person would not be willing to pay \$100 to avoid leaving at 7 PM, but rather some amount less than \$100 but greater than \$50. Even though she would pay more to avoid leaving at 7 PM vs. what she would pay to avoid leaving at 5 PM, she is disproportionately more sensitive to an extra two hours vs. four hours of work because she is willing to pay less than double to avoid leaving at 7 PM. This same logic was relied upon to test the diminishing sensitivity aspect of the prospect hypothesis in Experiment 1, as people were expected to prefer an extremely easy critical

deck, for example, less than twice as much as they preferred a moderately easy critical deck.

The prospect hypothesis also predicts that, overall, people would be more sensitive to the difference between harder than reference decks and the reference deck than they would have been to easier than reference decks and the reference deck (i.e., loss aversion). Imagine someone with a reference point of leaving the office at 5 PM. The prospect hypothesis predicts that this person will pay more to avoid leaving at 7 PM than he would pay to leave at 3 PM.

Experiments 2 and 3 aimed to test the prospect model in contexts more similar to traditional decision making research—where participants chose between prospects that varied with respect to their outcomes and the probabilities with which those outcomes would be realized (Kahneman & Tversky, 1979). Specifically, these experiments investigated the diminishing sensitivity aspect of the prospect model by testing whether people would be risk averse for gains and risk seeking for losses. This is analogous to the traditional prospect theory findings of reversed risk preferences as described above, where, for example, $V(-\$100, 0.5; \$0, 0.5) > V(-\$50, 1)$ vs. $V(\$100, 0.5; \$0, 0.5) < V(\$50, 1)$. Participants were presented with choices between exerting more (less) effort intensity than reference with certainty vs. a 0.5 gamble of exerting either the reference level of effort intensity or much more (less) effort intensity than reference. In other words, participants were presented with risky choices where all outcomes were framed as either losses or gains. The prospect hypothesis in both of these experiments aligns with the original claims of prospect theory (Kahneman & Tversky, 1979)—that people will be risk averse when choosing between prospects where outcomes are easier than reference

(i.e., when viewing outcomes as gains) but will be risk seeking when all outcomes are harder than reference (i.e., when viewing outcomes as losses).

In the ensuing discussion, it is often necessary to contrast theory and findings on mental effort phenomena against the theory and findings on decision making phenomena. The term *mental effort* will be relied upon to refer to the literature on mental effort (Kool & Botvinick, 2018; Kool et al., 2010). The term *risky decision* will refer to the literature on how people make decisions under risk (e.g., risky decision research, risky decision studies), particularly the phenomena and theory in the domain of prospect theory (Kahneman & Tversky, 1979), as discussed above.

Experiencing Costs

Risky decision studies have tended to use abstract, hypothetical costs, while mental effort studies have tended to use experienced costs in simple laboratory tasks involving attention and cognitive control. If one supposes a methodological continuum between these two fields—with experienced costs on one end of the continuum and described, hypothetical costs on the other—the present experiments strived to strike somewhat of a balance between the two. Experiment 1 started on the mental effort side of the continuum—it was a close replication of Kool et al. (2010; Experiment 1), where participants learned costs through experience and where cost experience was closely coupled with choice. The main difference between Experiment 1 and Kool et al (2010; Experiment 1) was that the present experiment established a reference level of effort intensity and, instead of choosing between high- and low-demand decks (as in Kool et al., 2010), participants chose between a reference deck and a deck that is more or less difficult than the reference deck.

Experiments 2 and 3 edged more toward the hypothetical-costs side of the continuum by relying more on explicit labels to signal effort intensity to participants and presenting choice in the context of risk. Decks were explicitly labeled with effort intensity outcomes and probabilities of those outcomes, which is more similar to how choices are framed in risky decision research (Kahneman & Tversky, 1979). These experiments used risk preferences to test for reference dependent effort intensity evaluation. Consistent with predictions based on prospect theory (Kahneman & Tversky, 1979), if evaluation of effort intensity is reference dependent, choices between outcomes more difficult than the reference should be risk seeking while the choices between outcomes easier than reference should be more risk averse.

Assumptions

The present series of experiments relied on a common set of assumptions to test the hypotheses described above. Most of these assumptions arose out of the need to rely on concepts in mental effort that might be analogous to concepts that are traditionally relied upon in risky decision research—namely, some type of currency that people value, the nature of the probability weighting function, and the process of setting up a reference point.

Probability of Switching as an Analogy to Money. Probability of switching in a task switching context is to mental effort intensity as money is to resources in traditional prospect theory research. Risky decision research uses money as an example, but the theory makes claims about how people value resources more generally (Kahneman & Tversky, 1979). The present experiments will use probability of switching to make inferences about how people value mental effort intensity. This relies on the assumption

that probability of switching is a linear proxy for effort intensity—performing task switching at 0.8 probability of switching ought to demand twice the effort intensity as performing task switching at 0.4 probability of switching. This assumption described in more detail and tested directly in initial pilot work prior to Experiment 1.

The Probability Weighting Function and Diminishing Sensitivity Co-Occur to Give Rise to Reversed Risk Preferences. Prospect theory predicts that people will be risk seeking when choosing between outcomes framed as losses and risk averse for those framed as gains (Kahneman & Tversky, 1979). As described above, this prediction is driven by the non-linearity implied by both diminishing sensitivity and the probability weighting function (specifically, people’s tendency to discount moderate probabilities).

The probability weighting function and diminishing sensitivity both contributing to reversed risk preferences can be thought of like a logical OR gate—reversed risk preferences are caused by the presence of the probability weighting function *and / or* diminishing sensitivity. All three traditional models claim that risk preferences would not be reversed because the probability weighting function and diminishing sensitivity are both absent. This is because the traditional models do not involve value functions that change shape around a reference point (i.e., diminishing sensitivity) and they do not involve a subjective probability weighting function. Whereas the prospect hypothesis predicts reversed risk preferences, because the theory claims both diminishing sensitivity and subjective probability weighting are present when making choices under risk. The present experiments revealed whether risk preferences were reversed or not, but—given the presence of reversed risk preferences—these experiments could not determine

whether reversed risk preferences were due to the probability weighting function, diminishing sensitivity, or both.

The mere presence of risk preferences is sufficient to rule out all traditional hypotheses. The present investigation is mostly concerned with only one of the causes underlying reversed risk preferences: the value function. However, these experiments could not rule out the possibility that reversed risk preferences were being caused by the simultaneous presence of the probability weighting function and absence of diminishing sensitivity. Therefore, it is an assumption that, if reversed risk preferences are present, it is the result of both the probability weighting function and diminishing sensitivity, as hypothesized by prospect theory (Kahneman & Tversky, 1979).

Establishing a Reference Point. The reference point is likely established early on (Baucells et al., 2011) and mediated through expectations (Abeler et al., 2011; Hack & Bieberstein, 2015). Thus, in the present experiments, the reference point was established by having participants experience the reference level of effort intensity upon their first exposure to the task (i.e., practice), and by reinforcing expectations about effort intensity via instructions. Because it is unclear whether experience is necessary for learning effort intensity, participants both experienced the effort intensity and learned about it in description form through instructions.

Pilot Experiment

Effort intensity is the central *independent* variable of the present investigation—the broad goal of the present experiments was to carefully manipulate effort intensity and observe how these manipulations influenced effort cost or utility. The purpose of the present pilot was to validate a way of manipulating effort intensity. Effort intensity can

be thought of as the volume level of a radio. The goal was to find some dial to turn that linearly increases effort intensity, just as turning a volume knob on a radio to level 20 causes the radio volume to be twice as loud as level 10.³

There also needed to be some way of measuring and quantifying the concept “volume” (i.e., effort intensity)—like a decibel meter, for example. If one observes that a one-unit increase in the volume knob results in a one-decibel increase on the decibel meter, one can make the claim that the volume knob linearly increases the volume of the radio. The pilot study was an effort to connect a manipulation of effort intensity to a measurement of effort intensity, much like the volume knob is connected to the decibel meter.

Finding a decibel meter for effort intensity is challenging because effort intensity is an extremely hard phenomenon to measure directly (Westbrook & Braver, 2015). Recall from the opening section that effort intensity was defined as how hard one is working on a task. Although imperfect, one of the best ways to measure effort intensity is through performance on said task—as task performance declines, it can be assumed that effort intensity has increased (Kool et al., 2010). Task performance is specifically defined as response times (RTs) and error rates on a given task—task performance is considered to decrease as RTs and error rates increase.

There are many knobs that can adjust task performance. The present experiments used an adjustment knob similar to one used by Kool et al. (2010) by using the

³ Typical volume knobs act as resistance against voltage supplied to speakers, which modulates the amplitude of the sound wave. In reality, voltage and resistance have a logarithmic (not linear) relationship with perceived loudness (i.e., decibels; Self, 2010). The reader is asked to leave aside the nuances of audio engineering for the sake of the metaphor.

probability of switching in a task switching context as a way to influence task performance. However, studies like Kool et al. (2010) only required rather coarse precision in the link between probability of switching and task performance—such studies were concerned with making an option either easy or hard, so these researchers only needed to find two positions for the volume knob, as it were, that corresponded to “loud” and “quiet”. The present studies required more fine grain precision between probability of switching and effort intensity (as measured by task performance).

A linear association between probability of switching and task performance is not a safe assumption because, while there is compelling evidence to suggest that a single task switch demands more effort than a single task repetition (Rogers & Monsell, 1995), it is unclear whether the same logic applies to longer runs of switches or repetitions. Imagine a deck in the DST for which each card drawn from it is a task switch (i.e., the deck is associated with a 1.0 probability of switching). The chooser knows with complete certainty what the upcoming task will be. There is evidence to suggest that switching might be less effort demanding when the switches are predictable (Reiman, Weaver, & Arrington, 2015), and theory and evidence converge to strongly suggest that the switch cost should decrease with more time to prepare for task performance (Braver, 2012; Rogers & Monsell, 1995). Therefore, the present pilot set out to test whether there was a linear relationship between probability of switching and task performance. If this relationship proved to be linear, it could be inferred that increasing probability of switching linearly increases effort intensity.

Eighteen undergraduate participants were recruited from Lehigh University to participate in the pilot and received partial course credit for their participation. One

participant was excluded for committing errors on more than 15% of trials. In short, the pilot experiment involved performing runs of 17 cued task switching trials. On each trial, participants were to perform one of two possible tasks (i.e., magnitude or parity) on a colored digit (1 through 9, excluding 5) presented at center fixation, where the color was randomly selected and determined which task should be performed on a given trial. The probability of switching in each run was parametrically manipulated between 0 and 1 in increments of 0.1. Whether a trial was a switch or repetition was determined for each trial independently according to the probability associated with the block. For example, in the 0.2 probability of switching block, the probability that *each* trial was a switch was 0.2 and the probability that it would be a repetition was 0.8. Participants were made aware of the probability of switching before each run began; runs for each probability of switching were randomly sampled without replacement, and, after all levels were sampled, the cycle repeated. The procedure lasted for approximately 45 min.

What is important for present purposes is assessing whether the relationships between probability of switching in a run and the performance measures (i.e., RTs and error rates) are positively and linearly associated. The means for each dependent variable on each level of probability of switching are plotted in Figure 3. As can be qualitatively seen from the figure, the relationships appear to be linear. A linear regression was fit separately for each dependent variable, which yielded significant, positive relationships between probability of switching in a run for both RTs ($b = 328, p < .01; R^2 = 0.05$) and error rates ($b = 0.05, p < .01; R^2 = 0.003$).

For each dependent variable, however, the mean at level 1.0 on probability of switching in a run ($M_{RT} = 955$ ms, $M_{Error} = .08$) was lower than that at the 0.9 level ($M_{RT} =$

970 ms, $M_{Error} = .10$). To test whether this decrease was significant, probability of switching in a run was used as a fixed factor in two ANOVAs (for the two dependent variables) and a follow-up test was conducted between the means at 0.9 and 1.0 levels of switching. These tests revealed that the mean at 1.0 was not significantly lower than at 0.9 for either RTs ($F < 1$) or error rates [$F(1, 160) = 2.51, p = .12$]. However, to avoid the possibility that participants would find the 1.0 probability of switching preferable to a lower level of switching, 0.9 probability of switching was the highest probability used and was assumed to be the most effort demanding.

The overall takeaway from this pilot experiment was that probability of switching appeared to be mostly linearly associated with effort demand. Each 0.1 increase in probability of switching was associated with a small but reliable increase in both RTs and error rates. As expected, there was some uncertainty in the data as to whether a probability of switching of 1.0 was harder or easier than 0.9. In the ensuing experiments, it was necessary to confront participants with choices between options that varied in effort demand. Given the results of the pilot, probability of switching in the experiments was varied between 0.1 and 0.9, with the assumption being that increases in probability of switching were associated with linear increases in effort demand.

Experiment 1

The goal of Experiment 1 was to test for the presence of a reference point in the evaluation of mental effort and to do so in a context similar to that which has been used in previous studies investigating mental effort (i.e., the DST). One prediction from the prospect model that distinguishes it from all traditional models is the idea that sensitivity to changes in effort intensity is strongest when closest to the reference point and

diminishes further from the reference point (i.e., diminishing sensitivity). This prediction was tested by making small modifications to the original DST.

The first change was the addition of prolonged practice of cued task switching at 0.5 switch probability to familiarize participants with the reference level of effort intensity. Participants also received instructions informing them that what they experienced in practice is the effort intensity that can be expected when performing such tasks. Participants then performed several blocks of the DST, where, in each block, they chose between the reference deck and a deck that is either more or less effort demanding than the reference deck (i.e., critical deck).

The two variables in the design were *direction* (harder than reference vs. easier than reference) and *magnitude* (moderate vs. extreme). Magnitude reflected to what extent the critical deck was easier or harder than the reference deck. For example, the critical deck might have been moderately or extremely easier than the reference. Magnitude was the critical variable for testing for diminishing sensitivity, because it revealed the extent to which people are more sensitive to changes in effort intensity near to vs. far from the reference point. Sensitivity to moderate critical decks should be disproportionately greater than sensitivity to extreme decks.

Method

Participants

The target sample size for this study was 70 participants.⁴ Data were initially collected from 104 participants. The initial sample size was so large because an alarmingly high number of participants were clicking through the experiment as fast as

possible or using automated scripts to perform the experiment, as evidenced by error rates near 50% and mean RTs around 200 ms.⁵ All participants were required to have IP addresses within the United States and have a completion rate of at least 90% on their previously-performed MTurk tasks. Overall, 32 participants were dropped from the analysis for having error rates higher than 15%, and 3 participants were dropped due to self-reporting impaired vision (one participant had impaired vision *and* error rates higher than 15%, so the total number of participants dropped was 34). These procedures left 70 participants remaining for analysis. Participants were compensated with \$1.50 for working for 30 min on the experiment.

According to data collected from the demographic survey, the mean age was 37.10 years ($SD = 10.65$), 33 of the 70 participants were female, 72.5% of participants were white, and 92.8% were non-Hispanic.

Materials

Tasks

As in the original DST (Kool et al., 2010; Experiment 1), participants made two types of responses on each trial. They first chose between options of varying effort demand, and they then performed an effort-demanding task.

The effort-demanding task used in this study was cued task switching. One of two possible judgments was made on a colored digit (1 through 9, excluding 5). The two possible judgments that could be performed were the magnitude and parity tasks. The

⁵ Upon realizing how many participants were likely using automated scripts to perform the experiment, a function was added to the experiment (and all subsequent experiments) that prematurely terminated the experiment if the participant failed to maintain at least a 70% accuracy rate through at least the first 2 blocks. This addition greatly improved the quality of data collection.

magnitude task involved judging whether the presented digit was greater than or less than five, while the parity task involved judging whether the presented digit was odd or even. Responses were mapped to the “J” and “K” keys on a keyboard. The color in which the digit was presented (i.e., blue or red) served as the task cue, signaling to participants which task to perform on each trial. The color-to-task mappings (i.e., task cues) were counterbalanced between participants.

The choice that participants made was framed as a choice between two decks of cards. Each time a deck was selected, a card was “flipped” upward from that deck, revealing the cued task stimulus presented against a card with a white background. The two decks varied in effort demand, and this was operationalized by varying the switch probabilities associated with each deck. Cards drawn from one of the decks resulted in a task switch at .5 probability (i.e., the reference deck); cards drawn from the other deck resulted in a task switch at either a higher or lower probability (i.e., the critical deck; see Design section below). The decks were presented on the lower left and right portions of the screen; the “D” and “F” keys were used as selection keys and were mapped to spatially compatible decks (e.g., “D” will select the left deck).

Apparatus and Stimuli

Running the present experiment (and all subsequent experiments in the present thesis) on MTurk meant forfeiting some control over stimulus presentation. The experiment was constrained such that no user with a device that has a viewport width smaller than 800 px or a viewport height smaller than 600 px was able to participate. This constraint was imposed to ensure that the experiment cannot be performed on a mobile device. Users with devices smaller than this threshold were redirected to a page where

they were informed that the experiment is not available for mobile devices, and that they need to perform the experiment on a desktop or laptop computer. The display area of the experiment was constrained to 800 px X 600 px, which is well within even the smallest pixel dimensions of most modern desktop or laptop display devices (e.g., 1280 px X 800 px). This 800 px X 600 px display area was horizontally and vertically centered within the Mechanical Turk worker's interface.

One consequence of display heterogeneity between participants is that the physical scale of stimuli (in centimeters, for example) was likely not uniform across displays. For example, stimuli presented at fixed pixel dimensions would have appeared physically larger on displays with smaller pixel dimensions (e.g., 1280 px X 800 px) relative to those with larger dimensions (1920 px X 1080 px). Thus, it is not possible to describe stimuli in terms of physical size, at least insofar as display resolutions varied across participants. These experiments also had no control over extraneous variables unique to each participant's browser and operating system. For example, MTurk's interface displays to the participant certain types of information about the experiment above the experiment's viewport, and the participant may have had other active tabs visible in the browser. Ensuing discussion of display of stimuli will reference the experiment's display area (generally referred to as the *screen*), which is the 800 px X 600 px area within the MTurk interface.

Using MTurk for data collection has long been popular for survey data, but it is increasingly popular for collecting decision and response time data, including a derivative of the DST that has been recently run on MTurk (Kool, Gershman, & Cushman, 2018). There have been several attempts to test the validity of running behavioral research on

MTurk, and effects have been replicated from tasks such as Stroop, task switching, flanker, and attentional blink, among others (Crump, McDonnell, & Gureckis, 2013). Another study specifically concluding that collecting RTs via JavaScript (through web browser) is reliable and comparable to laboratory collection methods (de Leeuw & Motz, 2016), and JavaScript is what was used to collect data throughout the present series of experiments.

The experiment was broken into two general phases: a practice block and the DST. The practice block involved successive cued task responses outside the context of choosing between decks. A target stimulus (i.e., a digit 1 through 9, excluding 5) was presented in the center of the screen in 25-point, Courier New font in either a blue or red color. The screen had a light grey background throughout the experiment.

Throughout the DST portion of the experiment, there were two decks presented on the lower half of the screen. The decks were 144 px in width by 216 px in height, which is consistent with the aspect ratio of a typical deck of playing cards. The decks were horizontally centered at the first and second thirds of the screen's width, and the decks were vertically positioned 5 px above the bottom edge of the screen. The background of the decks was a pattern that can be found on the back-face of a prototypical card from a deck of cards.⁶ A moderate drop shadow was applied to the decks to give them the illusion of depth.

⁶Kool et al. (2010) varied the color of the decks, but, unlike in their experiment, the present experiment presented these decks in multiple conditions per participant. Thus, to avoid the formation of any color-difficulty associations, both decks were displayed with an identically-colored, neutral pattern.

After a deck was selected, a card appeared directly above the chosen deck. The card had the same physical attributes as the deck except that it had a white background and did not have a drop shadow. A target stimulus with the same attributes as those from the practice phase described above appeared in the center of the card. If participants committed an error during the cued task prompt, the word “ERROR” appeared on the card above the target stimulus in a bold font-weight in a red font-color in Courier New font-family at 25-point font-size.

Procedure

The practice block was meant to both acclimate the participant with the nature of the task and to establish a reference point of effort demand. Each trial in the practice block began with a blank screen for 200 ms, after which the target stimulus appeared and remained on the screen until a response was made. If the participant made a correct response, the program advanced to the next trial. If the participant responded incorrectly, the word “ERROR” appeared above the target stimulus for 500 ms before the trial sequence repeated. This practice phase ran for 160 trials, which was approximately 5 min⁷.

The practice phase was followed by the DST phase (see Figure 4). Two decks were presented on the screen until a selection was made. After a deck was selected, a card with a target stimulus appeared above the selected deck and remained on the screen until a response was made. Following Kool et al. (2010), there was no time delay between

⁷ If an effort-intensity reference point can be established through experience, there is no precedent for how much experience it would take to do so. Five minutes was arbitrarily selected as it should be enough time for participants to gain extended familiarity with the level of effort intensity while leaving enough time for DST data collection.

trials. The trial sequence repeated following a correct response. Following an incorrect response, the word “ERROR” appeared above the target stimulus for 500 ms, after which the trial sequence repeated. Trials were grouped into four blocks that corresponded to each condition of the experiment (see Design section below). The location of the decks was constant within blocks and randomized between. For example, the deck associated with 0.5 probability of switching might have been on the left-hand side of the screen for one block and on the right-hand side for another. Participants performed each block twice. All four blocks were sampled without replacement twice in a row, such that participants would perform each block once before performing any given block again. There was a total of 100 trials per block, which led the overall experiment to last about 30 min.

Participants received instructions in the format of short videos before each phase with optional text instructions available. The instructions given before the practice phase described cued task switching and informed participants that the level of effort experienced in the practice is typical in these types of tasks and is what can be expected whenever performing such tasks. Participants were required to play the entire instructional video or scroll through all of the instructional text before they were allowed to advance to the practice phase, thus requiring them to read or hear this normative component in the instructions. The instructions before the DST phase were pulled and modified from Gold et al. (2015), which was a replication of Kool et al. (2010) that more

explicitly informed participants of differences in effort demand between decks.⁸ The main instructions about effort demand for decks in DST were as follows:

There are differences between the two decks: one deck will reveal numbers that change color less often, and the other deck will reveal numbers that change color more often. Remember: every time the color of the number changes, so does the rule. Responses throughout the rest of the experiment are broken down into groups. Every time the group changes, the deck will change also. So, in one group, the left deck might be a little easier than the right one, but this will change randomly between groups. If you develop a preference for one of the decks, feel free to continue to choose it for the rest of that group. Remember to try both decks in the beginning of each group before developing a preference, and try to avoid choosing decks based on where they are located.

There was a brief survey at the end of the experiment to collect demographic data. Participants were asked to report their age, gender identity, race, ethnicity, and whether they have impaired visual acuity and / or color vision. There was also an open-ended response where participants were able to give general feedback about the experiment.

Design and Hypotheses

The independent variables of the design were coded according to the probability of switching for the critical decks—namely, a 2 (direction: easier than reference vs. harder than reference) X 2 (magnitude: moderate vs. extreme) factorial design. The coding of probability of switching for each condition was as follows: extreme easy: 0.1; moderate easy: 0.3, moderate hard: 0.7; extreme hard: 0.9.

Constructing the hypotheses for the prospect and three traditional models started with the assumption that, if a critical deck had a probability of switching of 0.5 (i.e., the

⁸ Gold et al. (2015) failed to replicate Kool et al. (2010) when using the more subtle instructions pulled directly from Kool et al.'s (2010) procedure. Since the differences in effort demand between decks in the present experiment was smaller than that of these previous studies, the more explicit instructions from Gold et al. (2015) were used to create ideal conditions for observing the predicted effects.

same as reference), choice would be evenly distributed between the critical and reference deck (see Figure 5 for a schematic of the hypotheses for the different models). The models diverge on the shape of the function of choice over probability of switching for the critical deck. The hypotheses can be quantified by the difference in proportion of choice of the reference deck for each successive increase in critical deck probability of switching. In other words, the hypotheses make different predictions about the slope of choice over probability of switching for the critical deck. For example, the convex model predicts that the change in proportion selection of the reference deck will be smallest between the conditions extreme easy and moderate easy, and that the difference in choice will be largest between moderate hard and extreme hard conditions. The concave model makes the opposite prediction; the linear model predicts that the increase in proportion of choice of the reference deck will be the same across each successive condition.

The prospect model is unique in that it predicts larger changes in selection between the moderate conditions and reference (i.e., 0.5) than between the moderate and extreme conditions. This reflects the idea that people compare prospects to the reference point, and, because of diminishing sensitivity, people are more sensitive to changes near to rather than far from this reference point.

The extent of the increase in proportion selection of the reference deck between each successive condition (along the probability of switching scale) is the critical dependent variable for testing between the hypotheses. The difference scores for each condition are perhaps most easily understood looking at how they were coded:

$$EE_d = EM_p - EE_p$$

$$EM_d = 0.5 - EM_p$$

$$HM_d = HM_p - 0.5$$

$$HE_d = HE_p - HM_p$$

Where *EE* represents the extreme easy condition, *HM* represents the hard moderate condition, and so forth. The *d* subscript represents a difference term (i.e., the final dependent variable for each condition), and the *p* subscript refers to the observed proportion of choice of the reference deck for each condition. Because selection of the reference deck was predicted to monotonically increase as a function of critical deck intensity, the order of subtraction is such that the difference scores are expected to be positive (i.e., harder critical deck intensity conditions subtracted from easier ones).

Figure 6 displays the predictions for the difference-coded dependent variable broken down by model. Prospect theory is the only model predicting larger differences in choice between the moderate conditions and 0.5 than the difference between the extreme and moderate conditions. The prospect model hypothesizes a main effect of magnitude, where the moderate conditions have greater change in selection than the extreme conditions. This would be in support of the notion of diminishing sensitivity. This effect would be sufficient to rule out all other models, as there are crossover interactions predicted in the two non-linear models, thus eliminating the main effect of magnitude, and there are no differences between conditions expected at all for the linear model.

The prospect model also hypothesizes a main effect of direction, where the harder than reference conditions should have a greater change in selection than the easier than reference conditions. This would be in support of the notion of loss aversion. The only other model to predict this effect is the convex model, where proportion of choice of the

reference deck is expected to successively increase as the probability of switching for the critical deck increases.

Combining diminishing sensitivity and loss aversion predicts a two-way interaction between direction and magnitude such that differences in selection of the reference deck should be greater for moderate rather than extreme conditions, and this difference should be greater when the critical deck is harder than the reference relative to when it is easier. While the convex and concave hypotheses also predict interactions, the prospect hypothesis is unique because it is the only one to predict that the moderate condition is greater than the extreme condition for situations where the critical deck is easier *and* harder than the reference. The convex and concave hypotheses predict crossover interactions where the moderate condition is greater than the extreme for one level of the direction variable but not the other.

Results

Trial Trimming

MTurk workers are known to often be working on multiple tasks simultaneously or stop working on a task and come back some time later. To minimize the influence of these breaks on the data, all trials with RTs (for either cued responses or choice) greater than 10 s were dropped from the analysis, which resulted in less than 1% of the data being dropped. Once these extreme trials thought to be due to off-task activity were dropped, more standard trimming measures were applied. Trials were dropped where choice or cued RTs were more extreme than two standard deviations beyond participant-wise choice and cued RT means, which resulted in dropping 3% and 4% of the data,

respectively. Error trials and trials following error trials were dropped from the analysis, resulting in a loss of 8% of the data.

Choice Results

Perhaps the clearest way to get an initial sense of the choice results is by visualizing the mean proportion of choice of the reference deck as a function of critical deck probability of switching (see Figure 7). The condition means appear to be most aligned with the concave model (see the Appendix for a more quantitatively precise fitting of a concave model to the data). The formal 2 X 2 design is visualized in Figure 8. There was a main effect of direction [$F(1, 69) = 5.23, p = .03, \eta_p^2 = .07$], such that the change in selection of the reference deck was greater for decks easier than reference ($M = .07, SE = .04$) than for those harder than reference ($M = .03, SE = .04$). This main effect is in line with the concave hypothesis, suggesting that there is greater sensitivity between decks associated with lesser effort intensity rather than greater effort intensity. The main effect of magnitude failed to reach significance ($F < 1$).

There was also a significant two-way interaction [$F(1, 69) = 4.15, p = .045, \eta_p^2 = .06$]. Post-hoc follow-up tests revealed that the interaction was driven by the simple effect of direction for extreme magnitudes [$F(1, 69) = 4.14, p = .046, \eta_p^2 = .06$], such that the change in selection of the reference deck was significantly greater for the extreme easy condition ($M = .12, SE = .04$) than for the extreme hard condition ($M = -.02, SE = .04$). There was no simple effect of direction for moderate magnitudes ($p = .40$), as the change in selection of the reference deck for the moderate easy condition ($M = .02, SE = .03$) was not significantly different from the change in selection of the reference deck for the moderate hard condition ($M = .08, SE = .03$). This interaction is

also in line with the concave model, suggesting that people were significantly more sensitive to the difference between conditions associated with less effort intensity than those associated with more. A more complex analysis was run, taking into consideration between-participant variability in the impact of direction and magnitude on choice (see the Appendix). This analysis largely converged with that reported here, but it also revealed quite large individual differences in the pattern of effects, and so, overall, there was only tentative support for the concave model from these data.⁹

Individual Differences in Switch Costs

An exploratory analysis was conducted to investigate whether the choice results reported above varied as a function of individual differences in switch cost. An individual measure of switch cost for each participant was calculated by subtracting a participant's mean repetition RTs from the initial cued practice phase from that participant's mean switch RTs from the initial cued practice phase (the trimming procedures applied to data from the cued practice phase are discussed in the *Performance* section of the Experiment 1 Results). This switch cost measure was included as an independent variable in a

⁹ Due to the fact that participants at the beginning of each block must learn which deck was more difficult, there was a concern that such learning effects would confound the choice data. The analyses reported in this section were re-run on only the final fifty trials in each block (i.e., the second half of each block). The main effect of direction that supported the concave hypothesis remained present, but the two-way interaction between direction and magnitude was no longer present in this follow-up analysis. The fact that the main effect of direction remained significant is sufficient to conclude that the tentative evidence observed in favor of the concave hypothesis in the original analysis remains unchanged when excluding initial learning effects present in the data.

generalized linear¹⁰ model. The other independent variable in the model was the design of the 2 X 2 ANOVA described above condensed to one, numeric variable (*critical deck probability of switching*), with levels 0.1, 0.3, 0.7, and 0.9. The full model included the main effect of critical deck probability of switching, the main effect of individual differences in switch cost, and the interaction between the two predicting the binary outcome of selecting the reference deck (0: did not select reference deck; 1: selected reference deck). A logit link function was used to transform the binary outcome into continuous space for fitting the linear model, and all effects reported below were converted to odds ratios.

There was a main effect of critical deck probability of switching ($b = 1.24$; 95% *confidence interval* [CI] = 1.226, 1.254; $p < .001$), suggesting that each 0.2 increase in the critical deck probability of switching was associated with a 1.24 factor increase in the odds of selecting the reference deck. There was also a main effect of switch cost ($b = 1.06$; 95% $CI = 1.036, 1.075$; $p < .001$), suggesting that each unit increase in switch cost increased the odds of selecting the reference deck by a factor of 1.06.

Finally, there was a significant interaction between critical deck probability of switching and individual differences in switch cost ($b = 1.02$; 95% $CI = 1.009, 1.033$; $p < .001$). Following up on this interaction revealed that, when the probability of switching for the critical deck was high (i.e., 0.9), each unit increase in

¹⁰ The main choice results described above reveal that the relationship between chosen deck probability of switching and selection of the reference deck was not linear but rather concave. However, a linear model should be sufficient to examine whether choice depended on individual differences in switch costs, and so a linear model was used for the present analysis for simplicity.

individual switch costs was associated with a 1.10 factor increase in the odds of selecting the reference deck ($b = 1.10$; 95% $CI = 1.068, 1.134$; $p < .001$). However, when the probability of switching for the critical deck was low (i.e., 0.1), individual differences in switch costs were not associated with the odds of selecting the reference deck ($b = 1.01$; 95% $CI = 0.983, 1.042$, $p = .41$).

A qualitative examination of the observed means (see Figure 9) revealed that the impact of individual differences in switch costs on choice appeared to emerge in conditions where the critical deck was associated with 0.9 probability of switching (i.e., extreme hard condition). In the extreme hard condition, those with high switch costs selected the reference deck more often, whereas those with low switch costs appeared to select between the decks at chance.

Testing for Low-Demand Bias

Previous research using the DST revealed that, all else equal, people generally prefer a low-demand deck over a high-demand deck (i.e., low-demand bias; Kool et al., 2010). In the present experiment, the difficulty of the decks changed between blocks such that the reference deck was the harder of the two decks for half the blocks (i.e., easy condition) and it was the easier of the two decks for the other half of blocks (i.e., hard condition). We examined whether the present experiment replicated the low-demand bias by testing whether participants reliably chose the critical deck in easy conditions (when the critical deck is easier than the reference deck) and whether they reliably chose the reference deck in hard conditions (when the critical deck is harder than the reference deck).

The 2 (direction) X 2 (magnitude) design that was used to analyze choice above was collapsed across the magnitude variable, and two, one-sample t-tests were performed on each level of the direction variable (harder than reference vs. easier than reference), comparing each level against chance (0.5). These t-tests supported the presence of the low-demand bias, as participants were significantly more likely than chance to select the reference deck when it was the easier of the two decks ($M = .57, SE = .03; t(69) = 3.54, p < .001$), and they were significantly less likely than chance to select the reference deck when it was the harder of the two decks ($M = .41, SE = .02; t(69) = 2.81, p = .006$).

Shifting Reference Point

The choice analyses assume that, if a reference point exists when evaluating mental effort, it is established early on and is consistent over time (Baucells et al., 2011). But Baucells et al. (2011) also showed that, while early information has a robust influence on setting up a reference point, recent information can influence the reference point too. Participants were biased to select easier decks throughout the present experiment and, if the reference point updates based on recent information, the tendency to select the low-demand deck more often might have had the net impact of shifting the reference point toward easier levels of effort intensity. This shifting would have the effect of moving the conditions to the portion of the prospect hypothesis' value function that is concave, which would be consistent with the observed results.

The possibility of a shifting reference point was investigated with a 2 (direction: easier than reference vs. harder than reference) X 2 (lagged direction: previous block easier than reference vs. previous block harder than reference) within-subjects ANOVA

on the proportion selection of reference deck. The critical effect that would provide evidence for shifting reference points would be a main effect of lagged direction, where proportion selection of reference deck would be lower when the previous block was easier than reference relative to when it was harder than reference. However, neither the main effect of lagged direction or two-way interaction between lagged direction and direction were significant (all $F_s < 1$).

Performance Results

The main purpose of the performance analyses in the present experiment is to investigate the assumption that effort intensity linearly increases as a function of probability of switching. Because RTs and error rates serve as the measure for effort intensity, the goal was to assess the extent to which these dependent variables linearly increase with probability of switching. This investigation placed more emphases on RT effects, because error rates were expected to be low overall. To analyze performance data, the 2 X 2 design used to analyze choice above was flattened into a single factor called *chosen deck probability of switching* with levels 0.1, 0.3, 0.5, 0.7, and 0.9, which corresponded to the probability of switching for all different possible decks in the present experiment. The impact of transition was also considered, as switch costs are expected to decrease for decks where switching is more frequent.

For RT data, there was some qualitative evidence for a non-linear relationship between chosen deck probability of switching and RTs (see Figure 10). The association between chosen deck probability of switching and RTs seemed to be stronger for lower probabilities of switching than for higher probabilities of switching (i.e., a concave

relationship between chosen deck probability of switching and RTs). This suggests that the same concave pattern found in the choice data may also be present in the RT data.

To investigate the possibility that the relationship between chosen deck intensity and RTs was concave, a two-parameter linear model was compared to a two-parameter power function. The two models took the following forms:

$$\text{Linear: } RT = \alpha + \beta \cdot PW \quad (2)$$

$$\text{Power: } RT = \alpha \cdot PW^\beta \quad (3)$$

Where PW represents chosen deck probability of switching. Comparing the fit of the two models revealed that the power function (Equation 3) was a better fit to the data than the linear model (Equation 2; $\Delta AIC = 105.5$). The power parameter of the power model was fractional and significant ($\beta = 0.14, p < .001$), suggesting that the function was concave in shape.

To investigate for the presence of the switch cost and whether the switch cost depended on the chosen deck intensity, the fixed factors of transition and the two-way interaction between transition and chosen deck intensity were added to the linear model. The linear model was relied upon in this context for simplicity and ease of interpretation. This model revealed that all three fixed factors significantly impacted RTs. Each unit increase in chosen deck intensity was associated with a 244 ms increase in RTs ($p < .001$); switch trials were associated with a 255 ms increase in RTs over repetition trials ($p < .001$). Following up on the significant interaction ($b = -241 \text{ ms}, p < .001$) revealed that, when the chosen deck intensity was 0.1, switching tasks was associated with a 352 ms increase over repeating tasks ($p < .001$), but, when the chosen deck intensity was 0.9, there was only a 159 ms increase in RTs for switching vs. repeating

tasks. This suggests that the cost for switching diminishes in contexts where switching occurs more frequently.

For error rates, there was no qualitative evidence of a concave relationship, and so only a linear model was fit to the data. A model with chosen deck intensity, transition, and the two-way interaction predicting error rates revealed only a main effect of transition, where switching tasks was associated with a 0.02 unit increase in error rates over repeating tasks ($p < .001$). All other effects failed to reach significance ($ps > .3$). The predictions for error effects were more tentative than that for RTs because error rates in this paradigm were expected to be low overall. Indeed, errors in the present experiment were only committed on 3.85% of trials ($SD = .03$). The lack of error effects is consistent with Kool et al. (2010; Experiment 1), where the high-demand deck only had a marginally significantly higher error rate than the low-demand deck.

Choice Response Times

RTs for choices were also collected in the present experiment. These data were broken down by the same design used for analyzing choice data, namely a 2 (direction) X 2 (magnitude), and a repeated-measures ANOVA was performed on mean choice RTs; means and standard errors for choice RTs for each experiment can be seen in Table 1. This analysis revealed that choice RTs did not vary across any independent variable in the design (all $ps > .2$).

Cued Task Switching Practice

Performance data from the practice phase of cued task switching was also analyzed to test for the switch cost. A paired-sample t-test was performed on both error

rates and RTs, where error rates and RTs are expected to be significantly higher for switch trials relative to repetition trials.

The same trimming criteria applied to the performance data from the DST phase were also applied to these data. For both RTs and error analyses, the first trial of each block was dropped. All trials with RTs longer than 10 s were dropped, resulting in 1% of the data being dropped. RTs were trimmed for trials that were more extreme than two SDs beyond participant-wise means, resulting in 5% of the data being dropped. For RTs analyses only, error trials and trials following error trials were dropped, resulting in 14% of the data being dropped.

There was a significant switch cost for RTs [$t(69) = 13.34, p < .001$] as RTs on switch trials ($M = 1541\text{ ms}, SE = 56\text{ ms}$) were significantly slower than those on repetition trials ($M = 1063\text{ ms}, SE = 33\text{ ms}$). There was also a significant switch cost for error rates [$t(69) = 5.16, p < .001$], as error rates for switch trials were significantly higher ($M = .10, SE = .01$) than those on repetition trials ($M = .07, SE = .01$).

Experiment 1 Discussion

The version of the DST employed in Experiment 1 was highly similar to that used in previous research on mental effort (Kool et al., 2010). First, the present experiment replicated the basic finding from Kool et al. (2010)—participants reliably chose lower-demand decks over higher-demand decks. The present research extended the DST paradigm to investigate the *extent* to which lower-demand decks were preferred to higher-demand decks through a parametric manipulation of the probability of switching. Furthermore, the current design also included a reference deck that was consistent across all blocks and with which participants had extended practice prior to beginning DST

choice procedures. These two changes allowed for the observation of whether sensitivity to changes in effort intensity was greater when the critical deck was more similar in effort intensity to the reference deck (i.e., the moderate conditions) as opposed to when it was more dissimilar (i.e., the extreme conditions), which was predicted by the diminishing sensitivity feature of the prospect model.

In considering the pattern of the choice data, the main effect of direction and the interaction of direction and magnitude generally supported the concave hypothesis. For the extreme decks, participants appeared to be more sensitive to the decrease in difficulty between the reference deck and decks easier than reference than they were to the increase in difficulty between the reference deck and decks harder than reference. While both the prospect model and convex hypotheses predict main effects of direction, the direction of those predictions were opposite from the main effect found. Additionally, there was a cross over interaction found in the present experiment, the direction of which was only predicted by the concave model. The direction of this interaction was in line with the idea that sensitivity was greater for easier rather than harder than reference decks. The prospect hypothesis predicted that both moderate conditions would show greater sensitivity than both extreme conditions. This prediction was violated by the change in selection being greater between extreme and moderate easier than reference decks than the change in selection between the moderate easier than reference deck and 0.5.

The concave relationship between the critical deck probability of switching and selection of reference deck was most pronounced for individuals for whom switching is less costly (see Figure 9). Previous work using the DST has shown that people with higher switch costs show more demand avoidance overall (Kool et al., 2010; Experiment

5). The present results add nuance to this conclusion because high-switch-cost participants in the present study were significantly more demand averse in the extreme hard condition (when the critical deck was associated with 0.9 switch probability), but the high-switch-cost participants were no more demand averse than the low-switch-cost participants in the extreme easy condition. Previous studies in the effort discounting literature have tended to control for individual differences in effort ability rather than explore these differences (Apps et al., 2015; Bialaszek et al., 2017; Klein-Flügge et al., 2015). The present results suggest that perhaps everyone, regardless of cognitive effort ability, prefers an easy-difficulty task over a medium-difficulty task. Whereas perhaps only people for whom spending effort is costly have a strong preference for a medium-difficulty task over a hard-difficulty task while those for whom spending effort is not costly do not have as strong of a preference between the two. If one is highly skilled at a task, perhaps going from easy to moderate difficulty requires much more effort than going from moderate to extreme difficulty.

The present hypotheses rest on the assumption that effort intensity increases as a linear function of probability of switching. However, there was evidence to suggest that this assumption of linearity was violated in the present experiment, as the relationship between RTs and chosen deck intensity was best explained by a concave-shaped power function. This pattern could suggest that, in the present experiment, decks associated with 0.9 probability of switching did not require much more effort intensity than those associated with 0.7 probability of switching. If this were the case, then the assumption that probability of switching is linearly associated with effort intensity may have in fact been violated. The concern of this assumption being violated is in addition to the fact that

the mixed-model analysis (reported in the Appendix) found large individual differences in the pattern of effects, and so interpretations from the present data should be made with caution.

In summary, the results from Experiment 1 are tentatively in support of the concave hypothesis. These results were found in a context highly similar to past research on mental effort, where costs are experienced rather than described (Kool et al., 2010). For the extreme decks, the change in preference for the reference deck was significantly greater for decks easier than reference relative to those harder than reference. These findings add support to the idea that effort intensity is evaluated relative to rest rather than some reference-level of effort intensity (Chong et al., 2017; Nishiyama, 2016; Ostaszewski et al., 2013). The support is *tentative*, however, because, as discussed above, assumptions about the linearity between probability of switching and effort intensity may have been violated and there were large individual differences in the effects, which could be problematic for making inferences from choice data.

Experiment 2

One major way in which the prospect model differs from the traditional models is with respect to preferences toward risk. Prospect theory predicts that people are risk seeking when faced with a choice between losses and risk averse when faced with a choice between gains (Kahneman & Tversky, 1979). The traditional models do not hypothesize that effort intensity is evaluated relative to a reference point, and consequently these models do not predict that risk preferences would change across a reference point. The present experiment leverages these diverging predictions about risk preferences to test for reference-dependent effort intensity evaluation. In other words, if

people think about risk differently based on a reference point, it would be evidence in support of only the prospect hypothesis.

The design of Experiment 2 again built on the DST, but framed choices in a way more similar to research on risky decision making. Participants were faced with choices where the options were clearly labeled with their possible outcomes and the probability with which those outcomes would be realized. The reference point was established in the same way as in Experiment 1: by having participants practice cued task switching at the reference effort intensity level for five minutes and by conveying expectations about effort intensity via instructions.

Selection of either deck in this experiment resulted in a run of 17 cued task trials. One deck was riskless, and selection of this deck resulted in a known number of switches during subsequent cued task switching. The other deck was risky, and selection of this deck resulted in either more or fewer switches than the riskless deck. On some choice trials, both possible outcomes of the risky deck and the one outcome of the riskless deck involved fewer switches than a reference level of switching; on other choice trials, all outcomes involved more switching. The entire set of switching outcomes for a given choice is referred to as a *deck outcome combination* (see Table 2 for all deck outcome combinations used in Experiment 2).

On the prospect hypothesis, if people are comparing effortful prospects to a reference level of effort intensity, they should view a choice where all outcomes are easier than reference as being a choice between gains while viewing the opposite as a choice between losses. Thus, the prospect hypothesis predicts that people will be risk averse and choose the safe deck when faced with a choice between gains (i.e. easier than

reference decks), while being risk seeking and choosing the risky deck when faced with a choice between losses (i.e. harder than reference decks.) The traditional models predict that people will always be risk averse (i.e., convex), always be risk seeking (i.e., concave), or always be risk neutral (i.e., linear).

Experiment 2 tested the predictions of prospect theory in a risky context where all potential outcomes and their probabilities were known. This is in contrast to Experiment 1, where the effort intensities associated with decks needed to be learned from experience. While results from Experiment 1 tentatively supported the concave model, if results from Experiment 2 support the prospect model, it might suggest that people do not evaluate effort intensity relative a reference point when *only* experiencing effort (as in Experiment 1), but that they can do so from an arbitrary reference point when effort intensity is presented more descriptively and explicitly (as in Experiment 2).

Method

Participants

Seventy-six participants were recruited from MTurk according to the same criteria as in Experiment 1. A total of 6 participants were dropped from the analysis for having either impaired vision or error rates higher than 15%, leaving 70 participants for the final analysis. According to data collected from the demographic survey, the average age of the sample was 39.69 years ($SD = 12.39$ years). Participants identified as 49% female, 77% white, and 84% non-Hispanic.

Materials

Tasks

The tasks used in the present experiment were highly similar to those used in Experiment 1. Participants again made DST choices and cued task responses; what was different in the present experiment was the number of cued task switching responses made per selection and the nature of the DST choice itself. Each deck selection was followed by 17 cued-task trials. During DST choices, participants were explicitly choosing between a risky and a safe deck in the sense that the risky deck had two possible outcomes and the safe deck had only one. Deck outcomes corresponded to how many of the 16 possible trial transitions were task switches. The probability with which either outcome for the risky deck was realized after its selection was always 0.5, and participants were made aware of this probability and told that selecting the risky deck was analogous to flipping a coin to determine which of its outcomes is realized.

Apparatus and Stimuli

The apparatus and stimuli were largely the same as that used in Experiment 1. One notable exception was that the deck outcomes were displayed on the decks, instead of a neutral pattern being displayed on the decks (see Figure 11). On the top of each deck was the phrase, “Number of Switches:”, followed by its possible outcome(s). The gamble deck displayed two outcomes (e.g., “8” and “2”) on the upper and lower thirds of the deck’s face, vertically separated by the word “OR”. The safe deck displayed one outcome in the middle of its face. After a deck was selected, a black outline appeared around the realized outcome to signal to participants how many switches would be performed in the upcoming run of cued task switching. For the safe deck, this black outline marker was

obviously redundant (because participants should have known there is only one possible outcome), but the marker was implemented for the safe deck nonetheless to be consistent.

Procedure

The practice phase was similar to that of Experiment 1, except participants completed practice trials in 9 short, 17-trial blocks, yielding 153 total trials, which ran for approximately five minutes. The critical component of the practice, in addition to gaining basic familiarity with the task, was that participants established an effort reference point. Eight of the sixteen task transitions in the practice phase were switches, and participants were made aware of the number of switches to expect before beginning performance on the practice phase. The practice phase was accompanied by normative instructions telling the participants that the level of switching they are experiencing in the practice is typical in task switching studies and it is the level of difficulty that can be expected whenever performing such a task. Participants could not advance to the practice phase until they either played the entire instructional video or scrolled through all of the text instructions.

The DST phase followed the practice phase, and this phase ran for approximately 25 min (making the whole experiment last for about 30 min). See Figure 11 for a depiction of the trial sequence. At the beginning of each trial, decks were presented on the screen until selection. After selection, a black outline appeared around the selected outcome and remained present throughout all cued-task trials. A card “flipped” upward from the selected deck to reveal a cued task target stimulus and remained present until a response was made. After an incorrect response, the word “ERROR” appeared above the target stimulus for 500 ms. After either a correct response or error feedback, only the target stimulus disappeared (while the card itself remained present and blank) for 200 ms,

after which the next target stimulus appeared. After the 17th trial, the card, the deck attributes, and the black outline were cleared from the screen (leaving only borders around the blank deck faces) for 500 ms, after which new deck attributes appeared and the sequence repeated.

A *deck-outcome combination* refers to all outcomes that are possible during any one DST choice. This includes the one outcome for the safe deck and both outcomes for the risky deck. There was a total of four possible deck-outcome combinations that corresponded to the four within-subject conditions of the experiment (the details of these conditions are described below). Across DST selection prompts, these four conditions were sequentially and randomly sampled without replacement. This was done to ensure that the frequency and order of presentation for each condition were controlled for throughout the experiment. Thus, the structure of the decision phase in the present experiment (from subordinate to superordinate) was as follows: 17 cued-task trials per deck selection, four deck selections (one for each within-subject condition) per cycle, and 10 total cycles, leading the overall experiment to last about 30 minutes.

One concern about the procedure of Experiment 2 was that participants in Experiment 2 made far fewer choices during the DST phase (40 total choices per participant) than did the participants in Experiment 1 during the DST phase (800 total choices per participant). To compensate for the fewer choices in Experiment 2, a final phase was added on to Experiment 2 that was intended to double the number of choices each participant made and to collect these choices in a short period of time. This final phase differed slightly from the DST phase in that choice trials were no longer interleaved with cued-task trials. Instead, participants were informed that they would

make 40 consecutive choice trials—the black box still appeared around one of the outcomes after each choice, but, instead of performing 17 cued-task trials, the deck outcome combination reset and participants made a new choice. After all 40 choice trials had been completed, one outcome was randomly selected from the total 40 outcomes of the 40 choice trials. Participants then completed 17 cued-task trials, with the number of switches in the run of 17 cued-task trials determined from the randomly-selected outcome of the 40 choice trials.

This final phase, where participants performed 40 consecutive choice trials followed by one block of 17 cued-task trials, is referred to in the results as the *rapid fire phase*. This type of blocking of choices and performance is reflective of methods used in the effort discounting literature (Apps et al., 2015; Chong et al., 2017), where participants are sometimes told to make choices between outcomes that vary in effort demand and that, at the end of the experiment, they will have to exert the amount of effort determined from a randomly-selected outcome of their choices. Several analyses were run to determine whether the data from the rapid fire phase were suitable for analysis, and these analyses are discussed in more detail in the Results section.

Design and Hypotheses

The core design was a 2 (direction: easier than reference vs. harder than reference) X 2 (magnitude: moderate vs. extreme) within-subjects ANOVA on the proportion selection of the safe deck. The critical hypothesis hinges on the main effect of direction (see Figure 12 for predictions about the direction variable on all the hypotheses). The prospect model is the only hypothesis that predicts a main effect of direction, such that choices will be significantly risk averse for easier-than-reference

choices but significantly risk seeking for harder-than-reference choices. The concave model predicts that choices will always be risk averse, the convex model predicts that choices will always be risk seeking, and the linear model predicts that choices will not be significantly different from chance between the safe and risky decks.

Results

Trial Trimming

Due to the nested nature of trials in Experiments 2 and 3, the term *cued-task trial* will be used to refer to the runs of 17 trials where participants were making cued task responses, and the term *choice trial* will be used to refer to the 40 trials where participants were selecting between decks. Trimming procedures for cued-task trials are discussed first, followed by discussion of trimming procedures for choice trials.

Trimming procedures for cued-task trials were the same as those used for Experiment 1. All trials where cued RTs were longer than 10s were dropped from the analysis, resulting in a loss of less than 1% of the data. Cued-task trials were trimmed where cued RTs were more extreme than two SDs beyond participant-wise means, resulting in a loss of 5% of the data.

Because each participant only made 40 choices in the DST phase of Experiment 2 (compared to the 800 choices per subject in Experiment 1), it was important to take caution when trimming choice trials. The critical choice analyses were performed with and without the following trimming procedures. Choice trials were trimmed where choice RTs were longer than 10 s (dropping 3% of the data), after which choice trials were dropped where choice RTs were more extreme than 2 SDs beyond participant-wise choice RT means (dropping 4% of the data).

Choice Results

It is important to remember that the choices being made in the present experiment, unlike Experiment 1, involved risk preferences, where participants chose between safe and risky decks. The prospect hypothesis predicts participants would be more risk averse in the easier than reference condition and more risk seeking in the harder than reference condition.

As mentioned above, the choice data in the present experiment were sparse, and consequently the choice analyses were performed across different data trimming criteria in an effort to have as much clean choice data as possible and to ensure that the results of the analyses do not depend on the trimming criteria. There were three general trimming categories used for choice analyses: (1) no trimming, (2) the choice-trial-wise trimming described in the previous paragraph, and (3) choice-trial-wise trimming plus merging data from the rapid fire phase and trimming out participants based on choice quality criteria. The choice-based participant exclusion is discussed in the next section, and the impact of trimming on the analyses is discussed in the Critical Choice Results section (below). As a shorthand, these trimming procedures will be referred to as the no trim, trial trim, and rapid fire trim procedures, respectively.

Rapid Fire and Choice-Based Participant Exclusion

During the rapid fire phase, participants made successive choices between decks, followed by performing 17 cued task trials where the number of switches was determined by randomly selecting one of the outcomes of their choices. One concern is that, because this phase is very near the end, participants might try to get through this phase as fast as possible and not deliberate whatsoever during choice. Two predictable sources of

evidence of this behavior would be if choice RT means and standard deviations are extremely small (e.g., RTs less than 200 ms), and if there is a strong location bias to select the left or right deck (i.e., just pressing one key as fast as possible throughout the whole phase).

While choice RTs were fast overall in the rapid fire phase, there were four participants who were clear outliers for having extremely fast mean choice RTs (less than 200 ms) and small SDs—these participants were dropped from the rapid fire data. There were also four participants who exceeded 2 SDs beyond the group mean of location bias (suggesting a strong tendency to exclusively select the right or left deck), and these participants were dropped from the rapid fire data. During this data inspection, participants were also dropped for having a location bias more extreme than 2 SDs beyond the group mean of location bias in the *DST* phase, resulting in a loss of 8 participants. All rapid fire data were dropped for participants who were dropped from the *DST* phase (thereby removing all their data from analyses). There was a total of 7 participants whose *DST* data was acceptable but whose rapid fire data was not, so, of the 62 participants retained, there were 55 who had usable *DST* data *and* usable rapid fire data.

One way to assess whether the two choice phases are tapping into the same process is to test whether choices are correlated across the phases. Proportion selection of the safe deck was averaged for each participant and each phase, and there was a significant correlation of risk aversion between the phases ($r(68) = .59, p < .001$), thus lending support to the idea that the rapid fire data is indeed meaningful.

Critical Choice Results

Given that the rapid fire choice data appeared to be valid, all effects unless otherwise specified are from the rapid fire trim procedure described above, which includes choices from the rapid fire phase.

Cell means for data trimmed according to the rapid fire trim procedure are displayed in Figure 13. The predicted main effect of direction failed to reach significance ($F < 1$). Two one-sample t-tests revealed that neither of the levels of the direction variable were significantly different from chance ($ps > .2$). These null differences were reanalyzed with Bayesian t-tests, and these tests revealed that the observed means for the easier and harder than reference conditions were 3.18 and 5.16 times more likely to occur under the null hypothesis, respectively (i.e., positive but modest support for the hypothesis that these means are *not* significantly different from chance). There was an unexpected main effect of magnitude [$F(1, 61) = 4.06, p = .048, \eta_p^2 = .06$], where choices in the moderate conditions were significantly more risk averse ($M = .55, SE = .03$) than those in the extreme conditions ($M = .52, SE = .03$). The significance of the two-way interaction depended on trimming criteria. With the no trim procedure, the interaction was not significant [$F(1, 69) = 2.05, p = .16, \eta_p^2 = .03$]; with the trial trim procedure, the interaction was marginally significant [$F(1, 61) = 3.24, p = .08, \eta_p^2 = .05$]; with the rapid fire trim procedure, the interaction was significant [$F(1, 61) = 7.58, p = .008, \eta_p^2 = .11$].

Following up on the two-way interaction revealed a simple effect of direction in the predicted direction for extreme conditions [$F(1, 61) = 8.2, p = .006, \eta_p^2 = .12$], such that choices for easier than reference were significantly more risk averse ($M =$

.55, $SE = .03$) than those for harder than reference ($M = .49, SE = .03$). For the moderate conditions, choices for easier than reference conditions ($M = .54, SE = .04$) were not significantly different from those for harder than reference conditions [$M = .56, SE = .03; F(1, 61) = 1.06, p = .31, \eta_p^2 = .02$].

Two one-sample t-tests were used to test whether the two extreme conditions (i.e., both easier and harder than reference) were significantly different from chance. Recall that the prospect hypothesis predicts that choices will be significantly more risk averse than chance when faced with gains and significantly more risk seeking than chance when faced with losses. These t-tests revealed that neither of the extreme conditions were significantly different from chance ($ps > .1$). The data were reanalyzed with Bayesian t-tests to test for evidence in favor of the null hypothesis that these means are at chance. The mean for the extreme, easier than reference condition was not conclusively in support of either the null or the alternative hypothesis, as the Bayes factor (null / alternative) suggested that the observed data were only 2.95 times more likely under the null hypothesis. The mean for extreme, harder than reference condition, however, was strongly in support of the null hypothesis, as the Bayes factor (null / alternative) suggested that the observed data were 7.09 times more likely under the null than the alternative hypothesis.

Controlling for Between-Participant Variance. There was a high degree of variability in baseline risk preferences across participants. There was also a high degree of variability in the fixed effect of direction—some participants became more risk seeking for harder than reference choices, others become more risk averse, and for still others it did not seem to matter. One common way to analyze a fixed effect while

controlling for participant-level variance in the effect is to use a hierarchical model that allows the fixed effects to vary across participants (Barr, Levy, Scheepers, & Tily, 2013).

A logistic model with random factors of intercept, slope of magnitude, and slope of direction (i.e., no interactions or covariances) grouped by participant was fit to the data using the 2 (magnitude: easier than reference vs. harder than reference) X 2 (direction: moderate vs. extreme) design predicting choice (select risky deck vs. select safe deck). The random effect structure of this model fit the data significantly better than the model with the next-simplest random effects structure [$X^2(1) = 61.54, p < .001$]. The main effect of magnitude was still not significant after controlling for participant-wise variance, as participants were only 1.09 times more likely to select the risky deck for harder than reference choices relative to easier than reference choices ($b = 0.92, 95\% CI = [0.70, 1.22]; X^2(1, N = 63) = 0.4, p = .57$).¹¹

As an exploratory analysis, individual differences in switch cost were added to the model to assess whether these differences interacted with direction to predict choice. An individual measure of switch cost was computed by subtracting the mean RTs for repetitions from the mean RTs for switches for each participant in the practice phase. Since switch costs are computed from the *practice* phase, these switch costs are a useful predictor because they are independent from all measures collected during the DST phase.

The analysis including switch costs revealed that the main effect of switch cost was not significant; however, there was a marginally significant interaction between switch cost and direction [$b = 0.78; X^2(1, N = 63) = 3.48, p = .06$]. The nature of the

¹¹ All effects are reported as odds ratios.

interaction was such that, for easier than reference choices, individual differences in switch cost had no impact on risk preference ($p = .90$). However, for harder than reference choices, each unit increase in switch cost decreased the odds of selecting the safe deck by a factor of 0.81, although, this effect was not significant [$b = 0.81, X^2(1, N = 63) = 0.93, p = .34$]. Individuals one standard deviation above the mean on switch costs were marginally more risk seeking when faced with harder than reference choices rather than easier than reference choices [$b = .71; X^2(1, N = 63) = 3.19, p = .07$]. There was no difference in risk preferences based on the direction variable for those a standard deviation below the mean in switch costs ($p = .42$). These results might suggest that the risk preferences predicted by prospect theory (i.e., risk averse for gains and risk seeking for losses) are most pronounced for those who find switching more costly. However, this effect should be interpreted with caution because it is subtle.

Performance Results

Demand Selection Task Performance

Similar to Experiment 1, the choice design was flattened to a single factor that captured the intensity of the deck being performed. In the present experiment, decks might have multiple outcomes, so the intensity variable refers to specific deck outcomes rather than specific decks. The design was modeled with a linear regression where transition (repeat vs. switch), switches (numeric variable with range 2 – 14), and the interaction between the two predicts cued RTs. Another regression with the same predictors was used to model error rates.

The regression modeling RTs revealed that there was a significant main effect of transition ($p < .001$), where switching tasks was associated with a 269 ms increase in RTs. There was also a main effect of switches ($p < .001$), where each increase in switches per run was associated with a 4 ms increase in RTs. There was also a significant interaction ($b = -6 \text{ ms}, p < .001$). Following up on the interaction with simple slopes analyses revealed that switching tasks was associated with a greater increase in RTs at low ($b = 293 \text{ ms}, p < .001$) rather than high ($b = 245 \text{ ms}, p < .001$) number of switches per run. This interaction is in line with the idea that switch cost increases as switching becomes less frequent.

The regression modeling error rates revealed that there was a significant main effect of transition ($p < .001$), where switching tasks was associated with a 0.01 unit increase in error rates. There was also a main effect of switches ($p = .01$), where each increase in switches per run was associated with a 0.007 unit increase in error rates. The two-way interaction failed to reach significance.

The results from these two regressions must be interpreted with some caution as the predictors are obviously not independent from one another—number of switches in a run determines the relative frequency of transition. The effects, however, are relatively robust, and, if transition is dropped from the analysis, the number of switches in a run still positively predicts both RTs ($b = 21 \text{ ms}, p < .001$) and error rate ($b = .002, p < .001$; see Figure 14). Unlike Experiment 1, the bivariate relationship between RTs and switches in Experiment 2 did not appear to be curvilinear; thus, the manipulation check was successful—effort intensity increased as a linear function of number of switches in a run.

Choice Response Times

As in Experiment 1, choice RTs were recorded in the present experiment and were analyzed in the direction X magnitude design using a within-subjects ANOVA. This analysis revealed a main effect of direction [$F(1, 61) = 30.18, p < .001, \eta_p^2 = .33$], where choices were faster when deck outcome combinations were easier than reference ($M = 1583\text{ ms}, SE = 72\text{ ms}$) rather than harder than reference ($1716\text{ ms}, 80\text{ ms}$; see Table 1). All other effects failed to reach significance ($F_s < 1$).

Cued Task Switching Practice

As in Experiment 1, performance data from the practice phase of cued task switching were also analyzed to test for the switch cost. A paired-sample t-test was performed on both error rates and RTs, where error rates and RTs were expected to be significantly higher for switch trials relative to repetition trials.

The same trimming criteria applied to the performance data from the DST phase were also applied to these practice cued task switching data. For both RTs and error analyses, the first trial of each block was dropped. All trials with RTs longer than 10 s were dropped, resulting in 1% of the data being dropped. RTs were trimmed for trials that were more extreme than two SDs beyond participant-wise means, resulting in 5% of the data being dropped. For RT analyses only, error trials and trials following error trials were dropped, resulting in 14% of the data being dropped.

There was a significant switch cost for RTs [$t(69) = 13.34, p < .001$], as RTs on switch trials ($M = 1541\text{ ms}, SE = 56\text{ ms}$) were significantly slower than those on repetition trials ($M = 1063\text{ ms}, SE = 33\text{ ms}$). There was also a significant switch cost

for error rates [$t(69) = 5.16, p < .001$], as error rates for switch trials were significantly higher ($M = .10, SE = .01$) than those on repetition trials ($M = .07, SE = .01$).

Experiment 2 Discussion

Experiment 2 tested the predictions of the prospect model in a context that, relative to Experiment 1, was more similar to past research paradigms used to study prospect theory. While Experiment 1 was a near replication of work from the mental effort literature (Kool et al., 2010), Experiment 2 was novel in that the critical dependent variable involved risk preferences (i.e., choosing the safe vs. risky deck). The results of Experiment 2 provided tentative support for the prospect hypothesis. While the predicted main effect of direction failed to reach significance, the two-way interaction revealed that direction indeed had a significant impact across the two extreme magnitude conditions, but not across the two moderate magnitude conditions. As predicted by the prospect model, participants were significantly more risk averse (likely to choose the safe deck) when faced with a gamble that included an extremely easy outcome than they were when faced with a gamble that included an extremely harder outcome. However, although selection of the safe deck was significantly higher in the extremely easy vs. the extremely hard condition, neither of the conditions were significantly different from selecting at chance. This finding is being taken as tentative evidence in support of the prospect hypothesis for this reason and because it relied on post-hoc follow ups to an unplanned interaction, the significance of which depended on trimming procedures applied to choice data.

One possible interpretation of the interaction described above would be that participants may not have perceived there to be a meaningful difference in effort intensity

between the moderate conditions and reference. Recall that the safe option in the moderate conditions was only two switches more or fewer than reference, while the risky, non-reference outcome was four switches more or fewer than reference. In the pilot experiment, there were linear increases between RTs / error rates and increased probability of switching, but the difference in RTs and error rates between any successive 0.1 probability of switching conditions was fairly small. It is possible that two and four switches are small enough magnitudes to make participants feel indifferent about the possible outcomes. This lack of a noticeable difference in effort intensity would explain why differences in risk preferences seem to emerge only in the two extreme conditions—these differences in switches feel like they matter, perhaps leading participants to make choices more carefully. However, if participants were choosing more carefully in extreme conditions, one might expect choice RTs to be longer in these conditions, which was not the case.

It is interesting to note that choice RTs were significantly longer for harder rather than easier than reference conditions. It is difficult to interpret what this effect may mean. On one hand, it is possible that people would make slower choices when they are more sensitive to the differences between the outcomes. This possibility might imply that, for harder than reference conditions, the stakes are perceived to be high, and participants want to choose carefully to avoid increases in effort intensity. On the other hand, it is possible that people would make slower choices when they are *less* sensitive to differences between outcomes. This possibility might imply that people are less sensitive to outcomes harder than reference and have a harder time telling the difference between them, consequently they slow down during choice to more carefully consider the

difference between outcomes. The choice RT effect should be interpreted with caution, both because it was unplanned and because it does not clearly map on to the choice patterns.

Finally, it is important to note the high degree of between-participant variability in the fixed effects in the present experiment. Looking only at the cell means in the design would lead one to conclude that participants were choosing at chance between the options, but Bayesian tests revealed that evidence in favor of the idea that the means were *at* chance was weak. The multi-level modeling analysis also revealed large, between-participant differences in baseline risk preferences, meaning some participants prefer the risky deck on most trials while others prefer the safe deck on most trials. One possible source of the between-participant variability could be in the editing process (i.e., establishing the reference point). While participants were encouraged to take 8 switches as a reference point, it is possible that some thought of anything more than 2 switches as a loss, while others thought of anything less than 14 switches as a gain. This initial editing phase is critical for understanding risk preferences (Kahneman, 1992), and variability at this stage could account for the highly variable risk preferences observed in the present experiment.

In sum, Experiment 2 provided tentative support for the prospect hypothesis. It is important to take this support with caution, mostly because the effects were small and not significantly different from chance. Despite these cautions, the subtle effects that did emerge between the extreme conditions can only be accounted for by the prospect hypothesis, which lends support to the idea that people may be able to evaluate effort intensity relative to a reference point.

Experiment 3

If participants were evaluating effort relative to a reference point in Experiment 2, the effect was subtle and only detectable in the extreme conditions. One of the aims of Experiment 2 was to test effort evaluation in a context more similar to paradigms that are typically used to investigate prospect theory. This was done by clearly displaying to participants the possible outcomes of the choices and the probabilities with which those outcomes would be realized. The aim of Experiment 3 was to make the connection to risky decision research even stronger by explicitly framing deck outcomes around the reference point.

Experiment 3 was exactly the same as Experiment 2 except for how stimuli were presented to participants. Two concerns from Experiment 2 were that participants may not have felt that the moderate conditions were substantially different from reference, and that it was unclear whether they viewed eight switches as a reference level. Experiment 3 adjusted for these concerns by translating the number of switches to be performed in a run into an arbitrary “effort units” scale (ranging from 0 to 100). Each increase in number of switches was associated with about a six-unit increase on the effort units scale, thereby hopefully leading participants to perceive the differences as being larger.

The other concern, that participants may not have internalized eight switches as the reference point, was addressed by centering the effort units scale at 50 (which translates to 8 switches in a run). In other words, 50 was subtracted from the effort units, and this value was displayed as a deviation to participants. This means that deck outcomes with more switching than reference were displayed according to how much larger than 50 they were, while the opposite was true for those lower than 50. The font

color of the outcome was also manipulated based on whether it entailed more or fewer switches than reference—it was displayed in red for more switches than reference and green for fewer switches than reference.

Reframing number of switches as effort units should make the moderate conditions feel substantially different from reference. Coding the effort units relative to the reference point should strongly highlight the critical direction variable. These changes together should make the critical parts of the design more salient in Experiment 3. If risk preferences did not change as a function of whether outcomes are harder or easier than reference in the present experiment, it could be inferred that perhaps the reference point is not a meaningful construct in the evaluation of mental effort intensity.

Method

Participants

One hundred and thirty eight¹² participants were collected from MTurk in the same way as in Experiments 1 and 2. Six participants were excluded for committing errors on more than 15% of trials, and one was excluded for self-reporting impaired vision. This left 131 participants for the final analysis.

According to the data collected from the demographic survey at the end of the experiment, the mean age was 37.47 years old ($SD = 11.24$). A total of 70 participants identified as female, 77% of the sample identified as white, and 89% identified as non-Hispanic.

¹² The target sample size was 130, and this estimate came from a power analysis that used the observed effect size of difficulty in Experiment 2 to estimate the necessary sample size to reach 0.8 power to observe the effect of difficulty in Experiment 3.

Materials

Tasks

The tasks were identical to those used in Experiment 2.

Apparatus and Stimuli

The stimuli were nearly identical to that in Experiment 2 except for the appearance and framing of deck outcomes. Number of switches for the decks—the possible values for which between ranged 0 – 16—were translated into an arbitrary “effort units” scale that ranged from 0 (fewer switches) to 100 (more switches). The translation was simply linear to keep the scales proportional to one another (i.e., $EffortUnits = Switches * 6.25$). To make more salient the fact that 50 effort units (i.e., 8 switches) should be the reference point, outcomes were displayed according to their *deviation* from 50 effort units (i.e., $deviation(EffortUnits) = EffortUnits - 50$). For example, 12 switches in a run would be translated to 75 effort units, which would be displayed to participants as “+25”, because it is 25 effort units more difficult than 50.

The font color of the outcomes was also manipulated to make more salient whether an outcome was easier or harder than reference. If the outcome was harder than reference, it was displayed in a red font; if it was easier, it was displayed in a green font. The “lightness” of the color varied with how extreme the outcome deviated from 50 effort units. For example, outcomes with extremely-high effort units were presented in brighter red, and tasks with moderately-high effort units were displayed in darker red.

The goal of implementing both the font colors and effort units scale was to make key parts of the design more salient to participants. One possible interpretation of the

results from Experiment 2 was that participants simply did not perceive the moderate conditions as being a meaningful difference in difficulty from reference. Exaggerating the scale of the differences by relying on arbitrary effort units was one way to heighten sensitivity to outcomes in the moderate condition.

The colors were meant to strongly enforce the direction manipulation. One concern was that, without colors, participants might have mistakenly or subconsciously interpreted the “+” sign next to increased effort units as indicating something positive, while interpreting the “-“ sign as indicating something negative. It was important to establish that the “+” sign just means “more of a bad thing”, rather than have participants thinking that the sign indicates that the outcome is somehow desirable. Previous, unpublished work from our laboratory suggests that, when font color conveys valence of choice outcome, people are able to readily integrate this information into effort-investment decisions.

A 25-participant pilot was run to assess the clarity of the manipulations. Participants performed a shorter, 10-minute version of the experiment and answered several questions about the instructions at the end. When asked “Was it clear how effort units related to how difficult the task was?”, 60% of participants rated the instructions as “Very Clear”, and only 12% rated the instructions as “Very Unclear” or “Somewhat Unclear”. In the open comments response, participants were generally positive about the clarity of the instructions and did not suggest anything that was unclear or that needed to be changed.

Procedure

The procedure was largely identical to that of Experiment 2. Perhaps the only difference was the instructional video presented prior to beginning the DST phase. The beginning of the video explained the concept of effort units and drew an analogy to weight lifting—increasing effort units can be thought of as increasing the amount of weight on a barbell. This was done to strongly convey the idea that the “+” sign presented next to deck outcomes indeed signals an increase in something that should be aversive. In Experiment 2, participants were required to watch the entire instructional video prior to the practice phase before beginning, but this constraint was not imposed for the video prior to the DST phase. In Experiment 3, this constraint was imposed for *both* phases, because the instructions in both phases were critical to the manipulations.

Design

The design and analysis were identical to those used in Experiment 2. The choice analysis was a 2 (direction: easier than reference vs. harder than reference) X 2 (magnitude: moderate vs. extreme) within-subjects ANOVA on the proportion selection of the safe deck (i.e., proportion of risk averse choices). The critical effect is the main effect of direction. The prospect hypothesis is the only model to predict that choices in the easier than reference conditions will be significantly above chance in selecting the safe deck, while choices in the harder than reference conditions will be significantly above chance in selecting the risky deck (i.e., significantly below chance in selecting the safe deck). The prospect hypothesis also predicts that choices in the easier than reference condition will be significantly more risk averse than those in the harder than reference

condition (i.e., a main effect of direction). All other hypotheses do not predict a difference in risk preferences across the levels of the direction variable.

Results

Trial Trimming

The same trial-trimming procedures used in Experiment 2 for analysis of choice, RTs, and error rates were implemented in the present experiment. This included running the analyses across the same three trimming procedures described in Experiment 2. All trials where cued RTs were greater than 10 s were dropped from the analysis, resulting in a loss of less than 1% of the data. Trials were trimmed where cued RTs were more extreme than 2 SDs beyond participant-wise means, resulting in a loss of 4% of the data. When choice-based trial trimming was implemented, choice trials were trimmed when choice RTs were longer than 10 s (dropping 4% of the data) and when choice RTs were more extreme than 2 SDs beyond participant-wise means (dropping 5% of the data).

Choice Results

Just as in Experiment 2, the critical choice analyses were run when (1) not trimming any trials, (2) doing the choice-based trial trimming described above, and (3) doing choice-based trial trimming and including data from the rapid fire phase and making choice-based participant exclusions (described below). As a shorthand, these will be referred to as the no trim, trial trim, and rapid fire trim procedures, respectively.

Rapid Fire and Choice-Based Participant Exclusion

The same procedures that were used in Experiment 2 for excluding participants and inspecting the quality of the rapid fire data were used in the present experiment. Rapid fire data from four participants were dropped for having mean choice RTs less than

200 ms. Rapid fire data from nine participants were dropped for having a mean choice location bias (e.g., always selecting the left deck) that was more extreme than two SDs beyond the group mean. DST phase data from 12 participants were dropped because of this same choice location bias being present in the DST phase. These 12 participants were completely dropped from the analysis; of the 119 participants retained for analysis, there were 3 participants with unacceptable rapid fire data, so 115 participants had choice data from both DST and rapid fire phases.

Proportion selection of the safe deck was calculated for each participant for each phase and analyzed in a correlation. The choice data were highly correlated across the phases ($r(129) = .73, p < .001$), suggesting that the experiment is tapping into the same choice process across the two phases.

Critical Choice Results

Given that the rapid fire choice data appeared to be valid, all effects, unless otherwise specified, are from the rapid fire trimming procedure described above, which includes choices from the rapid fire phase.

Cell means for data undergoing the rapid fire trimming procedure are displayed in Figure 15. A main effect of direction was present in the data [$F(1, 118) = 15.44, p < .001, \eta_p^2 = .27$], but it was opposite the predicted direction based on the prospect hypothesis. Choices in the harder than reference conditions were significantly *more* risk averse ($M = .64, SE = .03$) than those in the easier than reference conditions ($M = .42, SE = .03$). Two one-sample t-tests (with Bonferroni alpha corrections for multiple comparisons) revealed that choices in the harder than reference conditions were significantly more risk averse than chance [$t(118) = 4.99, p < .001$], while choices in

the easier than reference conditions were significantly more risk seeking than chance [$t(118) = 2.93, p = .004$]. As expected, the main effect of magnitude failed to reach significance ($F < 1$).

The two-way interaction was not predicted, and its significance depended on the choice-based trimming criteria. The interaction was significant with the no trim [$F(1, 130) = 4.56, p = .04, \eta_p^2 = .03$] and with the trial trim procedures [$F(1, 130) = 4.18, p = .04, \eta_p^2 = .03$], but not with the rapid fire trim procedure ($F < 1$). For the first two trimming procedures, the interaction was driven by a simple effect of magnitude for the harder than reference direction conditions [$F(1, 130) = 5.02, p = .03, \eta_p^2 = .04$], where choices in the moderate hard condition were significantly more risk averse ($M = .63, SE = .03$) than choices in the extreme hard condition ($M = .59, SE = .03$). There was no simple effect of magnitude for the easier than reference conditions ($F < 1$).¹³

Controlling for Participant-Level Variance. Just as in Experiment 2, a generalized linear mixed model was conducted where magnitude, direction, and the interaction between the two (as well as the intercept) predicted the probability of choosing the safe deck. The results of this analysis converged with that of the ANOVA in suggesting that the main effect of direction strongly influenced choice. Specifically, participants were 4.42 times more likely to select the safe deck when deck outcome combinations were harder than reference rather than easier than reference ($b = 4.42, 95\% CI = [2.70, 7.03]; X^2(1) = 35.81, p < .001$). All other effects failed to reach significance.

¹³ Statistics were taken from the analysis conducted on the data with the no trim procedure. The pattern was consistent across the first two trimming procedures.

As in Experiment 2, the fixed effect of individual differences in switch costs (and all possible interactions) were added to the model. This analysis revealed that the main effect of switch cost and all interactions involving switch cost failed to reach significance ($ps > .2$).

Performance Results

Demand Selection Task Performance

The same analysis used to investigate performance data in Experiment 2 was used in the present experiment—namely two linear regressions where transition, number of switches, and the two-way interaction predicted both RTs and error rates.

For the regression modeling RTs, there was a significant main effect of transition ($p < .001$), where switching tasks was associated with an increase of 258 ms in RTs. There was a significant main effect of number of switches ($p < .001$), where each increase in number of switches per run was associated with a 6 ms increase in RTs. There was a significant two-way interaction ($b = -3\text{ ms}, p = .006$). Following up on the interaction with simple slopes analyses revealed that switching tasks was associated with a greater increase in RTs at lower number of switches per run ($b = 271\text{ ms}, p < .001$) than at a high number of switches per run ($b = 246\text{ ms}, p < .001$). This result is in line with the idea that switching becomes more costly as switching occurs less frequently.

For the regression modeling error rates, there was a significant main effect of transition ($p < .001$), where switching tasks was associated with a 0.02 unit increase in error rates. There was also a main effect of number of switches ($p < .001$), where each

increase in number of switches per run was associated with a 0.0007 unit increase in error rates. The interaction failed to reach significance ($b = 0.0008, p = .07$).

As in Experiment 2, these results must be interpreted with caution as these designs violate the multicollinearity assumption of the regression. Dropping transition from both analyses revealed that number of switches in a run still positively predicted both RTs ($b = 22\text{ ms}, p < .001$) and error rates ($b = 0.002, p < .001$; see Figure 16).

Choice Response Time

Choice response times were modeled with a 2 (magnitude: easier than reference vs. harder than reference) X 2 (direction: moderate vs. extreme) within-subjects ANOVA (see Table 1). This analysis revealed a marginal effect of direction [$F(1, 118) = 3.28, p = .07, \eta_p^2 = .03$], where choices were faster for deck outcome combinations easier than reference ($M = 1565\text{ ms}, SE = 47\text{ ms}$) rather than those harder than reference ($M = 1604\text{ ms}, SE = 48\text{ ms}$; see Table 1).

Practice Cued Task Switching

As in Experiments 1 and 2, data from the practice phase of cued task switching were analyzed to test for the switch cost. Less than 1% of trials were dropped for having RTs longer than 10 s, and 5% of trials were dropped for having RTs more extreme than 2 SDs beyond participant-wise means. For RT analyses only, 9% of trials were error trials or trials following error trials and were dropped from the analysis.

There was a significant switch cost for RTs [$t(130) = 20.25, p < .001$], as RTs on switch trials ($M = 1460\text{ ms}, SE = 34\text{ ms}$) were significantly slower than those on repetition trials ($M = 1042\text{ ms}, SE = 20\text{ ms}$). There was also a significant switch cost for

error rates [$t(130) = 6.18, p < .001$], as error rates were significantly higher for switch trials ($M = .06, SE = .01$) than for repetition trials ($M = .04, .003$).

Experiment 3 Discussion

Experiment 3 more explicitly framed the choice outcomes relative to a reference point and used an arbitrary effort units scale in an attempt to make differences in effort intensity feel more substantial to participants. These changes proved to have a large impact on choice, with people being significantly more risk averse (selecting the safe deck) for effort increases and more risk seeking (selecting the risky deck) for effort reductions. Risk preferences reversed depending on whether deck outcome combinations were easier or harder than reference, which was predicted by prospect theory but not any of the traditional models. However, the direction of risk preferences was exactly opposite prospect model's prediction.

Consider the fact that preferences toward risk undeniably depended on the direction variable. This fact conclusively rules out all traditional models. All traditional models claim that effort intensity is *not* evaluated relative to a reference point and that risk preferences should not vary based on the direction variable. The prospect model is the only model supported by the data in this sense, because this model does claim that effort intensity is evaluated relative to a reference point and that risk preferences should vary based on the direction variable. Furthermore, the finding that participants were significantly more risk seeking than chance for the easier than reference condition serves as strong evidence against the conclusion from Apps et al. (2015) that people are generally risk averse for effortful prospects. Experiments 2 and 3 together also conclusively show that comparing effort intensity to a reference point critically hinges on

the framing of choice. The two experiments involved the *exact same* effort demand and deck outcome combinations; all that changed was how these outcomes were presented to participants. Reasons for why results looked so drastically different across Experiments 2 and 3 are explored in the General Discussion.

Another notable aspect of the results was the lack of any magnitude effect. The findings suggest that participants were only sensitive to whether outcomes were easier or harder than reference (i.e., the direction variable) and not sensitive to how much easier or harder these outcomes were relative to reference (i.e., the magnitude variable).

Experiment 3 was different from Experiment 2 in that the scale and color of stimuli were changed to strongly emphasize the direction variable. Thus, the lack of an effect of magnitude could suggest that participants were only paying attention to whether deck outcomes were green or red, for example, and not paying attention to any information pertaining to the magnitude variable. This approach would lead all deck outcome combinations harder than reference (for example) to be treated the same regardless of whether those outcomes were moderately or extremely harder than reference. Overall, these results might suggest that making one part of the design more salient might reduce sensitivity to another part of the design.

In sum, the most notable aspect of the present findings was that the impact of the direction variable was strongly opposite the predicted direction of the prospect hypothesis. Possible explanations for the backwards findings are explored in the General Discussion. Risk preferences in Experiment 3 strongly depended on the gain and loss framing, but they did so opposite of what was predicted from the prospect hypothesis. People were more risk averse when faced with higher effort intensities (i.e., losses) and

more risk seeking when faced with lower effort intensities (i.e., gains). This suggests people can utilize a reference point to evaluate mental effort, but the way in which they do so is somewhat unclear.

General Discussion

The present series of experiments sought to test an implicit assumption in the mental effort literature—that the cost of mental effort is absolute and evaluated relative to rest (Bialaszek et al., 2017; Kool & Botvinick, 2018; Kool et al., 2010). This assumption was questioned because there is work demonstrating how people think subjectively in terms of gains and losses in other domains—such as when considering money or potential vacations (Kahneman & Tversky, 1979). The core ideas behind the notion of subjectively thinking about resources in terms of gains and losses are formalized in prospect theory (Kahneman & Tversky, 1979), and the central aim of the present investigation was to test whether people think about mental effort similarly to how prospect theory posits people think about other types of resources.

Two central concepts from prospect theory are reference-dependent evaluation and diminishing sensitivity (Kahneman & Tversky, 1979). Reference-dependent evaluation refers to the idea that—when computing the value of an outcome—the value is computed as a gain or loss relative to a reference point. A reference point in monetary terms is often one’s current net worth, and sums of money are evaluated as being gains or losses from one’s current net worth (i.e., reference point). Diminishing sensitivity refers to the idea that gains and losses near the reference point are associated with greater changes in subjective value than those further from the reference point. To take a monetary example, diminishing sensitivity is the idea that a gain from \$0 to \$100 is

associated with a greater increase in subjective value than a gain from \$1,000 to \$1,100. Reference-dependent evaluation and diminishing sensitivity are core features of prospect theory's value function (see Figure 2; Kahneman & Tversky, 1979), and these features are modeled by the value function being convex below the reference point and concave above it. The shape of prospect theory's value function is in contrast to that of the traditional models (see Figure 1), which do not change shape around a reference point because these models do not posit a reference point.

The general approach to testing whether people relied on a reference point to evaluate mental effort was to modify a paradigm traditionally used in mental effort research (i.e., DST; Kool et al., 2010). Two main modifications were made to the DST. The reasoning behind the first modification is made clear by referring to Figure 5. The competing hypotheses in the present investigation hinge on the *shape* of the relationship between effort demand (i.e., intensity) and effort cost. However, the original DST had participants select between only low- and high-demand decks (Kool et al., 2010). Using only two demand levels would be like having only two of the five points on any panel of Figure 5—any pattern of results would appear to be linear because a straight line is the best way to connect two points. Thus, more effort-demand levels were introduced to have more fine-grain sensitivity to the shape of the relationship between effort demand (i.e., probability of switching for the critical deck in Figure 5) and effort cost (i.e., proportion selection of reference deck in Figure 5).

The second main modification to the DST (employed in Experiments 2 and 3) involved introducing risk to the task. Prospect theory was developed as a theory of decision under risk (Kahneman & Tversky, 1979), and the theory has many predictions

regarding how risky decisions will unfold under different conditions. The prediction that was relied upon in the present investigation was that people are risk averse when faced with choices between gains and risk seeking when faced with choices between losses (Kahneman & Tversky, 1979). This prediction is driven both by reference-dependent evaluation and diminishing sensitivity¹⁴, and, consequently, it is not a prediction made by any of the traditional models. The DST was modified in Experiments 2 and 3 such that, instead of choosing between high- and low-demand decks where the decks are not labeled (Kool et al., 2010), participants chose between clearly labeled risky and safe decks. The risky deck had two possible outcomes, while the safe deck had one possible outcome, and all outcomes were made explicit to participants. Thus, this modification was meant to align the DST more with traditional risky decision research by introducing risk and making parameters of the choice explicit to the participants (Kahneman & Tversky, 1979).

Review of Experiments

Experiment 1 implemented the first of the two modifications to the original DST described above, namely establishing a reference point of mental effort intensity and introducing a wider variety of decks to choose from. Instead of simply choosing between high- and low-demand decks throughout the entire experiment (as in the original DST; Kool et al., 2010), Experiment 1 familiarized participants with a reference level of effort intensity and then had them choose between a deck associated with this reference intensity vs. one associated with either greater or lesser effort intensity. If effort intensity

¹⁴ As well as the probability weighting function.

is evaluated relative to a reference point, and if sensitivity to changes in effort intensity diminishes further away from the reference point, then participants should have been relatively more sensitive to decks more similar in effort intensity to the reference decks (i.e., moderate decks) rather than those more dissimilar (i.e., extreme decks). This hypothesis should have manifested as a main effect of magnitude, where sensitivity is relatively greater for critical decks moderately, rather than extremely, harder than the reference deck. Furthermore, that effect should have been larger for decks more difficult than the reference due to increased sensitivity to losses.

In contrast to the prospect hypothesis, however, there was no main effect of magnitude present in the data. There was instead a main effect of direction where sensitivity to the difference in effort intensity between the critical and reference decks tended to be strongest for easier critical decks and weakest for harder critical decks. This pattern of results is most in line with the concave model, as is perhaps most clearly demonstrated by Figure 7. The concave model posits that effort intensity is evaluated relative to rest, and that increases in effort intensity are associated with a diminishing cost. A diminishing cost, in the context of Experiment 1, explains why participants were highly sensitive to the difference between the critical and reference decks when the critical deck was associated with low effort intensity, and why this sensitivity between the decks diminished as the critical deck increased in effort intensity.

Experiments 2 and 3 aimed to test the prospect model in mental effort contexts more similar to those used in traditional risky decision research—where participants chose between prospects that varied with respect to their outcomes and the probability with which those outcomes are realized (Kahneman & Tversky, 1979). These

experiments sought to test one of the main risk-based predictions from prospect theory discussed above—namely that people will be risk averse when faced with gains and risk seeking when faced with losses.

Experiment 2 presented deck outcomes as the number of switches that would need to be performed in an upcoming run of cued task switching if the outcome was realized. If participants took eight switches as the reference point (equivalent to the number of switches learned as reference during the practice phase), then, on the prospect model, choices where deck outcomes are equal to or harder than reference should have been evaluated as losses while the deck outcomes equal to or easier than reference should have been evaluated as gains. Thus, participants should have been risk averse when faced with outcomes evaluated as gains, and risk seeking for those evaluated as losses. Experiment 2 provided only equivocal support for the prospect hypothesis, as risk preferences were aligned with the prospect hypothesis' predictions only after conducting post-hoc follow-up tests to an unplanned interaction for one of the three data trimming criteria. Support for the prospect hypothesis in Experiment 2 might be slightly stronger if assuming that participants were not sensitive to small differences in number of switches per run. If participants did not perceive the difference between outcomes in the moderate conditions to be meaningful, it would explain why the expected results were only found for the extreme conditions.

Experiment 3 followed up on Experiments 1 and 2 by making the gain / loss framing explicit to participants. The way in which the reference point is established and, consequently, how gains and losses are framed is highly contingent on context (Kahneman, 1992). It is possible that part of the reason for the messy results in

Experiments 1 and 2 was because the reference point was too subtle, leading to variability in how people were editing prospects (i.e., how they were thinking about gains and losses) during the experiments. Experiment 3 sought to control for this possibility by tightly constraining which deck outcomes should be perceived as gains and which should be perceived as losses. The approach to constraining perceived gains and losses was to heighten the salience of the direction variable by presenting outcomes that represented effort increases in a red font color and those that represented effort decreases in a green font color. Effort increases were presented as positive numbers to signal that the outcome would involve increasing something undesirable, while effort decreases were presented as negative numbers to signal that the outcome would involve increasing something desirable—much like adding weight to a barbell might be strenuous and aversive while removing it might be desirable.

The change in stimulus presentation in Experiment 3 proved to have a large impact on risk preferences, as participants' choices strongly depended on the direction variable. The direction of this effect was opposite of that predicted by the prospect hypothesis, however, as participants were significantly more risk seeking for deck outcomes easier than reference and significantly more risk averse for those easier than reference.

The following sections begin by discussing the main interpretation of the findings across all experiments, called the *value inversion* explanation. A general outline of how this explanation might account for the findings from all three experiments is discussed. The discussion then shifts toward covering alternate and perhaps less compelling

explanations, followed by a discussion of general limitations and broad suggestions for future work.

Value Inversion

The prospect model proposes that choices in the present experiments would have unfolded as a two-step process: (1) establish a reference point, and (2) evaluate outcomes relative to the reference point to make choices (Kahneman & Tversky, 1979). Results from the present experiments can be taken as evidence in support of the idea that people engage in this two-step process when deciding how to invest mental effort, although the details of the two-step process diverge from what was predicted from the prospect model. I am calling the following argument the *value inversion* account, and it is divided into two parts.

Establishing a reference point (i.e., step one of the two-step process above) depends on both how the reference point is framed and communicated and whether the reference point is *described or experienced*. This first half of the argument in support of value inversion can explain why choices appeared to only be reference dependent in Experiment 3, but not Experiment 1 and perhaps only weakly in Experiment 2.

Given that the reference point is established, evaluating mental effort unfolds in a way opposite from the claims of traditional prospect theory (as demonstrated in Experiment 3). This second half of the argument claims that the curves of the value function are inverted relative to that of the prospect model as applied to decisions about money and other types of resources, such as vacations or lives gained vs. lost (Kahneman & Tversky, 1979), which is why the account is referred to as value inversion.

Establishing a Reference Point

Establishing a reference point of effort intensity might depend on how information about the reference point is communicated. One major difference between Experiment 3 and the other two experiments is that the Experiment 3 stimuli very clearly and descriptively framed the reference point during choice. Experiments 1 and 2 instead left participants to infer the reference point through experience and subtle description. If establishing a reference point of mental intensity is only effective when the reference point is strongly framed, it would explain why choices only appeared to be reference dependent in Experiment 3.

It is well known that framing is a critical aspect of establishing a reference point (Tversky & Kahneman, 1981). A great example of the impact of framing comes from problems that are equivalent in terms of their end states but vary with respect to how the end states are framed. Consider the following two problems from Tversky and Kahneman (1989):

Problem 1 [my numbering] (N = 126): Assume yourself richer by \$300 than you are today. You have to choose between

a sure gain of \$100 [72%]

50% chance to gain \$200, and 50% chance to gain nothing [28%]

Problem 2 [my numbering] (N = 128): Assume yourself richer by \$500 than you are today. You have to choose between

a sure loss of \$100 [36%]

50% chance to lose nothing and 50% chance to lose \$200 [64%]

(p. 15).

The same end state (i.e., gain \$400) is greatly preferred in Problem 1 where it is framed as a sure gain and avoided in Problem 2 where it is framed as a sure loss. The impact of framing likely explains why, similar to the example above, risk preferences were reversed when outcomes were framed as gains vs. losses.

While the impact of framing on mental effort reference points seems clear, it is unclear whether reference points can be learned through experience. Some previous work has shown that the probability weighting function of prospect theory can be altered depending on whether probabilities are described or experienced (for a review, see Hertwig & Erev, 2009). A central feature of prospect theory's probability weighting function is that small probabilities are over weighted, leading people to be risk seeking and enter lotteries with small chances to win large sums of money (Kahneman & Tversky, 1979). This effect has been shown to reverse when lotteries are only *experienced*—meaning that, after learning about lotteries by only observing their outcomes and being naïve to the probabilities involved, people will be risk averse for lotteries where there are small chances to win large sums of money (Hertwig, Barron, Weber, & Erev, 2004). However, there is no analogous work looking at the difference between description and experience in setting up a mental effort reference point.

To discuss the potential impact of described vs. experienced information on establishing the reference point, two new terms will be introduced. A *given reference point* refers to the reference point that the experimenters assumed the participants would take, such as 0.5 probability of switching in Experiment 1 or eight switches in Experiments 2 and 3. An *individual reference point* refers to the idea of a participant

taking anything other than the given reference point as his or her reference point of mental effort intensity.

It is possible that only having participants learn about the given reference point through experience and subtle description in Experiments 1 and 2 was not sufficient for all participants to adopt the given reference point. This is in contrast to Experiment 3, where effort intensity outcomes were framed in ways similar to those that have been shown to be effective in establishing a reference point (Tversky & Kahneman, 1981). If participants in Experiments 1 and 2 did not adopt a given reference point, it would explain the lack of predicted results under the prospect hypothesis and the large individual differences in the pattern of effects. The predicted magnitude effect in Experiment 1 depended on there being most sensitivity to effort intensity levels close to the given reference point. The predicted direction effect in Experiment 2 depended on effort intensity levels harder than the collective point to be perceived as losses and those easier than the given reference point to be perceived as gains. The idea that participants might have instead adopted individual reference points is supported by the large individual differences in choice in Experiments 1 and 2. These individual differences were quantified in the more complex, multi-level modeling analyses (see the Appendix and Controlling for Between-Participant Variance from Experiment 2 results), which revealed large, between-participant variance in the impact of the experiments' independent variables on choice. While there were also large between-participant differences in Experiment 3, there was a large and robust impact of the direction variable on choice, suggesting that participants in Experiment 3 did indeed adopt the given reference point to evaluate outcomes against during choice.

If participants relied on individual reference points instead of the given reference point in Experiments 1 and 2, it is unclear what factors might influence how individual reference points were established. One possibility is that— in Experiments 1 and 2, where there was an absence of a strongly-framed and descriptively communicated given reference point—participants might have relied upon their individual degree of switch cost to establish their individual reference points. Establishing a reference point is thought to be tied to some status quo, such as net worth in monetary contexts (Kahneman, 1992), unless there is some clear expectation about a future state, such as expecting but failing to buy shoes (Kőszegi & Rabin, 2006). It is possible that Experiments 1 and 2 revealed important boundary conditions on how a given reference point can be established. If the more subtle framing of the reference point in Experiments 1 and 2 (relative to that of Experiment 3) was ineffective at establishing how much effort intensity one should *expect* to exert in the DST, then perhaps individual differences in switch costs served as a status quo with which participants established individual reference points. This idea that personal reference points might have been tied to some individual difference rather than experience in the experiment is also supported by the lack of any evidence in Experiment 1 of the reference point shifting as a function of recently-experienced effort intensity.

While individual differences in switch costs significantly impacted choice in Experiments 1 and 2, the nature of these effects did not strongly support the idea that participants were using individual switch costs to establish individual reference points. In Experiment 1, it seemed as though high-switch-cost individuals were more sensitive to differences in effort intensity than low-switch-cost individuals *only* for the extreme,

harder than reference condition. In Experiment 2, high-switch-cost individuals were only marginally more sensitive to the direction variable than low-switch-cost individuals, and the direction variable did not have a large impact on choice overall. If participants were indeed relying on individual reference points, it is not clear from these experiments what factors could explain the variability in constructing these individual reference points.

In sum, the evidence from the present experiments converged to support the idea that people are most likely to adopt a given reference point when it is conveyed descriptively and framed clearly during choice. When the given reference point is conveyed more subtly, such as through experience with a certain level of effort intensity, it is unclear what reference point people may be using to evaluate effort intensity against during choice. While there were large between-participant differences in all experiments, the idea that the given reference point was only adopted in Experiment 3 was supported by the robust impact of the direction variable in Experiment 3—where outcomes were framed relative to the given reference point—and by the absence of any clear impact of the direction variable in Experiments 1 and 2, where the given reference point was communicated only through experience and subtle instructions. Thus, Experiment 3 revealed that it is indeed possible for people to use a reference point of mental effort intensity, and Experiments 1 and 2 revealed important boundary conditions on using a reference point. The results from these experiments do not suggest that it is *impossible* to establish a reference point via experience with a mentally demanding task (i.e., in the absence of effort intensity framed relative to a reference point)—it is plausible, for example, that one might establish a reference point of how hard one is used to working after spending months on a particular project. These results instead suggest that, if it is

possible for experience to establish a reference point, the experience with mental effort intensity likely needs to be longer than a five-minute practice session.

Inverted Value Function

The value inversion account posits that the curves of the value function are inverted relative to traditional prospect theory when evaluating mental effort prospects (Kahneman & Tversky, 1979). This inversion would imply that the value function is convex above the reference point and concave below it. Inverting the value function reverses predictions with respect to risk—people should be risk seeking when faced with a choice between gains and risk averse when faced with a choice between losses. The present section seeks to explain how value inversion can account for the present findings, as well as explain why the value function might be inverted for mental effort.

Similar to traditional prospect theory, value inversion assumes that outcomes are evaluated relative to a reference point. Given the discussion in the previous section, it is assumed that participants only evaluated outcomes relative to the given reference point in Experiment 3. Therefore, in contexts where people evaluate mental effort intensity relative to a given reference point (as they did in Experiment 3), value inversion is mostly consistent with the data in the sense that participants were risk averse when choosing between outcomes defined as losses and risk seeking when choosing between outcomes defined as gains. No other competing hypothesis—the prospect model or any of the traditional models—predicted the observed pattern of results. For reasons that will be discussed in the following section, the value inversion account is not perfectly consistent with the observed results across all experiments, but, at present, inverting the curves of

the value function is the best explanation for the robust reversed risk preferences in Experiment 3.

The value inversion account claims the curves of the value function are inverted for mental effort, but why would such inversion take place for mental effort relative to something like money? Taking money as an example, consider how the boundaries on the scales of effort and money compare to one another. For effort, one can only work so hard, and one cannot work less hard than resting. The scale of effort intensity has these types of natural boundaries. Money, however, is far less constrained. Privileged few hoard billions, and many students incur hundreds of thousands in debt. There is, of course, a lower boundary on money at zero, assuming we leave aside the idea of going into debt. Traditional prospect theory does in fact posit that the value function changes shape to become steeply negative if a given loss threatens impoverishment (Kahneman & Tversky, 1979). This same notion of the value function bending steeply upon approaching a limit on the X-axis can be applied to mental effort. The value function on the value inversion account can be thought of as a type of logit function¹⁵ that bends steeply positive value when approaching minimum effort (i.e., rest) and steeply negative value when approaching maximum effort. Because the bounds on effort intensity are much tighter than those for money, it might explain why the value function bending near the boundaries seems to play a larger role for effort than for money. Thus, the difference in boundaries provides one reason why the value function for mental effort may be different from that of money.

¹⁵ Comparing the value function to a logit function here is meant as a demonstration tool rather than a precise mathematical description.

Prospect theory is, at its heart, a descriptive theory—the theory is concerned with describing *what* people tend to do in certain contexts, which is in contrast to concerning itself with *why* people do what they do or what they *should* be doing (Kahneman & Tversky, 1979). Perhaps a more descriptive approach to finding support for the idea that the value function is inverted for mental effort is to focus less on *why* it might be inverted for mental effort and to instead focus on looking for *whether* there is evidence of inverted value functions in related domains.

There are studies that extend prospect theory to how people value time, and this work might serve as a useful analogy to how people might value effort. As noted in the Introduction, effort is often conflated with time (Dunn et al., 2016). Evidence suggests that, contrary to utility models for money, the utility function for time is convex over moderate amounts of time (Festjens & Janiszewski, 2015; but see Festjens, Bruyneel, Diecidue, & Dewitte, 2015)—meaning that 1 hour, for example, is more than twice as valuable as 30 minutes. The utility function is shown to be concave, however, as time becomes long, reflecting the idea that increases in vast amounts of time are not particularly useful (i.e., resource abundance; Festjens & Janiszewski, 2015).

Because of the high degree of similarity between effort and time, this notion of a double-shaped value function from research on time value might also apply to effort. The function of effort utility over effort intensity may be convex for increases in smaller-to-moderate intensities and concave for increases in moderate-to-extreme intensities. This double shape would imply that writing a 30-page paper (i.e., medium effort) would feel more than twice as aversive as writing a 15-page paper (i.e., low effort), and that the

difference between 15 and 30 pages would feel much larger than the difference between 250 and 265 pages (i.e., high effort).

The notion of a double-shaped function is present in at least one study in the effort discounting literature (Klein-Flügge et al., 2015), but the shape of the curves is different from that described above. These authors fit a variation of a sigmoidal function to effort discounting data, where the shape of the function roughly aligns with the value function of traditional prospect theory—the sigmoidal function curves with a diminishing shape outward in both directions from a center-point on the X -axis. There were three main theoretical ideas behind the sigmoidal model’s specification. The authors argued it is intuitive that increases in small amounts of effort do not carry a large cost, which is modeled by the function taking an initial concave shape for low levels of effort. They also argued that increases in medium-to-high levels of effort are strenuous and thus these increments in effort carry an increasing cost (convex-shaped slope). Finally, they argued the cost asymptotes for impossibly high levels of effort, which is modeled by the asymptote as the sigmoid function reaches its upper limit. Even though this sigmoidal function resembles the traditional prospect value function in form, there is critically no notion of a reference point—just as in the traditional models, it is assumed that people take an implicit reference point of zero, comparing any level of effort to total rest. Klein-Flügge and colleagues, however, focused on physical effort (i.e., hand clenching), and so the extent to which the exact shapes of the discounting function would transfer directly to a mental effort context are unclear.

In sum, inverting the curves of the prospect theory value function is a parsimonious explanation for the observed risk preferences in Experiment 3. One reason

to believe that the value function might be inverted for mental effort relative to something like money is because of the difference in the boundaries of the scale between these two constructs—mental effort has narrower boundaries, and there is evidence to suggest that value functions may bend steeply near such boundaries (Kahneman & Tversky, 1979). Past research has suggested the value function for effort and time can bend in ways that are in contrast to the claims of prospect theory (Festgens & Janiszewski, 2015; Klein-Flügge et al., 2015). Although these studies are mixed with respect to how different value and discounting functions may be applied in different domains, they converge to suggest that time and effort are valued differently from money. The reversed risk preferences in Experiment 3 strongly suggest that effort is valued differently from other resources investigated by prospect theory, and an inverted value function is an elegant explanation for what might give rise to these differences. A reference dependent and inverted value function for mental effort might also explain why findings have been so discrepant in the effort discounting literature (Bialaszek et al., 2017; Chong et al., 2017; Ostaszewski et al., 2013). One example of an implication of an inverted value function would be a student with a reference point of writing a 10-page paper. This student would refuse a gamble between writing 10 pages vs. 30 pages and instead accept a safe offer to write a 20-page paper. Given a reference point at 10 pages, the choice should be framed as being between different degrees of loss, and, in contrast to prospect theory (Kahneman & Tversky, 1979), the value inversion account predicts people will be risk averse in such contexts.

Caveats of Value Inversion

While value inversion is the best explanation for risk preferences being opposite of the prospect model's prediction in Experiment 3, there are caveats to the explanation. Perhaps the most important of these caveats has to do with the implications of inverting the value function on the idea of diminishing sensitivity, and how these implications mismatch with the Experiment 3 results.

As opposed to traditional prospect theory's tenet of diminishing sensitivity (Kahneman & Tversky, 1979), inverting the curves of the value function in fact implies *increasing* sensitivity from the reference point. This increasing sensitivity prediction of the value inversion account is indeed somewhat inconsistent with the observed results. Increasing sensitivity would predict a larger difference between the moderate conditions and their respective extreme conditions, and a smaller difference between the moderate conditions and 0.5 (i.e., selecting between the decks equally). Instead, Experiment 3 revealed no difference between moderate conditions and their respective extreme conditions, while it revealed large differences between the moderate conditions and 0.5.

The value inversion account can accommodate the fact that the magnitude variable did not seem to impact choice in Experiment 3 if it is assumed that participants paid more attention to the differences in stimulus color between the outcomes rather than the outcomes' precise effort units. Recall that, in Experiment 3, outcomes meant to be thought of as losses or effort increases were presented in red font, while those meant to be thought of as gains or effort reductions were presented in green font. It is possible that participants relied only on parsing the hue of outcome stimuli in order to simplify choice. Participants were motivated to finish the study as quickly as possible, and it is

well known that people employ simplifying strategies when making fast decisions in order to translate difficult decisions into simpler ones (Tversky & Kahneman, 1974). Such a simplification might have been that all choices with outcomes presented in red font, for example, were perceived as being the same, even though the outcomes within that set of choices varied. There is also unpublished data from our laboratory consistent with the interpretation that, when stimulus color signals choice value, participants rely heavily on color cues to drive choice and might not parse more nuanced, reward-related features of stimuli in task-switching environments.

Another caveat of the value inversion account is that, while the reference point in Experiment 3 was communicated descriptively relative to Experiments 1 and 2, there was still an experienced component. Participants did experience the effort intensity following each choice, which is different from risky decision research, where choices are fully hypothetical and costs are never experienced (Kahneman & Tversky, 1979). This asymmetry leaves open the question as to whether the value function for mental effort would still be inverted if the mental effort were *fully* hypothetical. Strictly speaking, this caveat is perhaps more centered on limitations of claims that can be made about the value inversion account given the present methodology rather than a caveat of the value inversion account *per se*. A useful follow-up experiment would be to replicate Experiment 3 except eliminate all aspects of performing task switching. The task could be described in fully hypothetical terms, participants could make the same choices, and no performance would follow choice. If the results of this experiment are consistent with that of Experiment 3, it would be further evidence in support of the idea that, when a

clear reference point is established, representation of value for mental effort is inverted relative to the domains on which prospect theory has typically focused.

A final caveat of the value inversion account is that it is a post-hoc explanation of the Experiment 3 results. It would be prudent to simply replicate Experiment 3, which would ensure that the phenomenon of reversed risk preferences is reliable. Following a successful replication, one useful way to test value inversion as an *a priori* theory of how people value mental effort would be to look for evidence of value inversion in contexts other than the DST. There are other effort demanding tasks that could be adapted into a DST format (see Apps et al., 2015; Westbrook et al., 2013; Yantis et al., 2002), and it would also be useful to test the value inversion account in an effort discounting context.

Other Explanations

Aside from inverting the value function for mental effort, there seem to be few other explanations that might plausibly account for the Experiment 3 result of inverted risk preferences. One trivial possibility is that participants might just not have understood the stimuli in Experiment 3, and mistakenly believed that increases in effort units were associated with something desirable. In this sense, participants could be seen as being risk averse when faced with choices between “gains” of effort units and risk seeking when faced with choices between “losses” of effort units, which would be in line with the prospect model. This possibility seems highly unlikely, however, because the instructional video described in great detail how gaining effort units means doing more work. The majority of participants in the pilot study indicated that the instructions were clear, and so the idea that participants somehow reverse coded the stimuli seems unlikely.

The possibility that participants incorrectly perceived increases in effort units as a type of gain also seems unlikely given past prospect theory research looking at gains of undesirable outcomes (Kahneman & Tversky, 1979). This past work suggests that the coding of the *X*-axis of the value function (i.e., gains vs. losses) is more strongly tied to the valence of the outcome than to whether something is merely framed as being gained or lost. For example, people perceive increases in a death toll as a type of loss, even though, strictly speaking, “death” is a concept that is increasing in quantity (Kahneman & Tversky, 1979).

General Limitations and Future Work

The present investigation was the first to ask whether people think about mental effort in ways predicted by prospect theory, and so it is especially important to be mindful of all assumptions underlying both prospect theory and theories surrounding mental effort. There were several assumptions made in the present work that, if violated, may have undermined the logic behind the studies. Among these assumptions were that the reference point was established early on and maintained throughout the experiment, that switching frequency was positively and linearly associated with effort intensity, and that—in Experiments 2 and 3—risk preferences alone would reveal the nature of the value function.

Previous research on stock-selling decisions (Baucells et al., 2011) motivated the assumption that the reference point is established early on. But these researchers also concluded that recent information may also influence the reference point (Baucells et al., 2011), and other researchers believe that reference points can be set by aggregating over information sources (Helson, 1964; Lattin & Bucklin, 1989; Putler, 1992; Winer, 1986).

One notable limitation of the present methodology was that the given reference point was always set at the middle of the effort intensity spectrum (i.e., 0.5 probability of switching in Experiment 1, and 8 switches out of 16 transitions in Experiments 2 and 3). If the reference point is established early on, then practicing at 0.2 vs. 0.8 probability of switching at the beginning of the experiment, for example, should have a large impact on setting the reference point and, consequently, on subsequent choices. Future research should investigate whether practicing task switching at 0.2 probability of switching leads 0.5 probability of switching to feel like a loss, for example, while practicing at 0.8 probability of switching leads 0.5 probability of switching to feel like a gain. If such a result pans out, it might suggest that an effort-based reference point can indeed be established with early exposure to a task; otherwise, it might suggest that early exposure to a task is in fact not a primary factor in establishing a reference point.

It was also assumed that probability of switching is linearly and positively associated with effort intensity. Effort is a notoriously difficult construct to measure (Braver & Botvinick, 2015), but, as others have done in previous research (Kool et al., 2010), it was assumed that probability of switching would serve as a proxy for effort intensity. There was some evidence in Experiment 1 to suggest that there were larger differences in performance for easier than reference decks rather than harder than reference decks, suggesting that probability of switching might not scale linearly in difficulty. This might explain why, as evidenced by choice data in Experiment 1, people seemed to be more sensitive to differences between the reference deck and easier decks than between reference and harder decks. Probability of switching in a task switching context is only one way to manipulate mental effort, and switching between tasks draws

on *one* particular cognitive function (i.e., the ability to maintain and shift between competing representations performing similar tasks; Rogers & Monsell, 1995). As discussed earlier in the context of caveats of the value inversion account, there are other methods of investigating different types of mental effort intensity—such as the n-back task (Westbrook et al., 2013) and the rapid serial visual presentation task (Apps et al., 2015). Because mental effort intensity is difficult to operationalize and manipulate, future research should use a variety of tasks and seek converging evidence when testing whether the evaluation of effort intensity is reference dependent.

Another limitation of the present research is that Experiments 2 and 3 sought to make inferences about the value function through risk preferences without directly assessing the probability weighting function. In prospect theory, both the value and the probability weighting function independently contribute to the prediction that people will be risk averse for gains and risk seeking for losses (Kahneman & Tversky, 1979). Drawing inferences about the value function based on only risk preferences in mental effort contexts assumes that, as in monetary contexts, people will underweight moderate probabilities. There are relatively few studies that directly investigate preferences for mental effort under risk *per se*, but data from Apps et al. (2015) and Kool and Botvinick's (2018) interpretation of those data support the assumption that probabilities are underweighted for mental effort. Participants in Apps et al. (2015) were faced with risky vs. safe choices between prospects with mental effort outcomes, where the options were matched with respect to their expected value. The finding that participants were generally risk averse for these choices can be explained by suggesting that the value of the gamble was discounted because people discount moderate probabilities.

In addition to the uncertainty surrounding the specifics of the probability weighting function, the present research also assumed that subjective value of a choice is a weighted sum of the possible outcomes of the choice multiplied by their probabilities. This commonly-made assumption directly follows the equation at the heart of Kahneman and Tversky's (1979) prospect theory:

$$V = \sum v(X)\pi(P) \quad (4)$$

Where subjective value for a choice (V) is a linear combination of the outputs of the value function (v) and the probability weighting function (π). This general idea that the value of a choice is a weighted sum of its possible outcomes has deep roots in probability and decision theory—the earliest iteration of this idea is credited to Pascal's expected value theory in the 1650s, and expected utility theory extended this idea to become a dominant economic model of rational choice throughout the twentieth century (Bernoulli, 1738/1954; Neumann & Morgenstern, 1953). This idea that the value of a choice is a weighted sum of its possible outcomes is also prevalent in modern theories of how people balance reward and effort costs to allocate their cognitive effort (Kool & Botvinick, 2018; Shenhav, Botvinick, & Cohen, 2013). The present work extends this idea to how people might value mental effort, in the sense that choices among effort intensity prospects are valued according to their effort outcomes weighted by their probabilities. Nothing in the present results necessarily suggests that this assumption was violated, but it was not an assumption that was tested directly. Given that the results of Experiment 3 strongly suggest that *something* about how people value mental effort is different from how they value resources typically investigated by prospect theory, it seems prudent to take every

assumption—no matter how prevalent it may be in past research—into careful consideration.

Conclusion

Rebecca used to leave the office at 7 PM; Jack used to leave at 3 PM. Now, they both leave at 5 PM. How do they feel about their new workload? Findings from the present investigation suggest that—despite leaving the office at the same time—they might evaluate the work differently from one another. Rebecca is more likely to feel positively, perhaps viewing it as a gain, and Jack is more likely to feel negatively about it.

The present findings suggest that, when changes in effort intensity are clearly presented as losses and gains (as in Experiment 3), people think about them differently. If effort intensity is instead presented in terms of absolute states of work (as in Experiment 2), people might be less likely to consider changes in effort intensity as gains or losses. If, instead of being told to leave the office two hours earlier, Rebecca was simply presented with a new, somewhat easier project, she might be less likely to view the outcome as a gain.

While the present findings suggest people can think about effort intensity as gains or losses, the present findings also suggest that they might value these gains and losses in effort intensity differently from something like money, for example. People are typically risk averse when faced with losses and risk seeking when faced with gains (Kahneman & Tversky, 1979). The present findings, however, suggest that if, instead of simply being reassigned to leave at 5 PM, Rebecca was faced with a choice between leaving at 5 PM

for sure or a gamble between leaving at 3 PM vs. leaving at 7 PM (a choice between gains), she might be more likely to select the gamble. The opposite is true for Jack.

The finding that risk preferences are reversed for mental effort suggests that the value function for mental effort might be inverted relative to something like money. For money, increases in states of wealth are associated with diminishing marginal utility—gaining \$100 is less than twice as valuable as gaining \$50. But the opposite might be true for mental effort—reducing one’s workday by four hours might be *more* than twice as valuable as reducing it by two hours. Whereas working for four extra hours might be more than twice as aversive as working for two extra hours.

Combining ideas from psychology and economics has fundamentally changed our collective understanding of human judgment and decision making (Kahneman & Tversky, 1979). Similar changes have started to take place in the study of mental effort, where great empirical and theoretical progress has been made by thinking about effort as an economic resource (Kool & Botvinick, 2018). The present research showed that, like money, people may think about effort in relative terms. But the exact way in which they do so remains unclear. More work is needed to fully reveal how people navigate the effort economy.

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Table 1.

Choice RTs as a Function of Direction and Magnitude across all Experiments.

Average Milliseconds (Standard Errors)

		Experiment 1		Experiment 2		Experiment 3	
		Moderate Magnitude	Extreme Magnitude	Moderate Magnitude	Extreme Magnitude	Moderate Magnitude	Extreme Magnitude
Easy Direction		311 (15)	302 (15)	1574 (72)	1590 (77)	1552 (48)	1577 (48)
Hard Direction		311 (15)	307 (14)	1719 (80)	1712 (82)	1605 (48)	1599 (51)

Table 2.

Deck-Outcome Combinations for each Within-Subject Condition in Experiment 2

Number of switches for safe deck vs. [number of switches for outcome 1 of risky deck, number of switches for outcome 2 of risky deck]

Reference point of effort intensity is 8 switches

Direction	Magnitude	
	Medium Magnitude	Extreme Magnitude
Easy Direction	6 vs. [8, 4]	5 vs. [8, 2]
Hard Direction	10 vs. [8, 12]	11 vs. [8, 14]

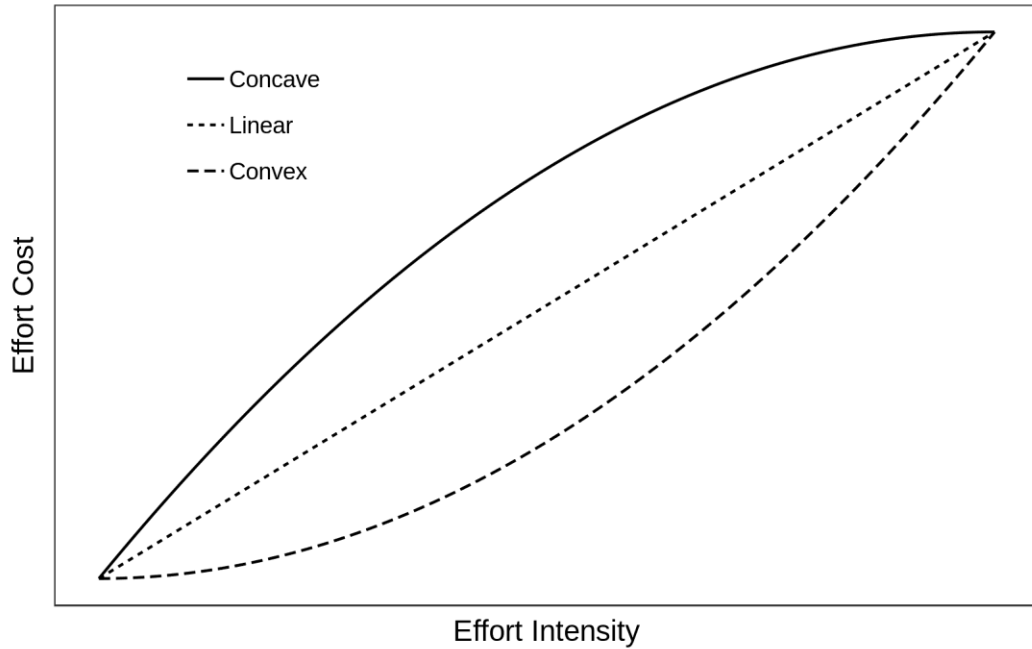


Figure 1. The relationship between effort intensity and effort cost on the three traditional models. Concave represents diminishing sensitivity, where each unit increase in effort intensity is associated with a progressively smaller increase in effort cost. Convex represents increasing cost, where each unit increase in effort intensity is associated with progressively larger increases in effort cost. Linear represents an effort cost that scales in fixed units with effort intensity.

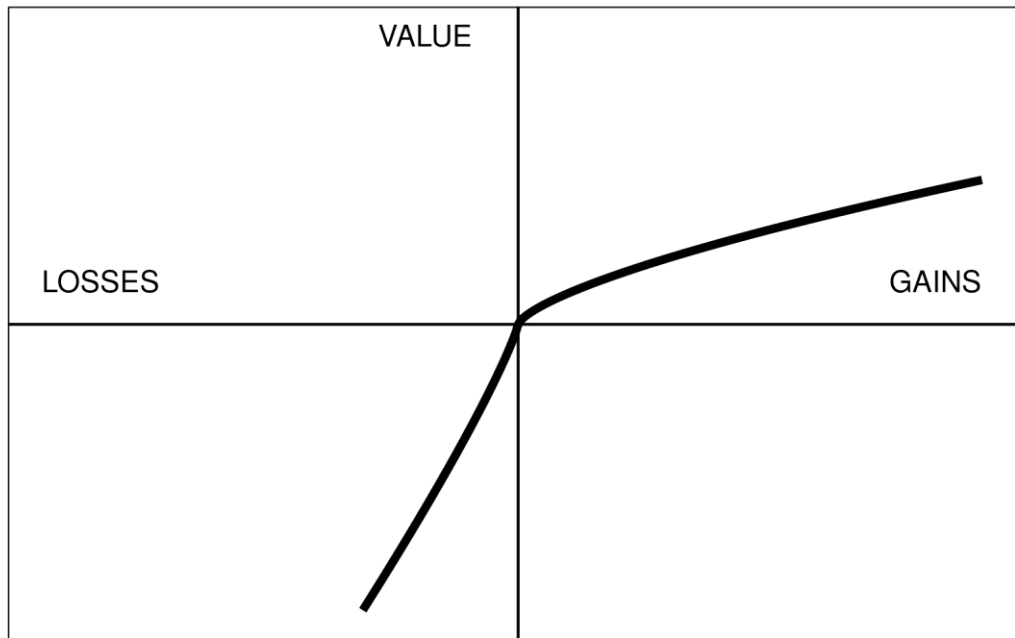


Figure 2. The hypothetical value function of prospect theory (Kahneman & Tversky, 1979). The value of an outcome is evaluated relative to a reference point (i.e., the origin). The function to the right of the reference point is convex, and the function to the left is concave. The shapes of these functions represent diminishing sensitivity to amounts that are more distant from the reference point. The slope of the function is also asymmetrical on either side of the reference point, where the slope is steeper (i.e., more negative) for losses and less steep for gains. This feature of the model captures loss aversion, which is the idea that people find a loss to be more aversive than they find the equivalent gain to be desirable.

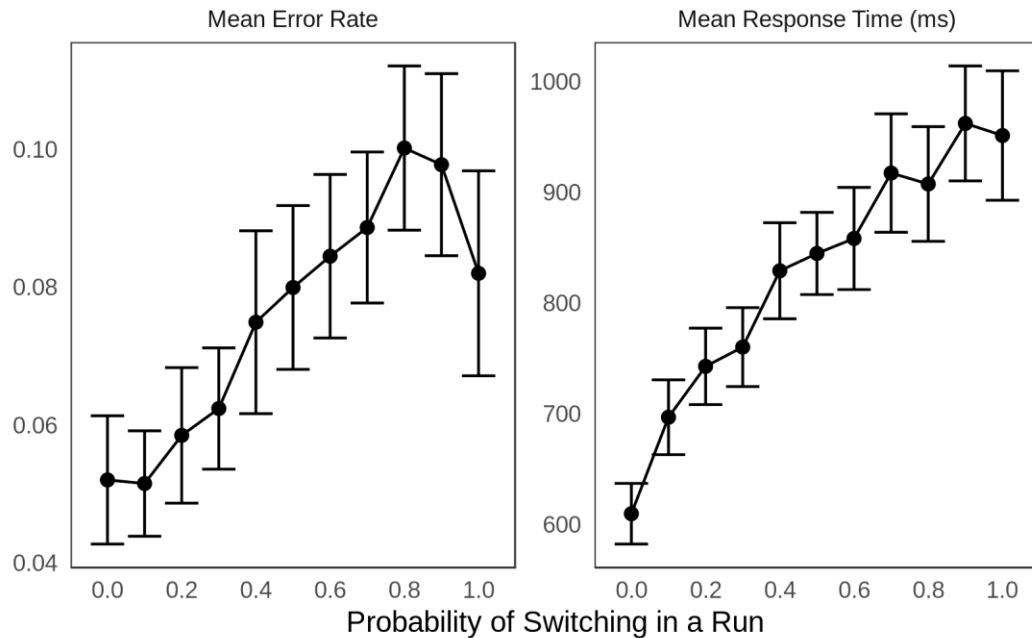


Figure 3. Performance data by probability of switching in a run of cued task switching trials from the pilot experiment. Performance generally worsened as the probability of switching in a run increased. The exception to this trend is between 0.9 and 1.0 on probability of switching in a run for mean error rates, where participants were generally more accurate when the task switched each trial compared to when the task only switched with 0.9 probability.

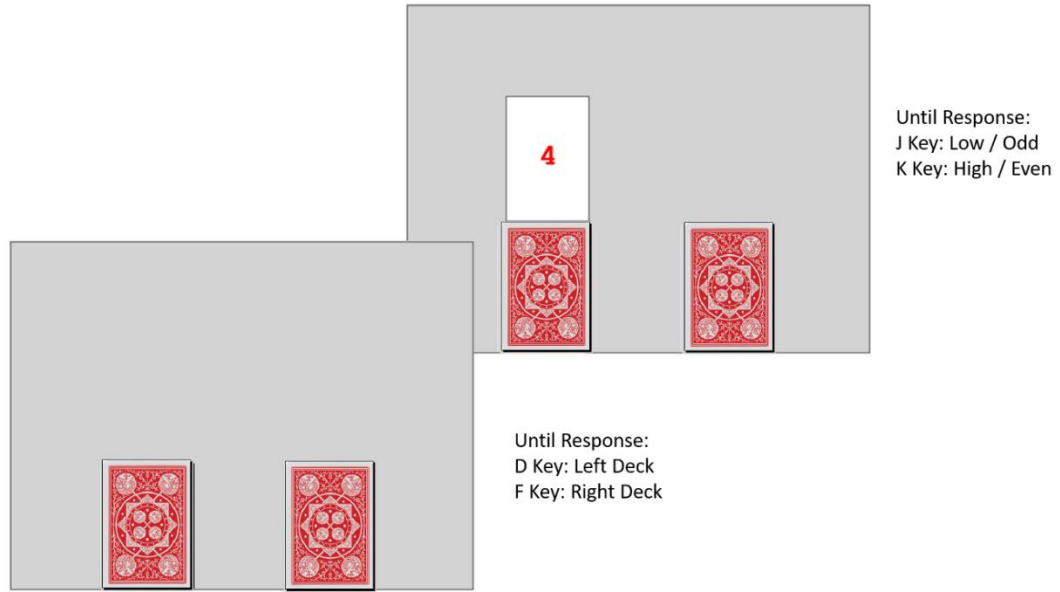


Figure 4. DST trial sequence for Experiment 1. Participants first select a card from one of the two decks presented on the screen. After selection, a card “flips” from the chosen deck revealing a target stimulus, upon which participants must make an appropriate judgment given the color cue and digit. For example, if the color red was the cue for the magnitude task, participants would make a “lower than five” judgment on the target stimulus above.

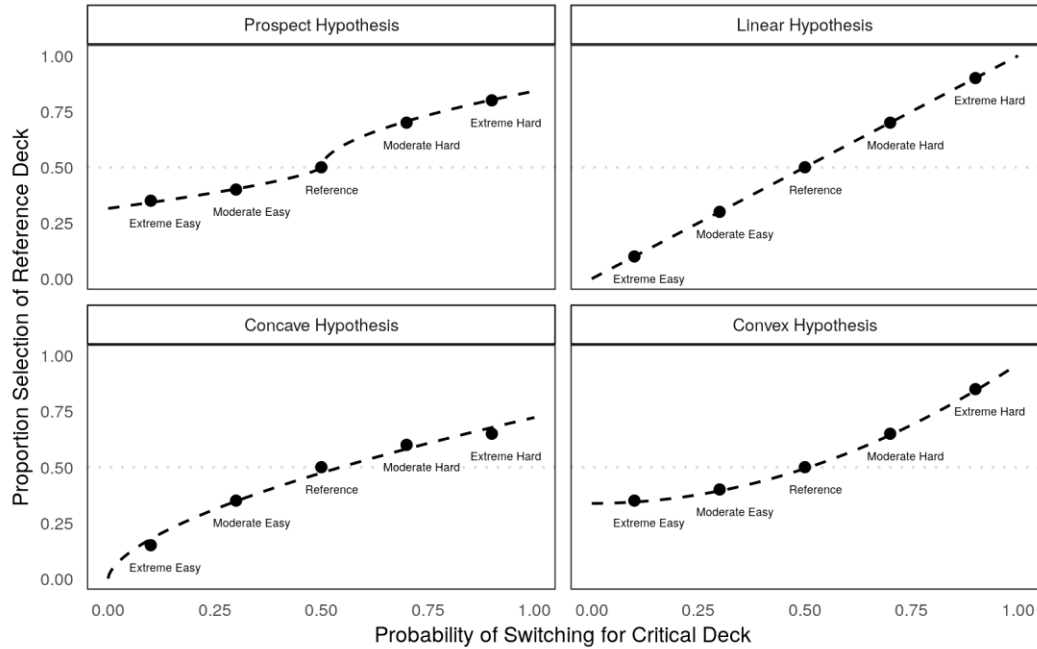


Figure 5. Proportion selection of the reference deck in Experiment 1 under each hypothesis. The “Reference” condition can be thought of as the hypothetical scenario where choices are evenly distributed between two decks associated with 0.5 probability of switching. Each hypothesis predicts that proportion selection of the reference deck will increase as the proportion of switching for the critical deck increases—the critical differences reside in the slope of this increase across the hypotheses. The prospect model is the only hypothesis to predict larger changes in selection around the reference (i.e., 0.5) than at the extremes.

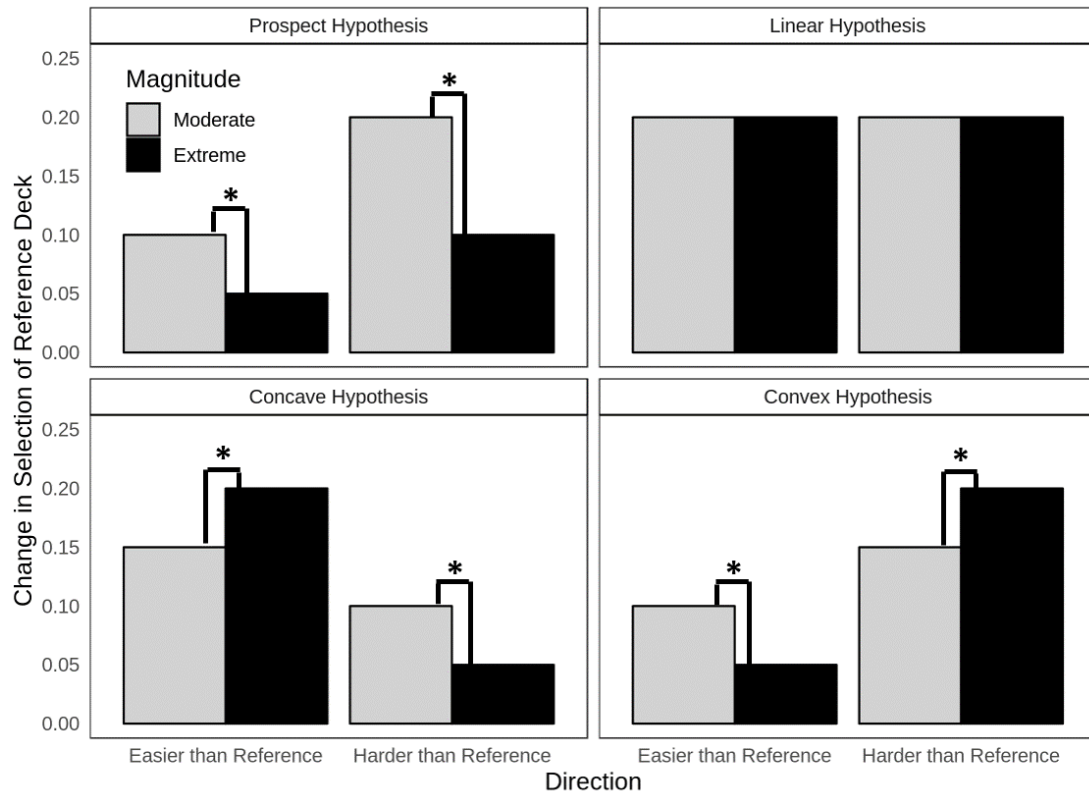


Figure 6. Change in selection of reference deck as critical deck increases in difficulty in Experiment 1. The extreme (black) bars represent the difference in selection of the reference between the extreme conditions and their respective moderate condition. The moderate (grey) bars reflect the difference in selection of the reference between the moderate conditions and .05. A significant interaction is predicted on all hypotheses except linear. The prospect hypothesis is the only model that does *not* predict a crossover interaction. In other words, it is the only hypothesis that predicts larger change in selection of the reference deck for the moderate conditions compared to their respective extreme conditions.

* Predicted significant difference

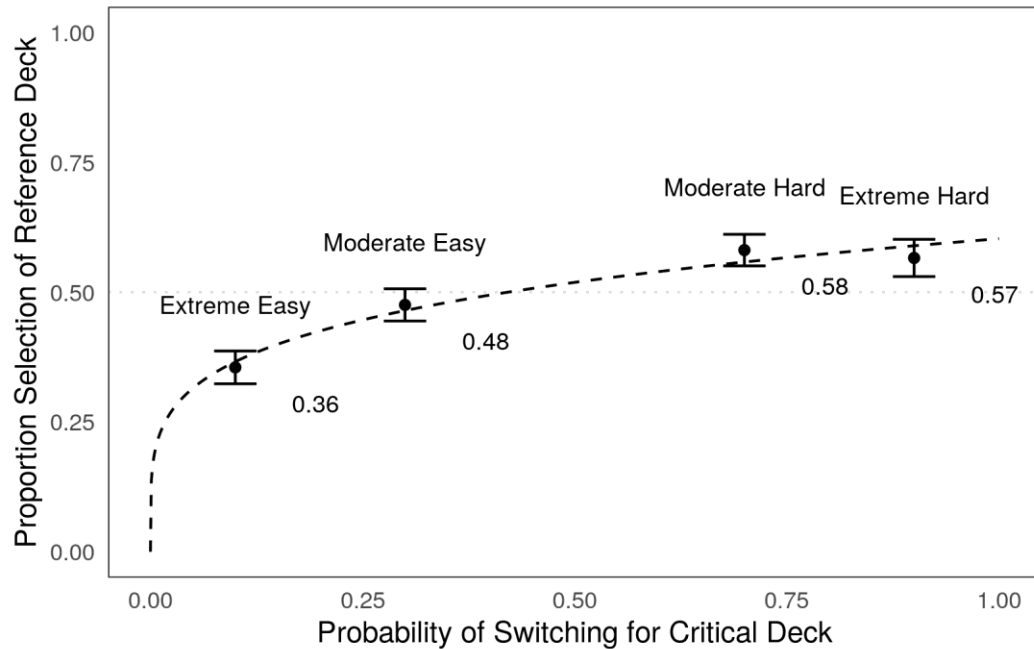


Figure 7. Observed proportion selection of the reference deck as a function of proportion of switching for the critical deck in Experiment 1. The black, dashed line represents a basic power function fit to the condition means for demonstration purposes. The grey, dotted line represents selecting between the two decks at chance. Points represent condition means and error bars represent standard errors. Condition means appeared to be most in line with the concave hypothesis, as differences in choice are larger between the easier conditions and diminish as conditions increase in difficulty.

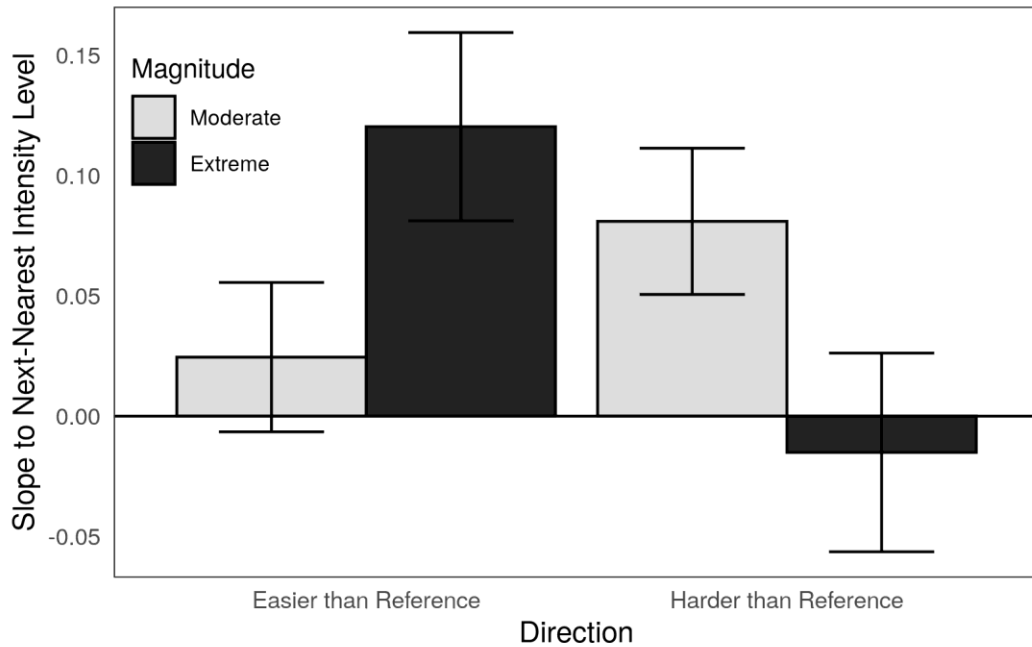


Figure 8. Slope of choice over probability of switching for the critical deck for the four, magnitude X direction conditions in Experiment 1. Consistent with the concave model, the largest slope was between the two easiest conditions (i.e., extreme and moderate easier than reference). The slope tended to diminish as the critical deck increased in direction, as the smallest slope was between the two hardest conditions (i.e., extreme and moderate harder than reference).

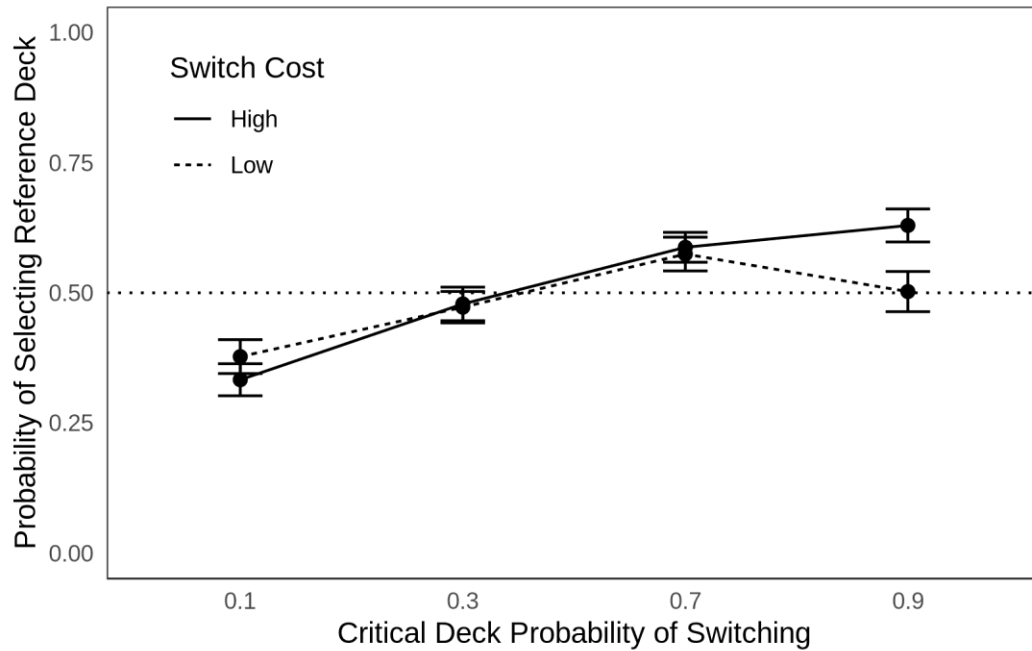


Figure 9. Probability of selecting the reference deck as a function of the critical deck probability of switching and individual differences in switch cost in Experiment 1.

Switch cost represents a median split on an individual measure of switch cost calculated in the cued practice phase of Experiment 1. The horizontal, dotted line represents selecting between the reference and critical decks at chance. Points represent means and bars represent standard errors.

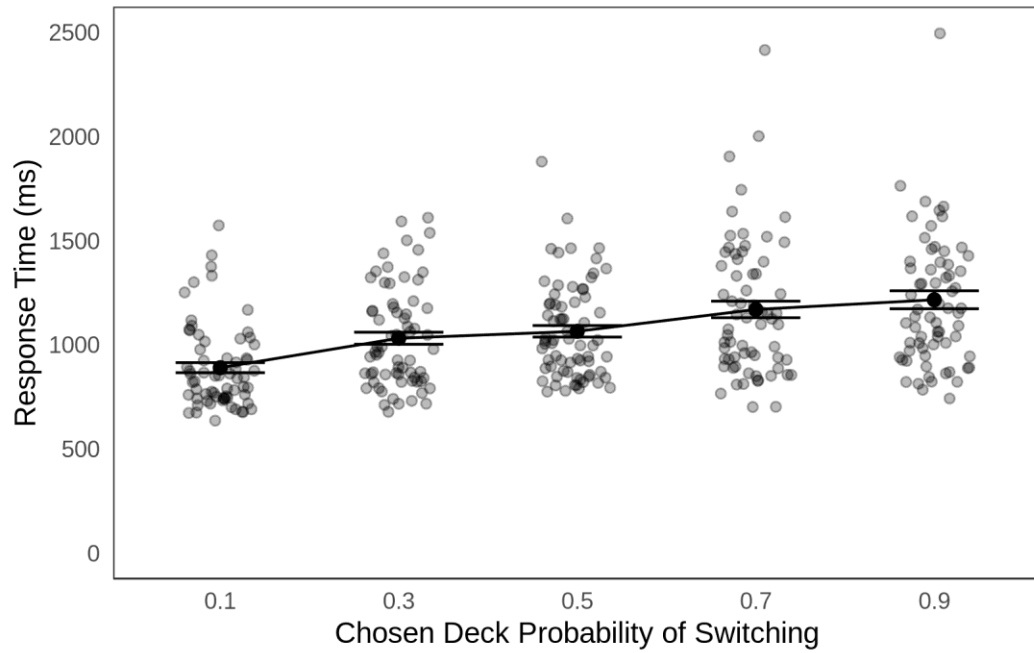


Figure 10a. Response time as a function of chosen deck probability of switching in Experiment 1. Each point represents participant-wise means. The line and the points along the line represent condition means. Bars reflect standard errors.

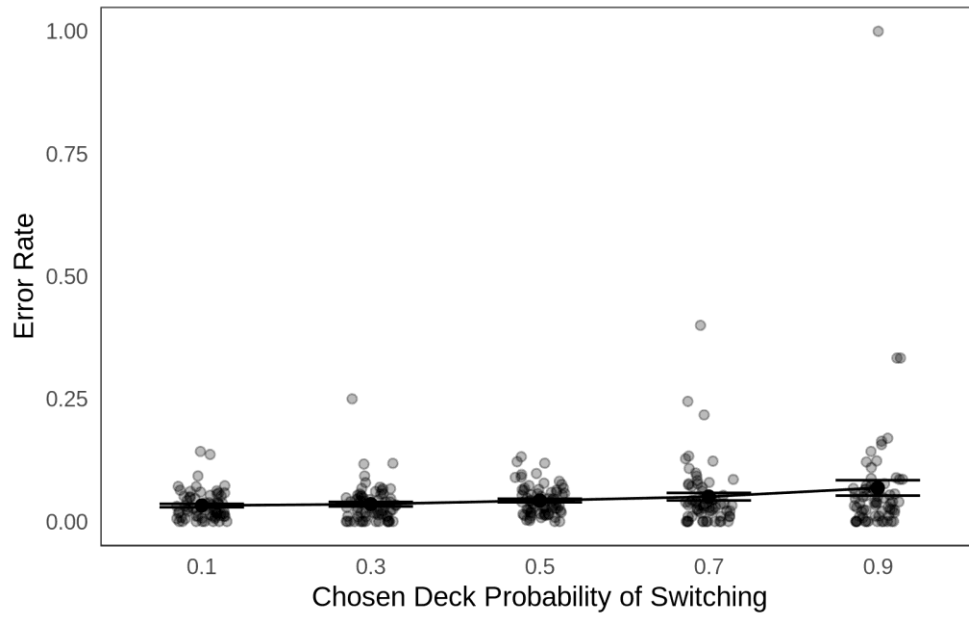


Figure 10b. Error rate as a function of chosen deck probability of switching in Experiment 1. Each point represents participant-wise means. The line and the points along the line represent condition means. Bars reflect standard errors.

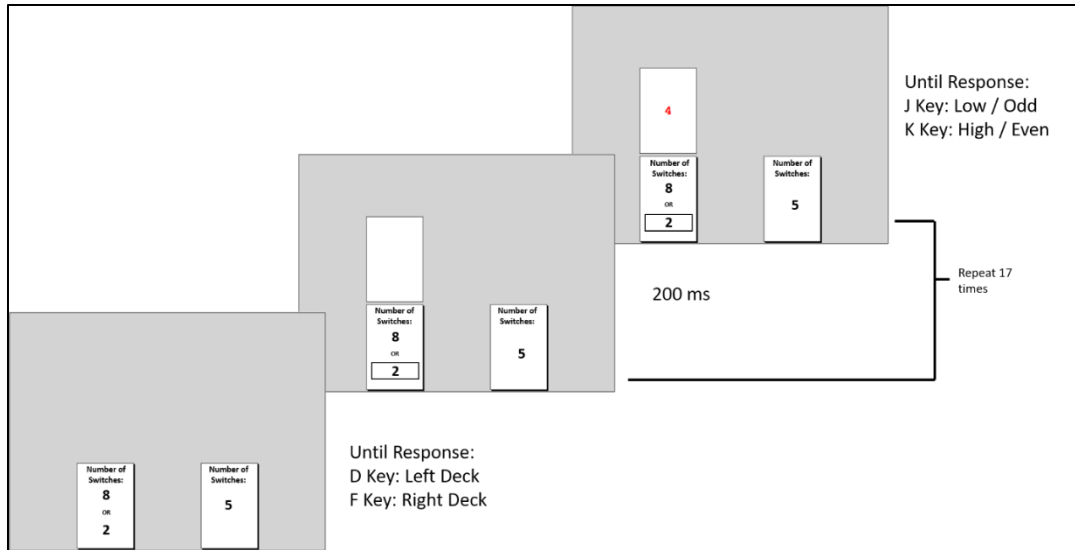


Figure 11. Trial sequence for Experiment 2. The possible outcomes for each deck are labelled on the face of each deck as digits. After a deck is selected, a black box appears around the selected outcome signaling to participants how many out of the upcoming 16 trial transitions will be task switches. After 17 cued task trials, the deck outcomes are cleared from the screen for 500 ms, leaving only the white decks, before the trial sequence repeats (not pictured here).

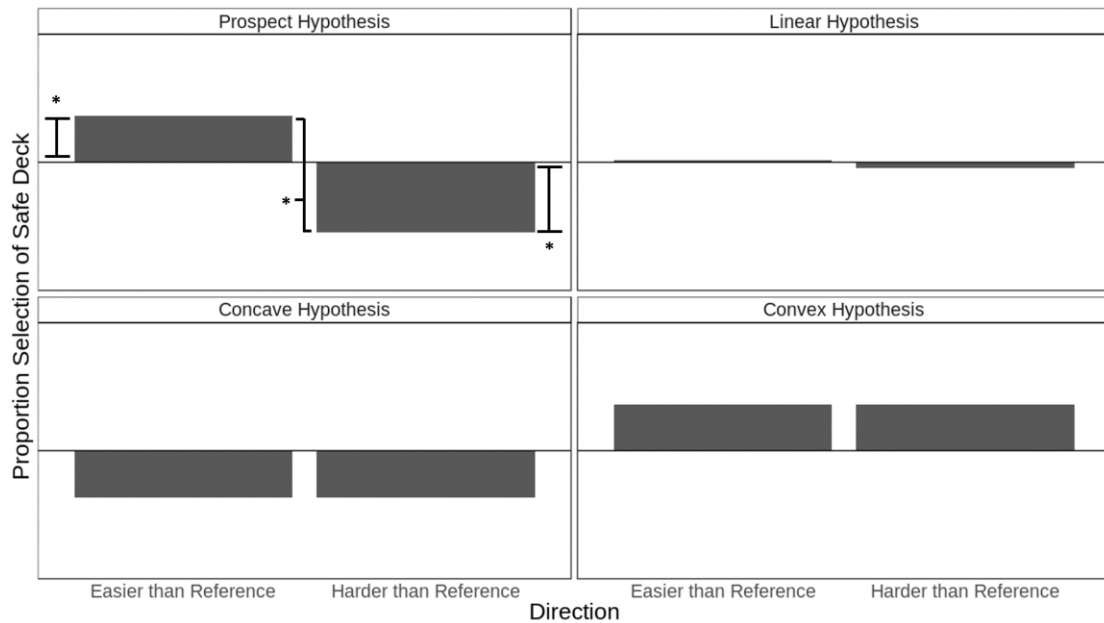


Figure 12. Proportion of Safe Deck Selections as a function of deck outcome direction by hypothesis in Experiment 2. The black, horizontal line represents risk neutral, or selecting between the decks at chance. The prospect hypothesis is the only model to predict that risk preferences diverge as a function of whether the deck outcomes are harder or easier than the reference point. When deck outcomes are easier than or equal to the reference point, selection of the safe deck should be significantly higher than chance; when deck outcomes are harder than or equal to the reference point, selection of the risky deck should be significantly higher than chance. All other hypotheses either predict that risk preferences are higher (i.e., convex) or lower (i.e., concave) than chance, or not significantly different from chance (i.e., linear).

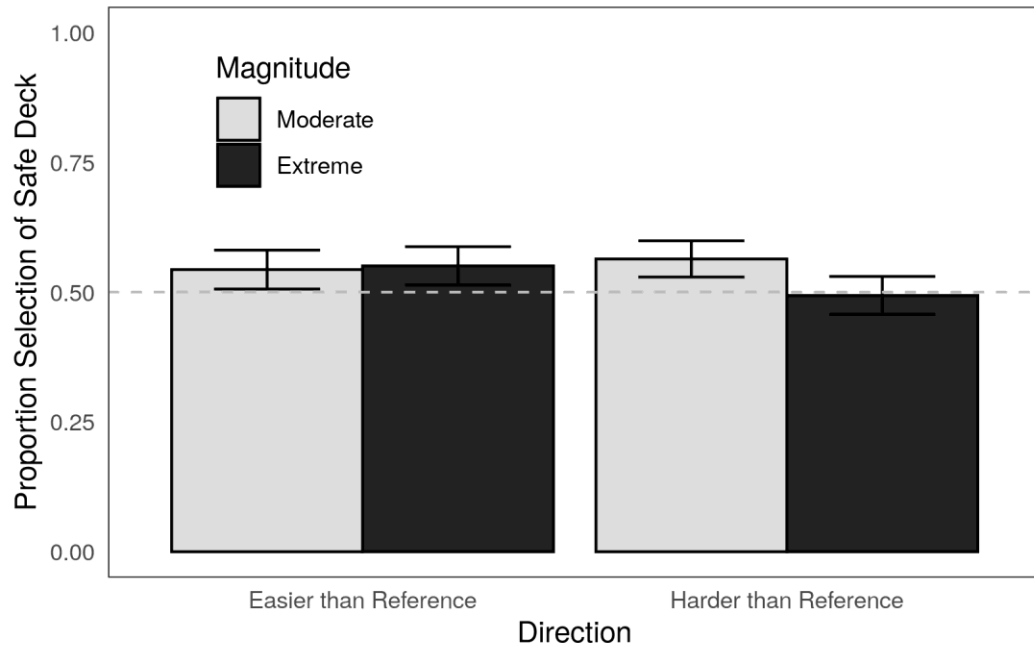


Figure 13. Proportion selection of safe deck as a function of deck magnitude and direction in Experiment 2. The dashed, grey line represents selecting between decks at chance.

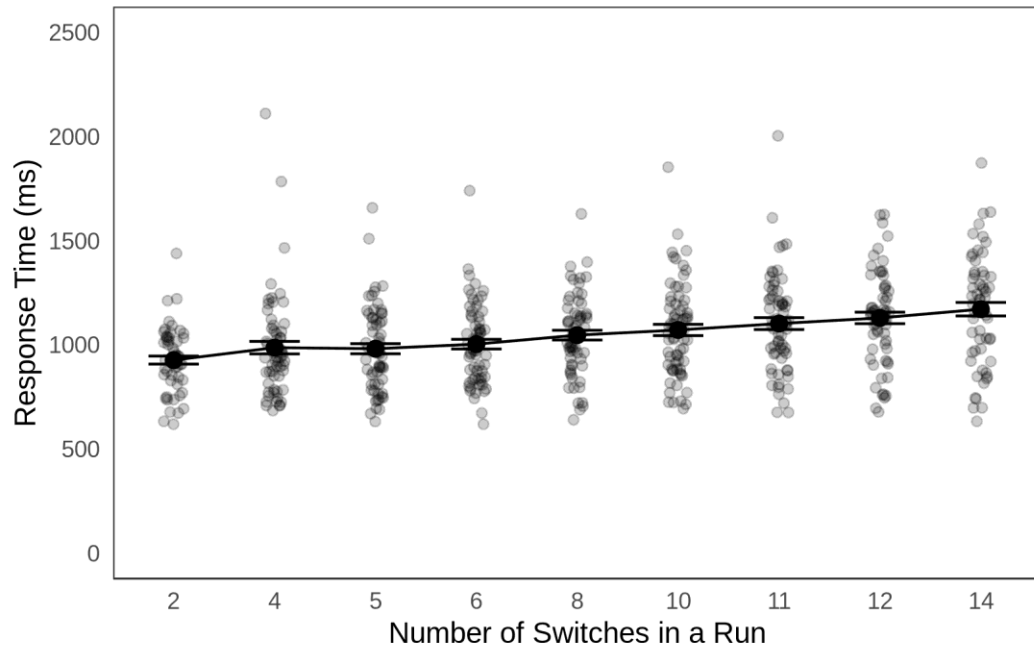


Figure 14a. Response time as a function of number of switches in a run in Experiment 2.

Each point represents participant-wise means. The line and the points along the line represent condition means. Bars reflect standard errors.

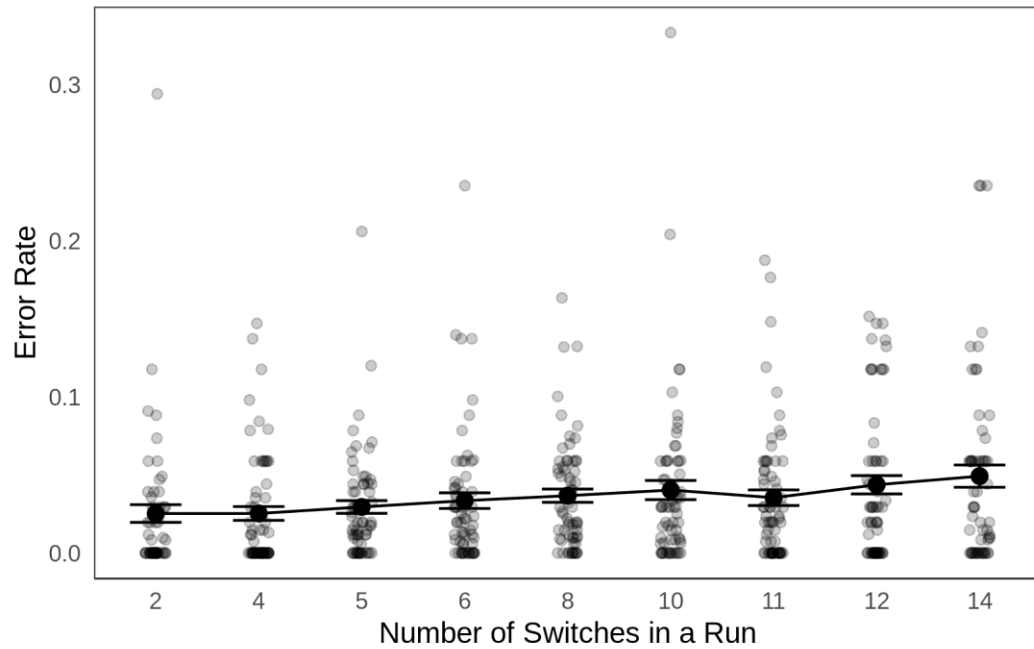


Figure 14b. Error rate as a function of number of switches in a run in Experiment 2. Each point represents participant-wise means. The line and the points along the line represent condition means. Bars reflect standard errors.

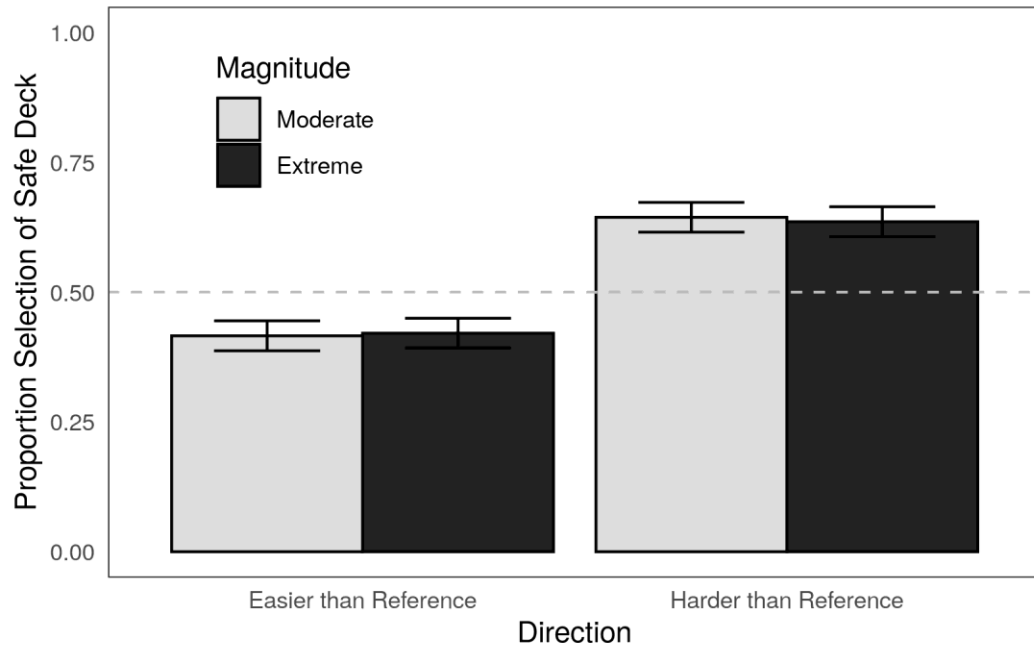


Figure 15. Proportion selection of the safe deck as a function of deck magnitude and direction in Experiment 3. The dashed, grey line represents selecting between decks at chance.

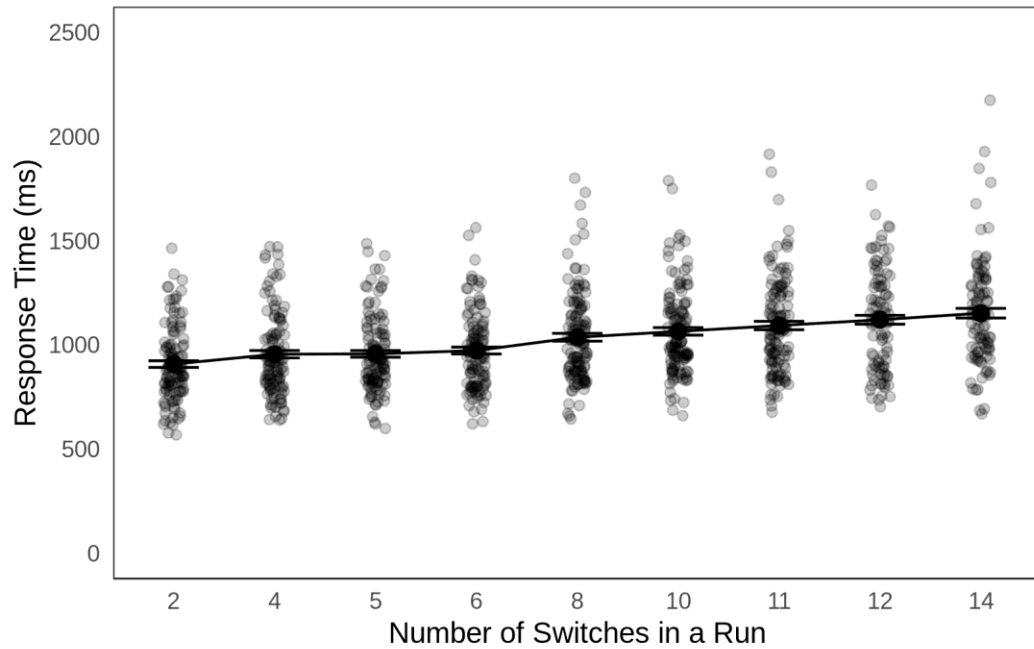


Figure 16a. Response time as a function of number of switches in a run in Experiment 3.

Each point represents a participant-wise mean. The line and the points along the line represent condition means. Bars reflect standard errors.

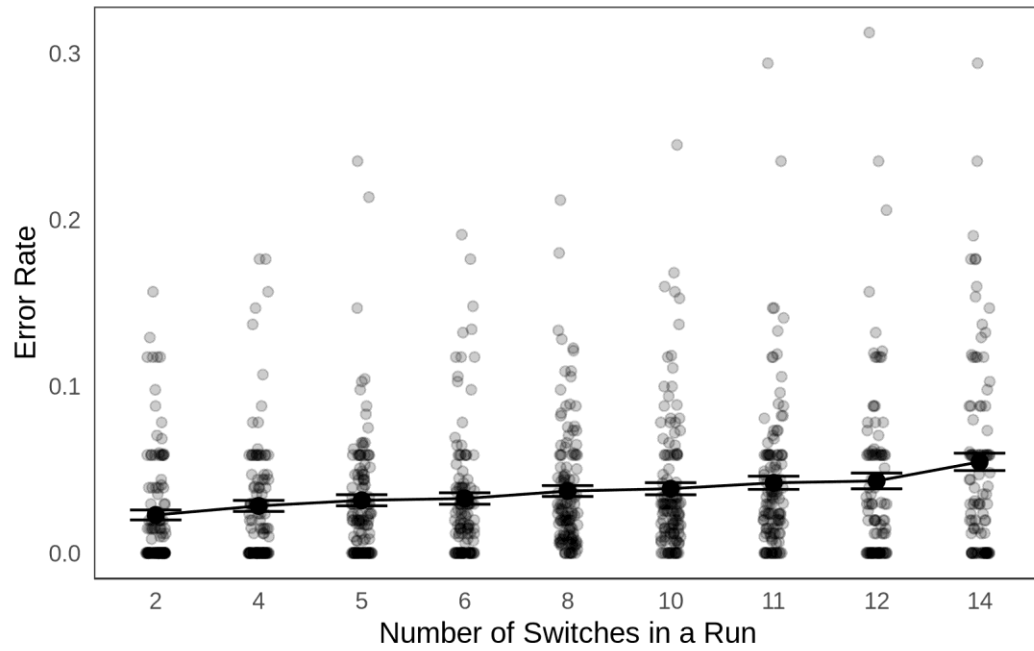
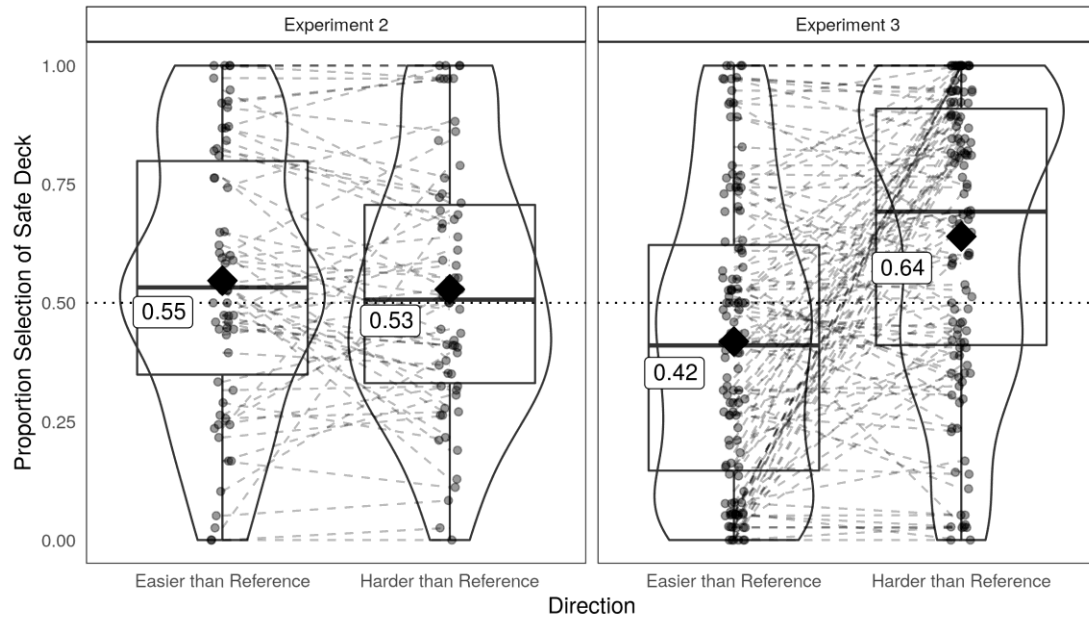


Figure 16b. Error rate as a function of number of switches in a run in Experiment 3. Each point represents a participant-wise mean. The line and the points along the line represent condition means. Bars reflect standard errors.



Black diamond reflects condition means. Horizontal, black, solid lines reflect medians.

Figure 17. Proportion selection of safe deck by experiment, direction, and participant.

Black diamond and number labels reflect condition means. Dashed lines and points represent individual participant data.

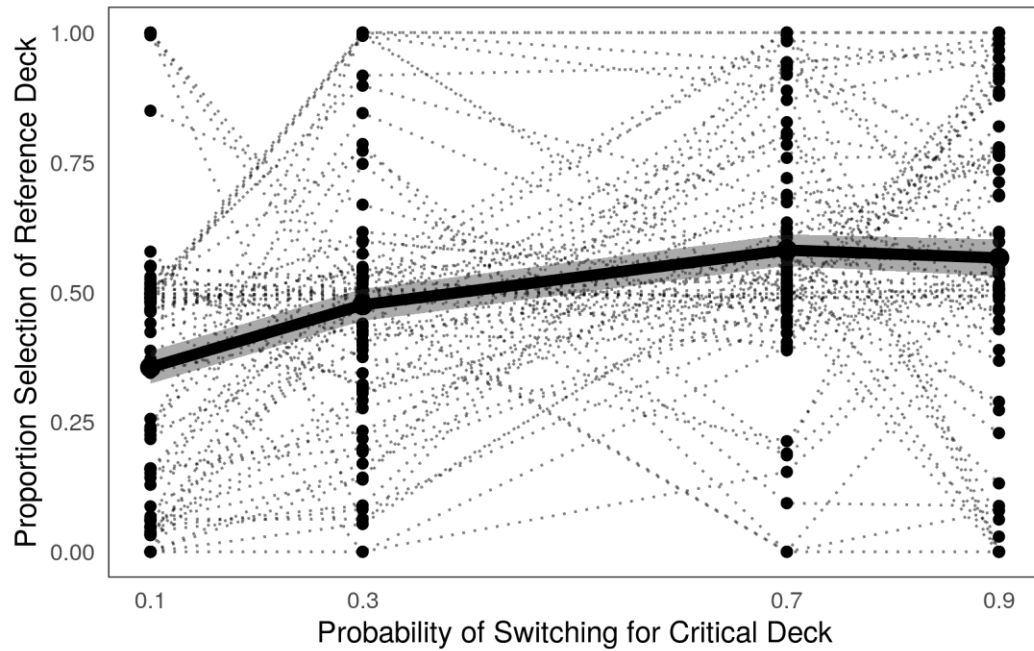


Figure 18. Proportion selection of reference deck as a function of probability of switching for critical deck in Experiment 1. Small, dotted lines represent participant-wise proportions. The thick, solid line represent mean proportions for each condition. The grey band around the thick, solid line represents ± 1 standard error around the mean.

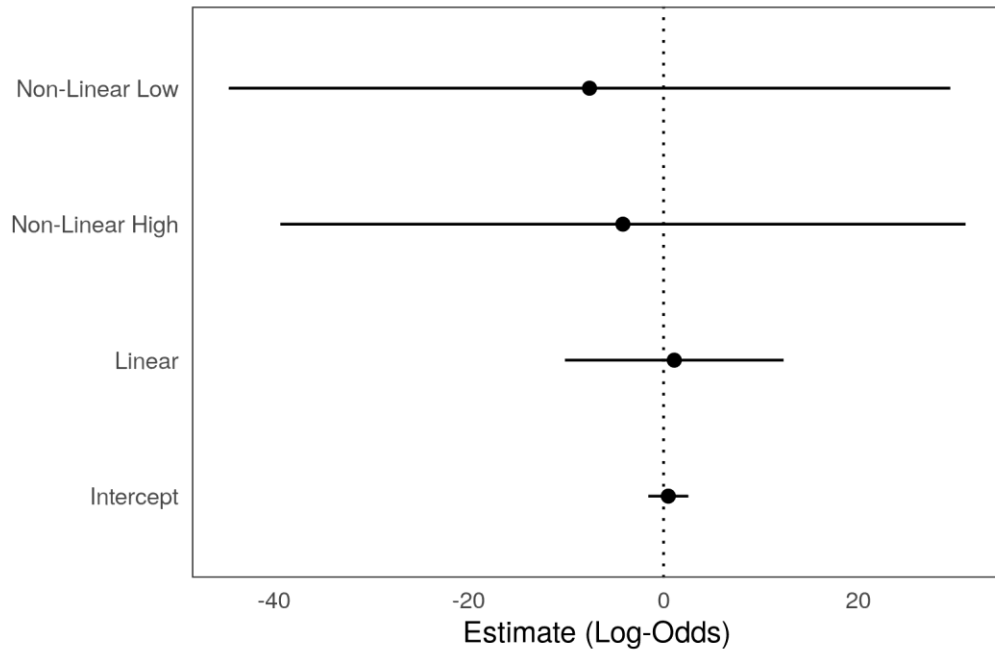


Figure 19. Fixed and random effects estimates from the MARS model fit to the entire data in Experiment 1. “Linear” represents the linear parameter; “Non-Linear High” represents the parameter for the non-linear term for observations where the probability of switching for the critical deck was greater than 0.5; “Non-Linear Low” represents the parameter for the non-linear term for observations where the probability of switching for the critical deck was less than 0.5. Lines around each dot represent ± 1 between-participant standard deviation around the fixed effects estimates. The vertical, dotted line represents the null hypothesis.

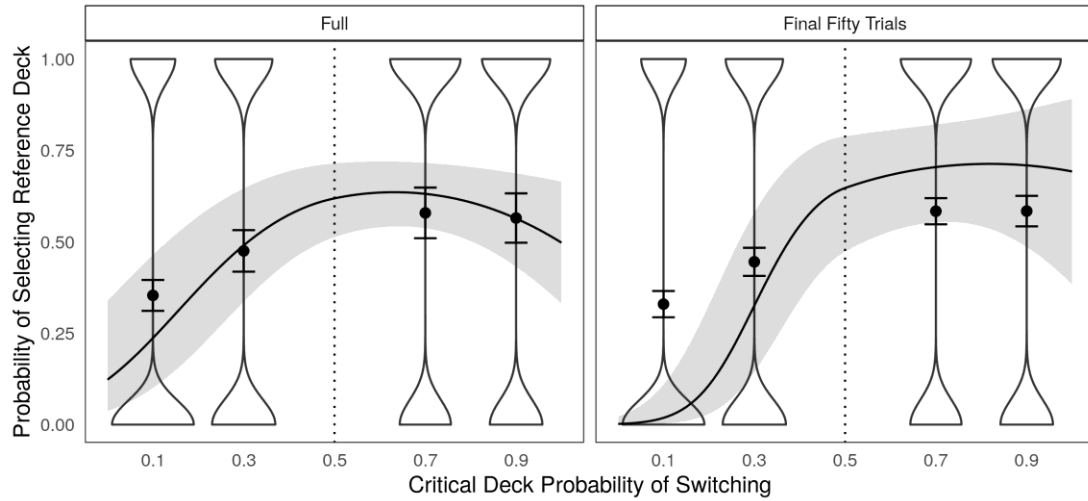


Figure 20. MARS model predictions across conditions in Experiment 1. The left panel represents the model that was fit to the entire data; the right panel represents the model that was fit to only the final fifty trials in each block. The violin objects represent density of responses at the binary observation level. The dots represent condition means; error bars around the dots represent ± 1 standard error around the condition means. The lines reflect predictions from the MARS model; the bands around the lines reflect 95% prediction intervals around the predictions (not taking random effects into consideration).

Appendix

The 2 (direction: harder than reference vs. easier than reference) X 2 (magnitude: moderate vs. extreme) ANOVA on choice data reported in the text for Experiment 1 has two main drawbacks. The first drawback has to do with how the competing hypotheses differ from one another in the context of the Experiment 1 paradigm. All hypotheses predict that the probability of selecting the reference deck will monotonically increase as a function of the probability of switching for the critical deck—the hypotheses diverge on the *nature* of this monotonic function (e.g., linear vs. concave vs. convex vs. prospect). To compensate for this fact, the dependent variable of the 2 X 2 ANOVA was converted to a difference score so that the hypotheses would make different predictions about the relative order of the conditions.

The other main drawback of the 2 X 2 ANOVA was its inability to control for participant-level variance. Given observations nested within a group (such as participants), statistical experts have argued that it is becoming increasingly popular and more prudent to control for the impact of group-level variance when attempting to analyze the influence of some fixed effect on an outcome (Barr et al., 2013). Plotting the participant-wise choice proportions across critical deck difficulty conditions revealed a wide degree of between-participant variability (see Figure 18). Therefore, even though the ANOVA found support for the concave hypothesis, it is unclear whether this conclusion is generalizable after controlling for differences between participants.

The best statistical analysis to address the two concerns above is a multivariate adaptive mixed-effects regression spline (MARS; Friedman, 1991). MARS is a non-parametric analysis where the shape of the regression is derived from the data. The

MARS model in the present analysis was specified such that the model could take the form of any of the competing hypotheses, and the significance of the parameters in the model would reveal which hypothesis best explained the data. This adaptability addressed the first concern described above—the data did not need to be transformed or aggregated in order to test among hypotheses. The second concern described above was addressed by using a mixed-effects model such that the fixed effects were free to vary between participants, allowing the between-participant variance in the fixed effects to be partialled out of the estimation of the fixed effects themselves. Finally, a logit link function was used to transform the binary dependent variable (choose reference deck vs. do not choose reference deck) to boundless space (i.e., log odds). This transformation is the same as is used in logistic regression, and it allowed for the data to be modeled at the observation level.

The MARS model took the following form

$$\begin{aligned} \text{logit}\left(P(Y_{ij} = \text{SelectReference})\right) \\ = \beta_{0j} + \beta_{1j}X_{1ij} + \beta_{2j}R_{Lij}X_{1ij}^2 + \beta_{3j}R_{Hij}X_{1ij}^2 + \epsilon_{ij} \end{aligned}$$

Where the left-hand side of the equation represents the log odds of selecting the reference deck on a given trial (i) for a given participant (j). X_1 reflects the probability of switching for the critical deck, and ϵ reflects residual variance in Y not explained by the model. All β s reflect parameters estimated by the model. The two non-linear terms of the model (involving X^2) are controlled by the dummy-coded R variables— R_L is set to 1 when the probability of switching for the critical deck is less than 0.5 and is set to 0 otherwise, while R_H is set to 1 when the probability of switching for the critical deck is greater than 0.5 and is set to 0 otherwise. Note how this R variable allows the model to estimate two

separate non-linear coefficients (β_2 and β_3) depending on whether the probability of switching for the critical deck is greater than or less than 0.5. Thus, the model can be concave, convex, or linear separately both to the left and right of the reference point (0.5).

Random effects were estimated by allowing the fixed effects to vary between participants. This functionality is explained by level two equations of the model, where

$$\beta_0 = \gamma_{00} + u_{0j}$$

$$\beta_1 = \gamma_{10} + u_{1j}$$

$$\beta_2 = \gamma_{20} + u_{2j}$$

$$\beta_3 = \gamma_{30} + u_{3j}$$

Where each β estimate can be represented as a mean estimate for that parameter (γ_{p0}) and a random effect for a given participant (u_{pj}). Random effects are sampled from a normal distribution for each participant

$$u = N(0, G)$$

Where the mean is 0 and the variance G is given by the variance-covariance matrix

$$G = \begin{bmatrix} \sigma_0^2 & \dots & \dots & \dots \\ \sigma_{01}^2 & \sigma_1^2 & \dots & \dots \\ \sigma_{02}^2 & \sigma_{12}^2 & \sigma_2^2 & \dots \\ 0 & 0 & 0 & \sigma_3^2 \end{bmatrix}$$

Where all variance and covariance parameters were included except for covariance related to σ_3^2 (σ_3^2 is the non-linear term for observations less than 0.5 on probability of switching for the critical deck). The best fitting model was determined by specifying the most complex model allowed by the data and incrementally eliminating random effect variances from the model until the model converged and contained only significant

random effects (Barr et al., 2013). Covariance related to σ_3^2 was not included because the full model with all random effects included failed to converge. The model specified here fit the data significantly better than the model eliminating covariance related to σ_2^2 [$X(2) = 90.86, p < .001$].

The MARs model revealed extremely large between-participant variability in the fixed effects (see Figure 19), therefore, effects should be interpreted with caution. After controlling for random effects, only the parameter for the non-linear term below the reference point approached significance [$b = 0.0005, 95\% CI = 0.00003, 0.007, X^2(1) = 2.74, p = 0.098$]¹⁶. The nature of this effect suggests that the model below 0.5 on critical deck probability of switching was an upside-down, U-shaped function, which is in line with the concave model. Neither the linear parameter nor the non-linear parameter above 0.5 on critical deck probability of switching were significant ($ps > .3$).

A follow-up MARS model was run using only the final 50 trials from each block. Since the DST, as implemented in Experiment 1, requires participants to learn deck intensities at the beginning of each block (Kool et al., 2010), it is possible that some of the participant-level variability described above was due to differences in learning the deck intensities. Since the present investigation was not focused on learning processes, the first 50 trials were removed as an exploratory way of assessing whether the data in the

¹⁶ These confidence intervals were estimated using the Wald method, which is known to make incorrect assumptions about the residual distribution around fixed effects (Bolker et al., 2008). However, more accurate methods of estimating confidence intervals proved to be computationally infeasible. For this reason, the reported p-value, which was estimated using a likelihood ratio test, is likely to be a more accurate reflection of the parameter's significance than the confidence intervals.

latter half of each block told a cleaner story than the overall data. Running the analysis this way, the parameter for the non-linear term below the reference point was significant [$b = 4.37 \times 10^{-11}$, 95% $CI = 2.16 \times 10^{-14}, 8.87 \times 10^{-8}$, $X^2(1) = 7.77$, $p = 0.005$]. All other parameters failed to reach significance. The direction of the parameter for the non-linear term below the reference point was consistent with the prior analysis, in that the model takes an upside-down U-shape below the reference point, which is consistent with the concave model. However, there was still a large amount of between-participant variability in the effect [in log-odds: $b = -23.85$, $\sigma_b = 64.83$]. See Figure 20 for a contrast in model predictions between the model fit to the full data vs. the model fit to only the final 50 trials in each block.

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Technical Skills

LANGUAGES: R | Python (Pandas, NumPy, Scikit-Learn) | SQL | Bash | JavaScript / JQuery

MACHINE LEARNING: Hierarchical logistic regression (lme4, nls) | Decision Tree & Random Forest | K-Means | NLP (Word2Vec) | Deep Learning (Basic Familiarity)

GENERAL DATA SCIENCE: Interactive Data Visualization (R Shiny, ggplot2) | Git / GitHub | Big Data (Spark) | Data Wrangling (dplyr, data.table, requests, RegEx, BeautifulSoup) | Jupyter Notebooks

Education

PH.D. CANDIDATE IN COGNITIVE PSYCHOLOGY | LEHIGH UNIVERSITY | MAY 2020

M.S. IN COGNITIVE PSYCHOLOGY | LEHIGH UNIVERSITY | DECEMBER 2016

B.A. IN PSYCHOLOGY | STOCKTON UNIVERSITY | MAY 2013

Experience

THE DATA INCUBATOR FELLOW

SEPTEMBER 2019 – NOVEMBER 2019

- Selected (< 2% admittance rate) among 3000 + scientists and engineers with advanced degrees for skills in statistics, data analysis, and programming to participate in a rigorous, eight-week data science fellowship.
- Built a web application with R Shiny that utilized Python to scrape and cluster data from over 8,000 breweries to make branding recommendations to brewery owners

LEHIGH UNIVERSITY RESEARCH ASSISTANT

SEPTEMBER 2014 – AUGUST 2019

- Used brain imaging and machine learning to improve intelligent tutoring systems.
- Built a browser-compatible version of a lab-based study using JavaScript and JQuery to collect data from 270 subjects in a 40-minute experiment in two days.
- Introduced novel paradigm into the literature to investigate how effort discounts reward during choice.
- Implemented a workflow in R to preprocess which applied statistical models to visualize data from eye tracking experiments to save dozens of hours in manual experimenter labor.

TEACHING ASSISTANT

SEPTEMBER 2014 – AUGUST 2019

- Presented research at over 13 conferences to articulate complex analyses to non-technical audiences.
- Led over six recitations and mentored undergraduates in a variety of Psychology courses, including statistics and research methods.

- Developed an R workshop and taught ten peer graduate students how to use data science tools for research.
- Served as statistical mentor for six undergraduate students working on honors theses.
- Selected as the statistical consultant for faculty member in the department to replace simple ANOVA with hierarchical regression to control for systematic sources of uninteresting variance.

Achievements

PUBLICATIONS

- Braun, D., Arrington, C. M. (2018). Assessing the role of effort and reward in task selection using a reward-based voluntary task switching paradigm. *Psychological Research*, 82, 54-64.